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VLSI PHYSICAL DESIGN

ADVANCE NOTES
ASSIGNMENTS
INTERVIEW QUESTIONS

2023

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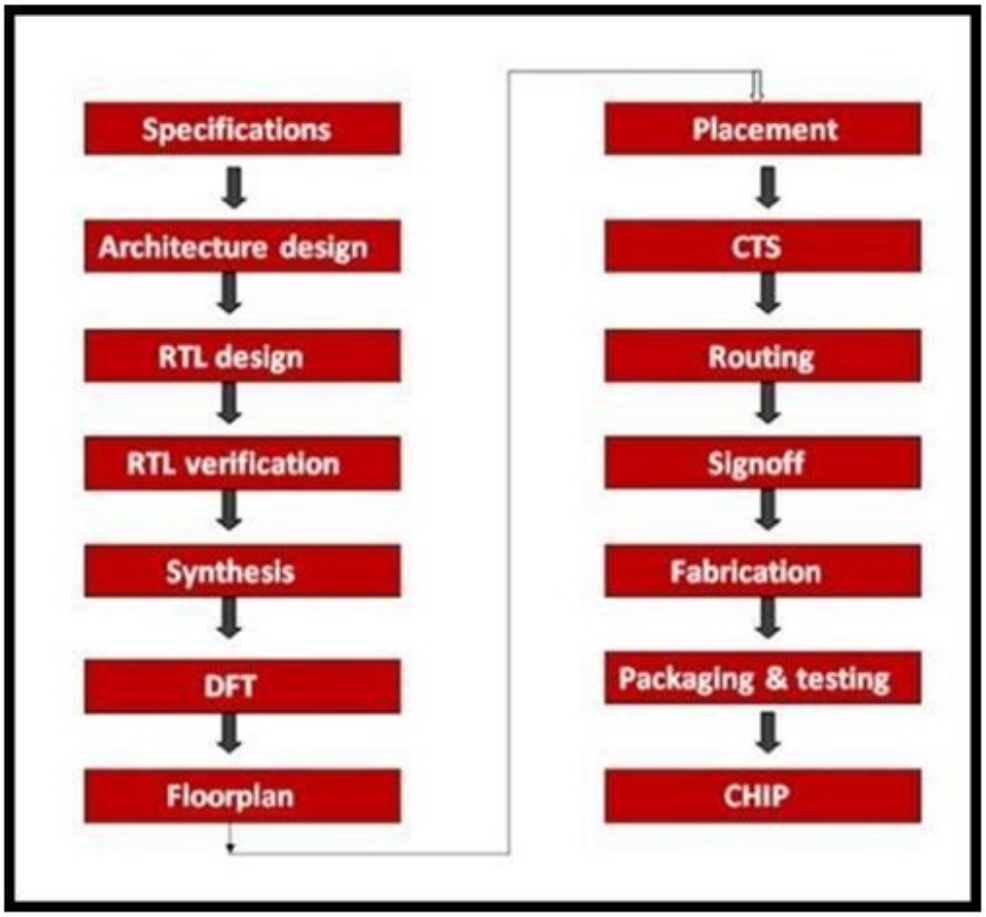
ASIC Design Flow



Specifications: -

In this stage the features information which is expected from the customer is collected by some marketing team.

Architecture design: -

The architecture team will design an architecture based on the specifications. The architecture is like a block diagram we can find the all the details which are using in the design (like Processors,memories) and how they are connected. This architecture team will estimate the block area , how much power is required and cost for the design.

RTL design: -

Register transfer level(RTL) constructing a digital design using combinational and sequential circuit using hardware description language like verilog or VHDL.The above architecture is converted into verilog or VHDL code. This code describes how data is transformed as it is passed from register to register .

RTL Verification:-

It is a functional verification of RTL design. After the RTL design by applying test cases we verify the design in verification stage. If any mistakes are found, then the design is resend to the RTL designing team.

Synthesis:-

It is a process of converting the RTL code into gate level netlist. Up to RTL verification the design is technology independent. In synthesis process the design is converted into technology dependent gate level netlist.

DFT:-

Design for testability (DFT) is a technique which facilitates a design to become testable after production. In this stage we put extra logic like Scan chain insertion along with the design logic ,The DFT will make the testing easy at post production process. At this stage an ATPG (Automatic Test Pattern Generator)file will generated.

Floorplan:-

The floorplan is the process of determining die area/core area. In this stage we determines the size of the die and create the site rows for placement of standard cell . In this step port/pad Placement , Macro placement and power planning is also done.

Placement:-

Placement is the process of automatically assigning correct position to standard cells inside core area with no overlapping. By global placement, standard cells will be placed inside core area roughly. By the detailed placement the standard cells will place in site rows (legalize placement). after placement stage we check for congestion and Timing.

CTS (clock tree synthesis):-

In this stage we build the clock tree by using inverters and buffers. In the chip clock signal is essential to the flip flops. To supply the clock signal from clock source we built the clock tree. It is the process of balancing the clock skew and minimizing insertion delay in order to meet timing and power.

Routing: -

Before the routing stage the connection between the macros, standard cells, clock, i/o port are logical connections. In this stage we connect all the cells physically with the metal layers. Routing is divided as two parts 1) Global routing 2) detailed routing. The global routing will tell for which signal which metal layer is used. In detailed routing the physical connections are done.

Timing Signoff: SPEF extraction and Timing fixing using Tempus/Prime_time is done at this stage

Signoff :

After the routing the physical layout of chip is completed. In signoff stage all the tests are done (DRC, LVS and LEC) to check the quality and performance of the layout before tape out. After this the design is converted into GDS II file.

Fabrication: -

GDS II file information is used to fabricate the chip. The total design is converted into chip by the manufacturing process in foundry.

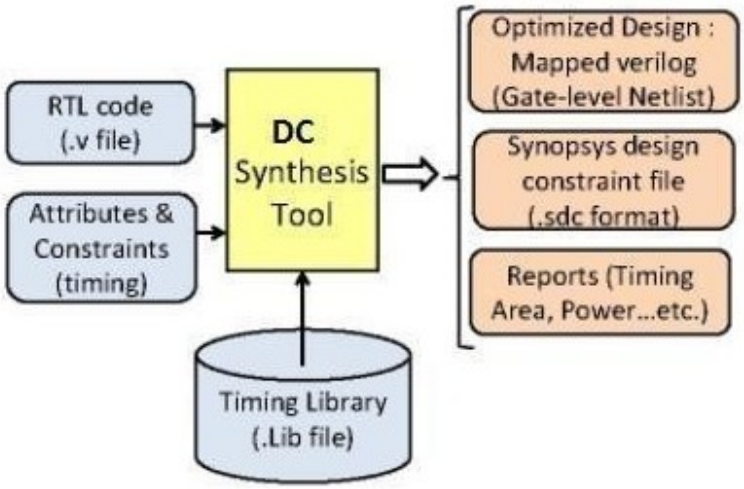
Assignment: Know all the tools for each and every step by different vendors.

Si No	Design Phase	Synopsis	Cadence	Mentor Graphics / Siemens
1	Functional Verification			
2	Synthesis			
3.	Placement and Routing			
4.	SPEF extraction			
5.	STA			
6	Physical verification			
7	Power Sign-off			

Synthesis

Synthesis transforms the simple RTL design into a gate-level netlist with all the constraints as specified by the designer. In simple language, Synthesis is a process that converts the abstract form of design to logic gates.

Input Files for Synthesis:



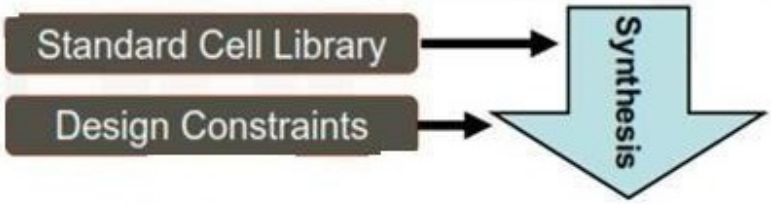
1) RTL Code (.v) : Design Written in Verilog/VHDL in behavioral style modelling.
2) Logical Library file (.lib) : The logical library is also called a timing library or functional library or power library as it contains the functionality, time and power information of standard cells of particular technology.
3) Constraint file. (.sdc) : It provides design Timing, area and power constraints to synthesis tool (Mainly clock information).

Synthesis steps

Synthesis takes place in multiple steps:

1) Elaboration: Elaboration is the process of expanding your HDL description to represent all instances of all modules(Verilog) or entities(VHDL) into unique objects
2) Converting RTL into simple logic gates (Technology independent Schematic) .
3) Mapping those gates to actual technology-dependent logic gates available in the technology libraries.
4) Optimizing the mapped netlist keeping the constraints set by the designer intact.

```
module counter(
     input clk, rstn, load,
     input  [1:0] in,
     output reg [1:0] out);
always @(posedge clk)
  if (!rstn) out <= 2'b0;
  else if (load) out <= in;
  else out <= out + 1;
endmodule
```




```
module counter ( clk, rstn, load, in, out );
  input [1:0] in;
  output [1:0] out;
  input clk, rstn, load;
  wire   N6, N7, n5, n6, n7, n8;

  FFPQ1 out_reg_1 (.D(N7),.CK(clk),.Q(out[1]));
  FFPQ1 out_reg_0 (.D(N6),.CK(clk),.Q(out[0]));
  NAN2D1 U8 (.A1(out[0]),.A2(n5),.Z(n8));
  NAN2D1 U9 (.A1(n5),.A2(n7),.Z(n6));
  INVD1 U10 (.A(load),.Z(n5));
  OA211D1 U11 (.A1(in[0]),.A2(n5),.B(rstn),.C(n8),.Z(N6));
  OA211D1 U12 (.A1(in[1]),.A2(n5),.B(rstn),.C(n6),.Z(N7));
  EXNOR2D1 U13 (.A1(out[1]),.A2(out[0]),.Z(n7));
endmodule
```

Activate W
Go to Setting

Output from Synthesis tool:

1) Verilog Netlist/VHDL Netlist (.v/.vhd)
2) Constraint file (.sdc)

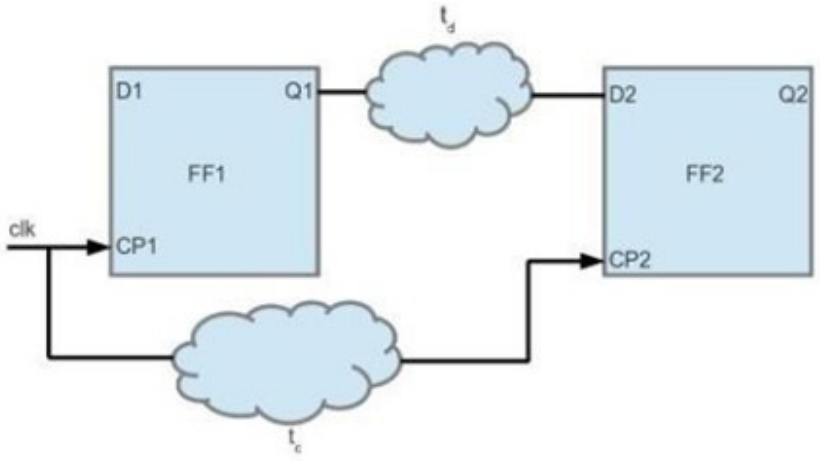
Checks performed During synthesis

1) Static Timing Analysis: Setup and hold check
2) Power Analysis
3) Area Analysis.

Assignment :

For timing Analysis : Following information is required to do STA . Kindly mention where you get following information

1	Flip-flop setup time and hold time	
2	Flip-flop delay	
3	Combinational circuit delay.	
4	Wire delay	
5	Clock frequency/clock information	



INPUT FILES FOR PD

Following input files are required to start PD:

SI no	File Name	Extension	Lender
1	Verilog Netlist	.v	Synthesis team
2	Constraints	.sdc	Synthesis team
3	Logical libraries (Std cells and macros)	.lib (.db)	Vendor
4	Physical libraries (Std cell and macros)	.lef (.Fram_views)	Vendor
5	Technology File	.tf	Foundry
6	TLU + file	. TLUP	Foundry

NETLIST

Netlist: Format is .V

1. It contains logical connectivity of all cells (Std cells, Macros).
2. It contains list of nets Connecting std cells and Macros
3. Each cell has its own instance/cell name and library/ref name

SDC

SDC :Format is .SDC :
These Constraints are timing Constraints .
These Constraints are mainly used to meet timing requirements of design .
Constraints are

1. CLOCK DEFINITIONS: To create different types of clocks
2. Clock uncertainty.
3. Setting Input Delay
4. Setting Output Delay
5. Driving Cell
6. Setting load on output ports

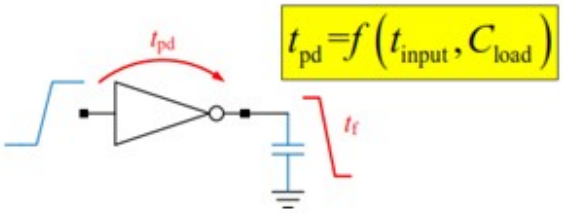
. Timing Exceptions

7. Multi cycle path
8. False path
9. Half cycle path
10. Disable timing arcs
11. Case Analysis

LOGICAL LIBRARIES

Logical libraries: format is .lib

1. Timing information of Standard cells, macros.
2. Functionality information of Standard cells.
3. Timing DRV like max transition and max capacitance.
4. In timing information look-up table is used for output transition, Cell delays, Setup, hold time.
5. Cell delay is Function of input transition and output load. Cell delay is calculated based on lookup tables.
6. It also has wire load model to calculate resistance and capacitance of wires
7. Functionality is used for Optimization Purpose.
8. It also Contain Power information of Std cells.



Look-Up Table

```
lu_table_template(delay_template_5x5) {
  variable_1 : input_net_transition;
  variable_2 : total_output_net_capacitance;
  index_1 ("1000.0, 1001.0, 1002.0, 1003.0, 1004.0");
  index_2 ("1000.0, 1001.0, 1002.0, 1003.0, 1004.0");
}
```

```
cell (INVX1) {
  pin(Y) {
    timing() {
      cell_rise(delay_template_5x5) {
        values ( \
          "0.147955, 0.218038, 0.359898, 0.922746, 1.76604", \
          "0.224384, 0.292903, 0.430394, 0.991288, 1.83116", \
          "0.365378, 0.448722, 0.584275, 1.13597, 1.97017", \
          "0.462096, 0.551586, 0.70164, 1.24437, 2.08131", \
          "0.756459, 0.874246, 1.05713, 1.62898, 2.44989"); }
```

Above is the table for calculating rise cell delay, Index-1 is input transition values and index-2 is output load capacitance values.

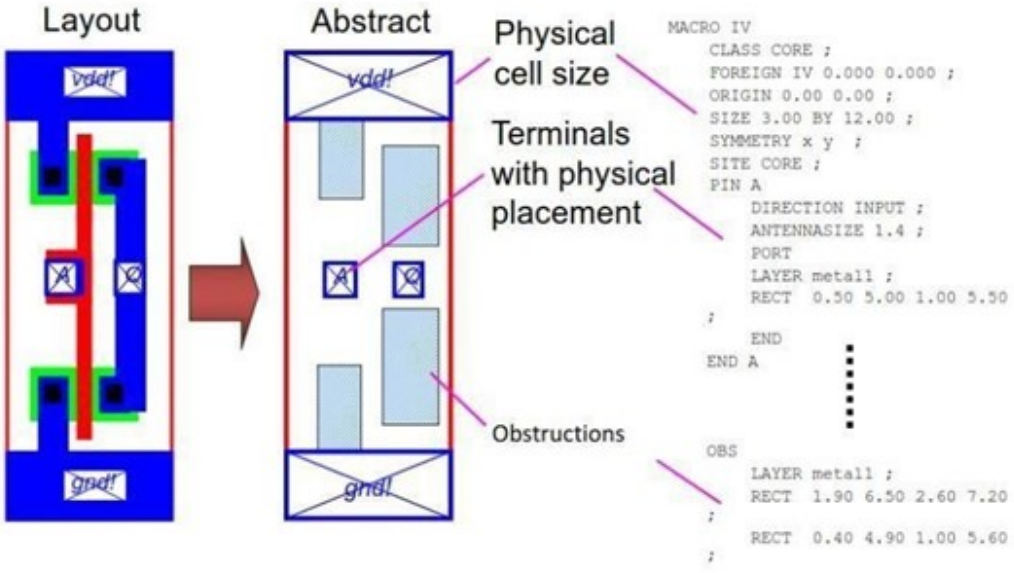
Wire load models:

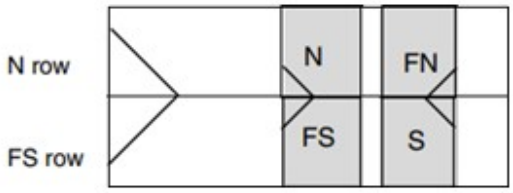
```
library (myLib) {
  wire_load("WLM1")
    resistance: 0.0006 ; // R per unit length
    capacitance: 0.0001 ; // C per unit length
    area : 0.1 ; // Area per unit length
    slope : 1.5 ; // Used for linear extrapolation
    fanout_length(1, 0.002) ; // for fo=1, Lwire=0.002
    fanout_length(2, 0.006) ; // for fo=2, Lwire=0.006
    fanout_length(3, 0.009) ; // for fo=3, Lwire=0.009
    fanout_length(4, 0.015) ; // for fo=4, Lwire=0.015
    fanout_length(5, 0.020) ; // for fo=5, Lwire=0.020
    fanout_length(6, 0.028) ; // for fo=6, Lwire=0.028
  }
} /* end of library */
```

PHYSICAL LIBRARIES (Layout exchange format)

Physical libraries: format is .lef (. Fram views for synopsys)

1. It contains physical information of standard cells, macros, pads.
2. Contain the name of the pin, pin location (Co-ordinates) , pin layers, direction of pin (in, out, inout), uses of pin (Signal, Power, Ground) , height and width of the pin and cell.
3. Size of the cell (Height and width)
4. Symmetry of cell.
5. Obstruction layers and its co-ordinates





X	N and FS orientations should be tried.
Y	N and FN orientations should be tried.
X Y	N, FN, FS, and S orientations should all be tried.

TECHNOLOGYFILE:

1. It contains manufacturing grid definition and site/unit tile definition
2. It contains Name, Number conventions of layer and via
3. It contains Physical, electrical characteristics of layer and via
4. In Physical characteristics Min width, Min Spacing, pitch of layers are present.
5. In Electrical characteristics Max Current Density is present.
6. Colors and pattern of layer and via .
7. Physical Design rules of layer and via
8. Tech file used by the Cadence tool is .techlef format and .tf format by Synopsys tool.

```
LAYER MET1
  TYPE ROUTING ;
  PITCH 3.5 ;
  WIDTH 1.2 ;
  SPACING 1.4 ;
  DIRECTION HORIZONTAL ;
  RESISTANCE RPERSQ .7E-01 ;
  CAPACITANCE CPERSQDIST .46E-04 ;
END MET1

LAYER VIA
  TYPE CUT ;
END VIA
```

TLU PLUS

TLU+ files: format is .TLUP:

1. R,C parasitics of metal layer per unit length.
2. These (R,C parasitics) are used for calculating Net Delays.
3. We can get TLU+ file from ITF file
4. For Loading TLU+ files we have to load two files .
5. Those are Max TLU+, Min TLU+

MAP file.

1. MAP file maps the layer and via names of TLU+ file and .tf file .

UPF- File (Unified Power Format)

1. UPF is designed to reflect the power intent of a design at a relatively high level.
2. UPF scripts describe which power rails should be routed to individual blocks, when blocks are expected to be powered up or shut down.
3. It describes how voltage levels should be shifted as signals cross from one power domain to another and whether measures should be taken to retain register and memory-cell contents if the primary power supply to a domain is removed.

Note : To Store Library and design Information ICC2 uses NDM format .
INDM is formed by compiling .lib , .lef and .tf files .

Assignment

1. Open Netlist (.v) and Understand Hierarchical design
2. Open .lib and for cell AND2X4_HVT note following information
   a) Leakage power
   b) Internal Power table
   c) Dynamic power table
   d) Rise/fall output transition table
   e) Rise/fall cell delay table
   f) On output pin Note max_capacitance and max_transition
3. Open. lef file for AND2X4_HVT cell and note
   a) Its dimension
   b) Allowed Orientation
   c) Its pins and Pin layers
4. Open .tf and Note
   a) Site row information
   b) Manufacturing grid
   c) For all routing metal layers note
      i) Min spacing
      ii) Min width
      iii) Pitch
      iv) Routing direction

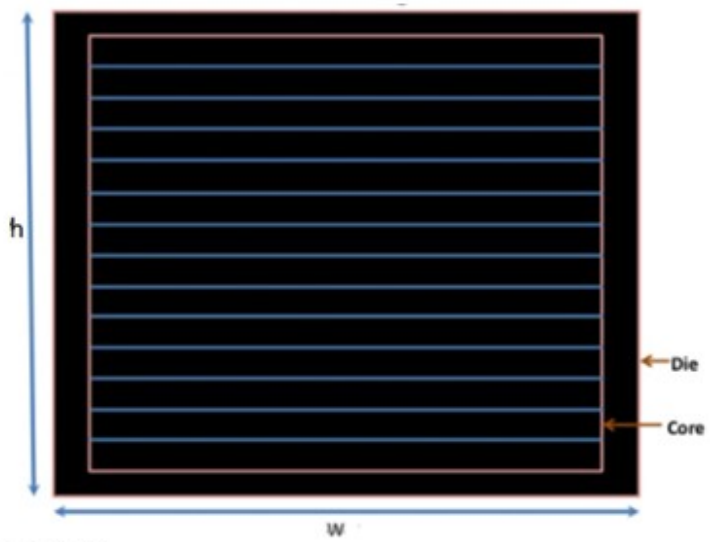
Floor Planning

Floor-plan design is an important step in physical design of VLSI circuits to plan the positions of a set of cell modules on a chip in order to optimize the circuit performance. *Floor planning* is the process of creating an area for macros and standard cells to be placed

Floor Planning Steps :

1. Decide core width and height for die size estimation.
2. Placement of IO pads/Ports .
3. Creating Voltage area.
4. Placement of macros.
5. Adding physical only cells
6. Power planning (pre routing)

1) Decide core width and height for die size estimation.



Core area depends upon :

1. **Aspect ratio:** Aspect ratio will decide the size and shape of the chip .It is ratio of height and width of core area .

   Aspect ratio = width/height

2. **Core utilization:** - Utilization will define the area occupied by the standard cells, macros, and other cells. If core utilization is 0.8 (80%) that means 80% of the core area is used for placing the standard cells, macros, and other cells, and the remaining 20% is free.

   core utilization = (macros area + std cell area)/ total core area

2) I/O Pad placement:

I/O pad placement is done at full chip -level and port placement is done at block level design .

3) Voltage area Creation :

Multi voltage design is used to have trade-off between power consumption and performance

In multi-voltage design different blocks operate with different voltages. We use level-shifters when signals crossing from one power domain to another.

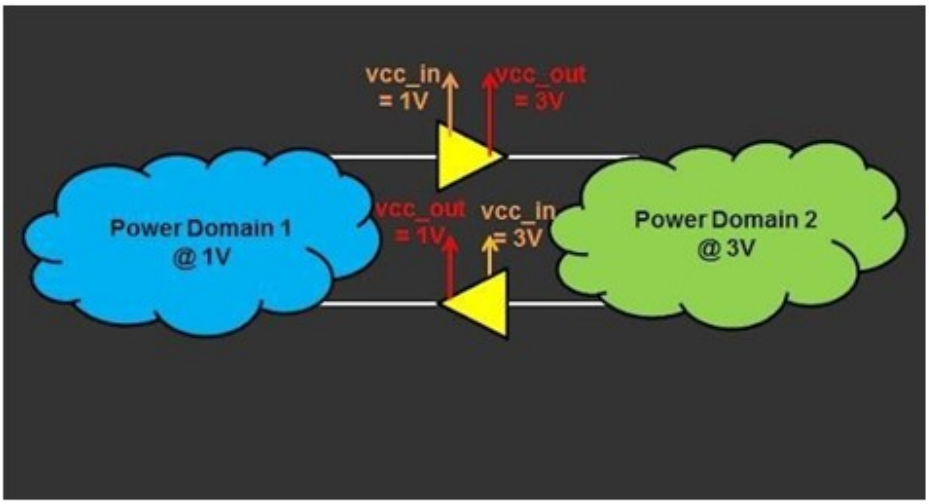


Fig : Different power domain with level shifters

4) Macro Placement:

Macros may be memories, analog blocks. Proper placement of macros has a great impact on the quality and performance of the ASIC design. Macro placement can be manual or automatic. Generally manual macro placement is Preferred

Note :

Types of macros:

- **Hard macros**: The circuit is fixed. We can't see the functionality information about macros. Only we know the timing information.
- **Soft macros:** The circuit is not fixed and we can see the functionality and type of gates present in it, also we know the timing information.

Guidelines to place macros:

1) Placement of macros are the based on the fly-lines (Fly-lines shows the connectivity b/w macro to macro and macro to ports/pins) so we can minimize the interconnect length between IO ports , macros and other cells.

macros to IO pin:

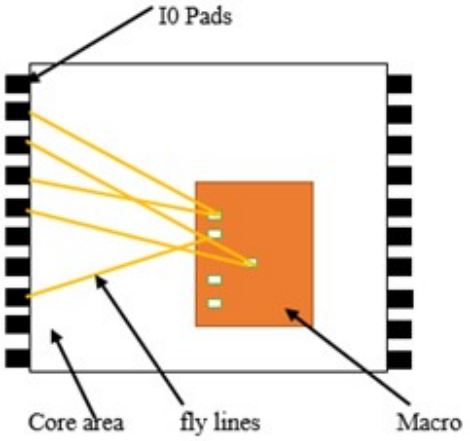


Fig a: macro to IO ports fly lines

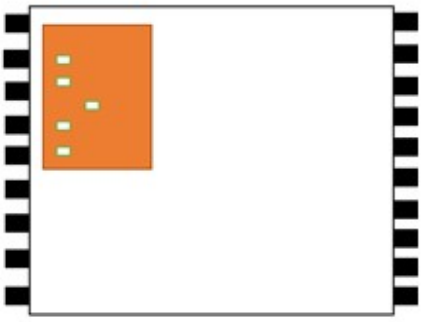


Fig b: macro placed at the core boundary

macros to macros fly lines:

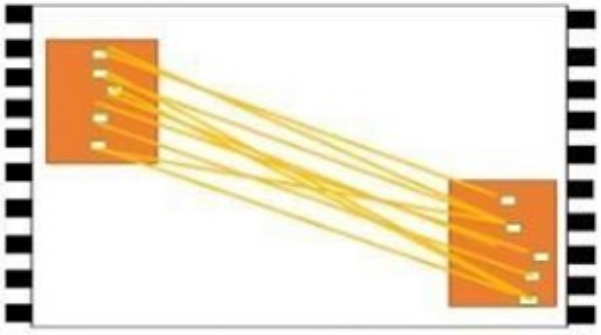


Fig a: macro to macro fly lines

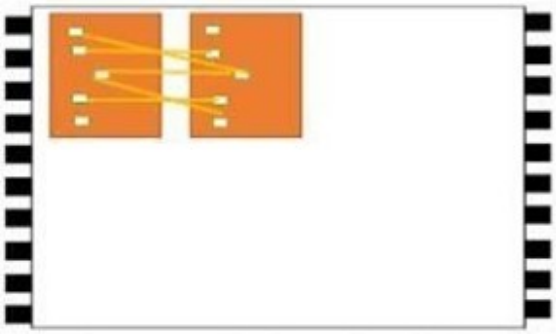
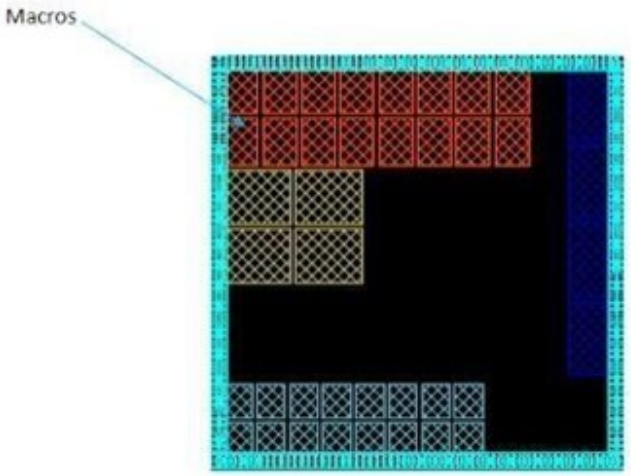
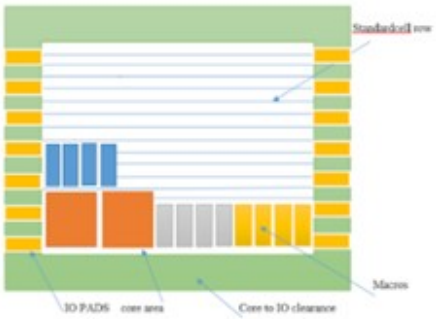


fig b: macros placed near to each other

2) Place the macros around to the boundary of the core, leaving some space between macro to core edge so that during optimization this space will be used for buffer/inverter insertion

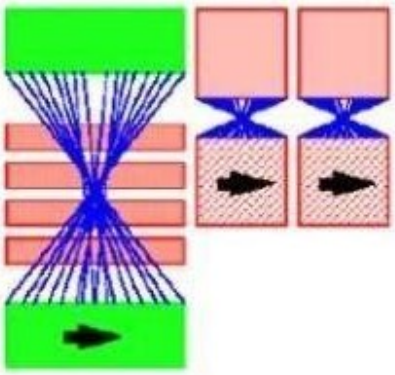
3) Place the macros of same hierarchy together.



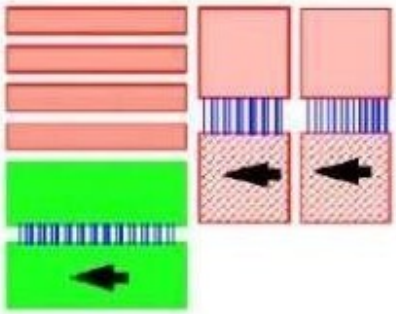


4) Keep sufficient channel between macros
channel width = (number of pins*pitch)/ number of layers either horizontalor vertical
Eg. Let's assume If there are two macros having 50 pins and the pitch values are 0.6 and the total number of horizontal and vertical layers are 12. Means M0 M2 M4 M6 M8 M10 are horizontal layers and M1 M3 M5 M7 M9 M11are vertical layers.
Channel width = ((50+50)*0.6)/6 = 10

5) Avoid crisscross connection of nets after macro placement.

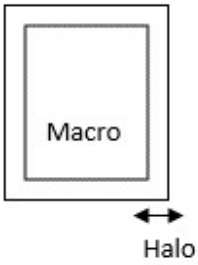


Inefficient floor plan



Efficient floor plan

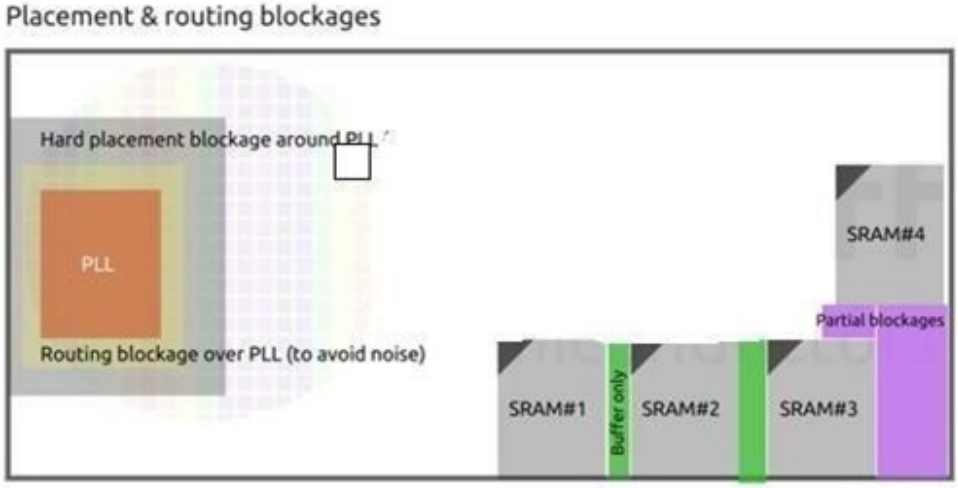
6) Apply keep-out margin/Halo around the four sides of macros so standard cells will notsit near to Macro pins. This technique avoids the congestion and to protect macro pins.



ICC command for keepout margin:
create_keepout_margin -outer {10 10 10 10} macro_name -type soft/hard

7) **Use Placement Blockages near macros to avoid congestion**

Note: Blockages: Blockages are applied to the specific location where the placing of cells is blocked. If the macros moved from one place to another place, blockages **will not move.**



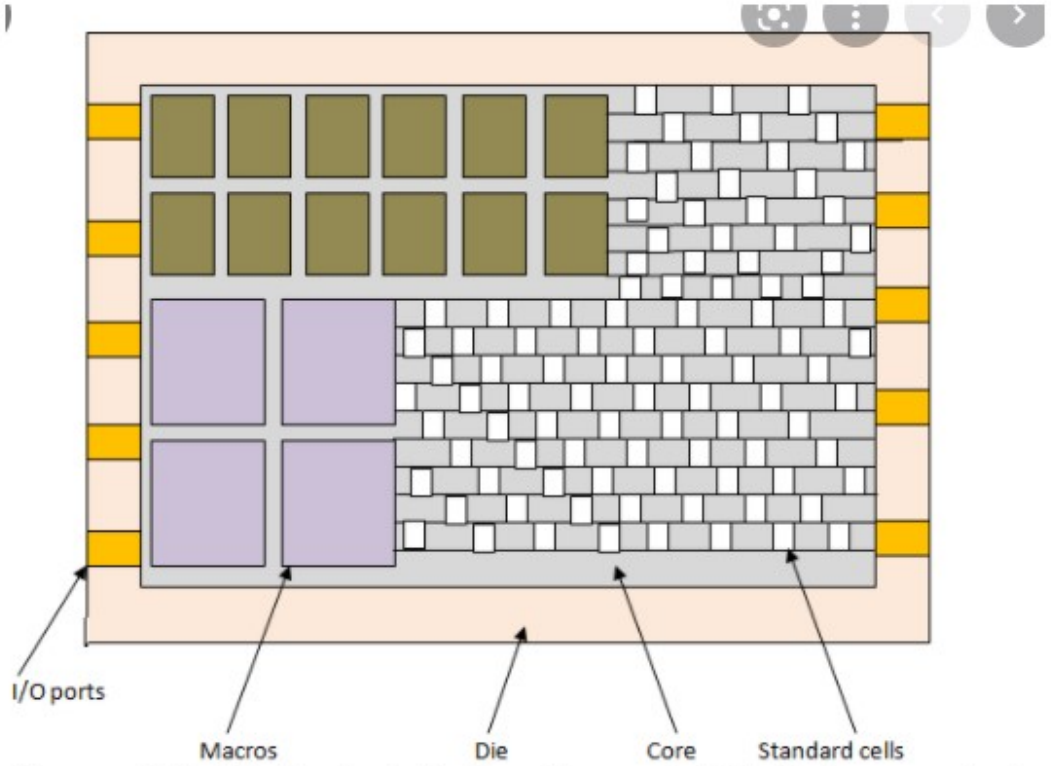
Blockages are of three types. a) Soft b) Hard c) Partial

a) **Soft blockages:**
- Prevents from the placement of std cell and hard macro within the specified area during coarse placement but allows placement of buffer/inv during optimization, legalization and clock tree synthesis.

b) **Hard blockages:**
- No standard cells, macros and buffer/inv can be placed within the specified area during coarse placement, optimization, and legalization.
- Used to avoid routing congestion at macros corners.

c) **Partial blockages:**
- Partial blockages limit the cell density in the specified area.
- Ex: If partial blockage is 40 % , Then in that area the cell density is 60% (blocked % is 40)
- To allow unlimited usage of a partial blockage area specify a blockage percentage to zero.

5) The standard cell rows created for standard cell placement.



- The area allotted for the standard cells on the core is divided into rows and columns where standardcells are placed.
- The height of the row is equal to the height of the most of standard cell and width varies. the height of standard cells varies according to multiple standard cell row height. there may be double-height cells, triple-height cells, etc.
- The standard cells will sit in the row with proper orientation.
- Width of standard cells is also multiple of standard cell columns.

6) Physical-Only Cells :

a) Tap cells:

- A tap cell is a special nonlogic cell with a well tie, substrate tie, or both to avoid latch-up Problem (nwell is connected to VDD and psub is connected to ground).
- Tap cells are placed in regular intervals in standard cell row and distance between two tap cells given in the design rule manual.
- Before global placement (during the floor-planning stage), we can insert tap cells in the block to form a two-dimensional array structure.

b) Tie –Cells

- These are special-purpose cells whose output is constant high or low.
- In design if input of cell needs to be connected to VDD (logic 1) or ground (logic 0), but we do not connect them directly to supply rails, as glitches can damage the transistor so we use tie high and tie low cells. Output of tie cells are connected to the gate of the transistor.
- Tie Cells are inserted during placement stage.

c) Filler Cells

- To have n-well and Substrate continuity we Use filler cells
- If there is continuity b/w nwell and implant layer it is easier for foundry people to generate them and the creation of a mask is a very costly process so it is better to use only a single mask.
- If nwell is discontinuous the DRC will be flagged to place cells further apart i.e maintain the minimum spacing because there is a **well proximity effect.**
- After routing and after timing sign-off we add Filler cells.

d) Decap Cells

- If standard cells and macros are not getting sufficient power due to IR-drop, They may go to metastable state.
- Decap filler cells has small capacitors which are placed between VDD and GND all over the design when the logic cells draw a high amount of current, this capacitor provides extra charge to that cells to reduce IR drop. when logic cellsare not drawing any current, the de-cap is charged up to maximum capacitance.

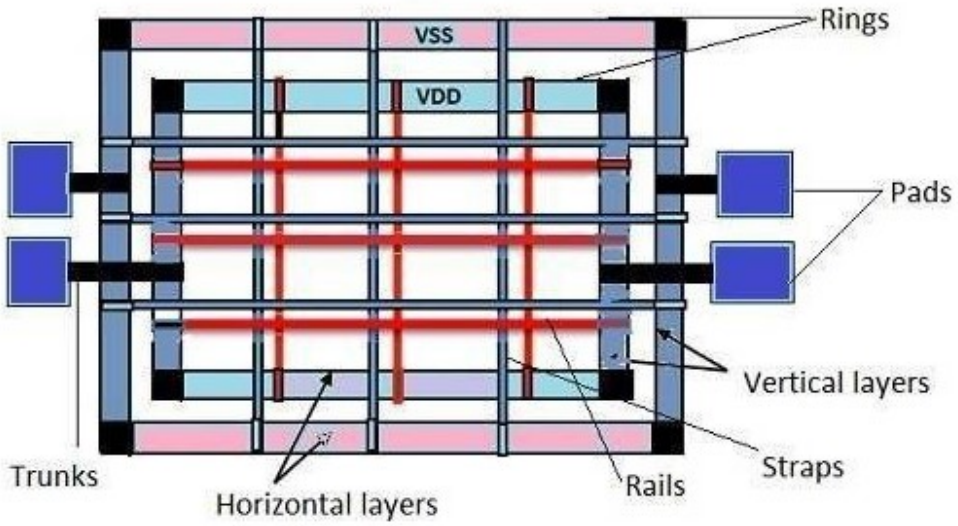
e) Endcap Cells

- Before placing the standard cells, we can add boundary cells to the block. Boundary cells consist of end-cap cells, which are added to the ends of the cell rows and around the boundaries of objects such as the core area, hard macros, hard blockages, and voltage areas.
- End-cap Cells are used to protect the gate of a standard cell placed near the boundary from damage during manufacturing and to avoid the base layer DRC (Nwell and Implant layer) at the boundary.

7) Power planning (pre routing):

What is power planning?

- In power planning power grid network is created to distribute power to each part of the design equally.
- It is used to provide power to macros and standard cells within the IR-drop limit.
- Power planning can be done manually as well as automatically through the tool.



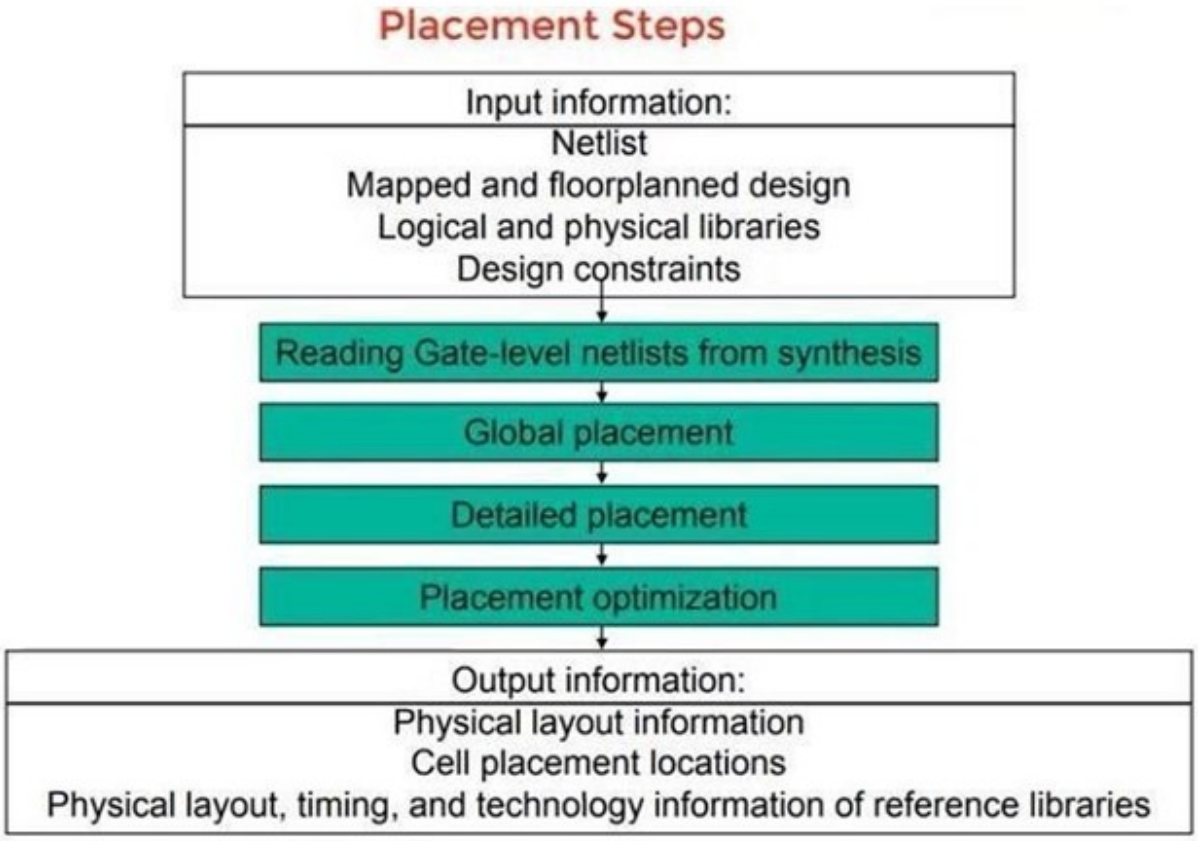
- **Rings:** It Carries VDD and VSS around the chip
- **Stripes:** It Carries VDD and VSS from Rings across the chip
- **Rails:** It connects VDD and VSS to the standard cell VDD and VSS.
- **Trunk:** The connection between Pad and Ring
- **Pad:** Interface from IC to the outside world.

Placement

Placement is the process of placing the standard cells inside the core boundary in an optimal location. The tool tries to place the standard cell in such a way that the design should have minimal congestions and the best timing. Every PnR tool provides various commands/switches so that users can optimize the design in a better way in terms of timing, congestion, area, and power as per their requirements.

Steps in Placement :

i) **Pre-placement Stage**
ii) **Initial Placement / Global Placement / Coarse Placement**
iii) **Legalization**
iv) **Tie Cell insertion**
v) **Scan-Chain Reorder**
vi) **HFNS (High Fanout Net Synthesis)**
vii) **Iteration for Congestion, Timing, DRV, and Power Optimization**



i) **Pre-placement Stage:**

- Perform checks on port placement
- Perform checks on end-cap cells and tap-cells placement
- Perform Power planning check
- Perform checks on macro-placement and use blockages at required places

- See that all macros are fixed and all macros have sufficient keep-out margin
- Verify whether all required input files are sourced
- Set local density limit (G-cell density)
- Make clock ideal and use path-grouping if required

ii) Initial Placement / Global Placement / Coarse Placement

- During the coarse placement, the tool determines an approximate location for each cell according to the timing, congestion and multi-voltage constraints.
- The placed cells don't fall on the placement grid and might overlap each other. Large cells like RAM and IP blocks act as placement blockages for standard cells.
- Coarse placement is fast and sufficiently accurate for initial timing and congestion analysis.

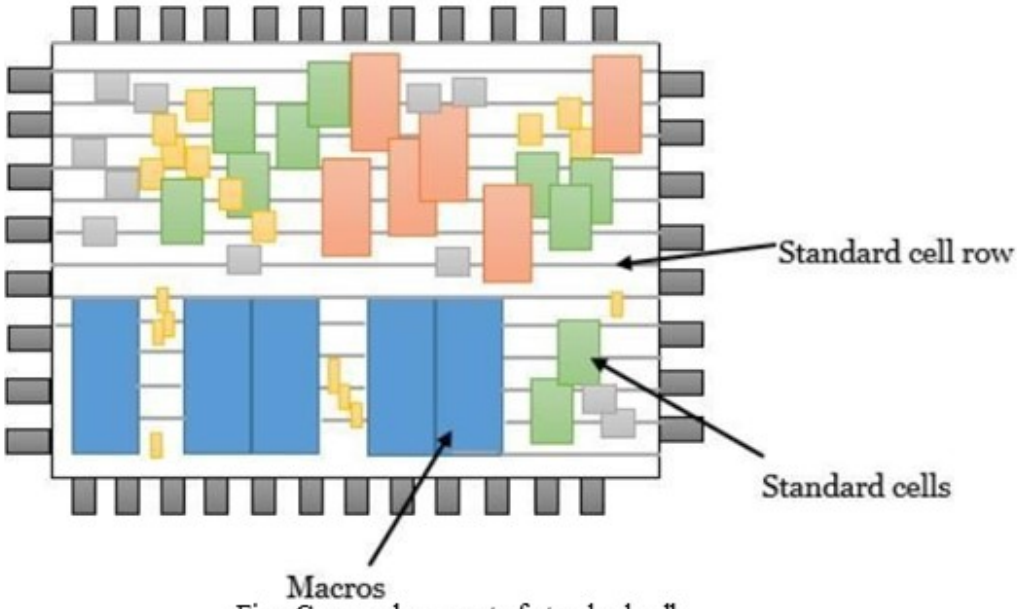


Fig : Coarse placement of standard cells

iii) Legalization:

- During legalization, the tool moves the cells to legal locations on the placement grid and eliminate any overlap between cells.
- These small changes to cell location cause the lengths of the wire connections to change, possibly causing new timing violations.
- Such violations can often be fixed by incremental optimization, for example: by resizing the driving cells.

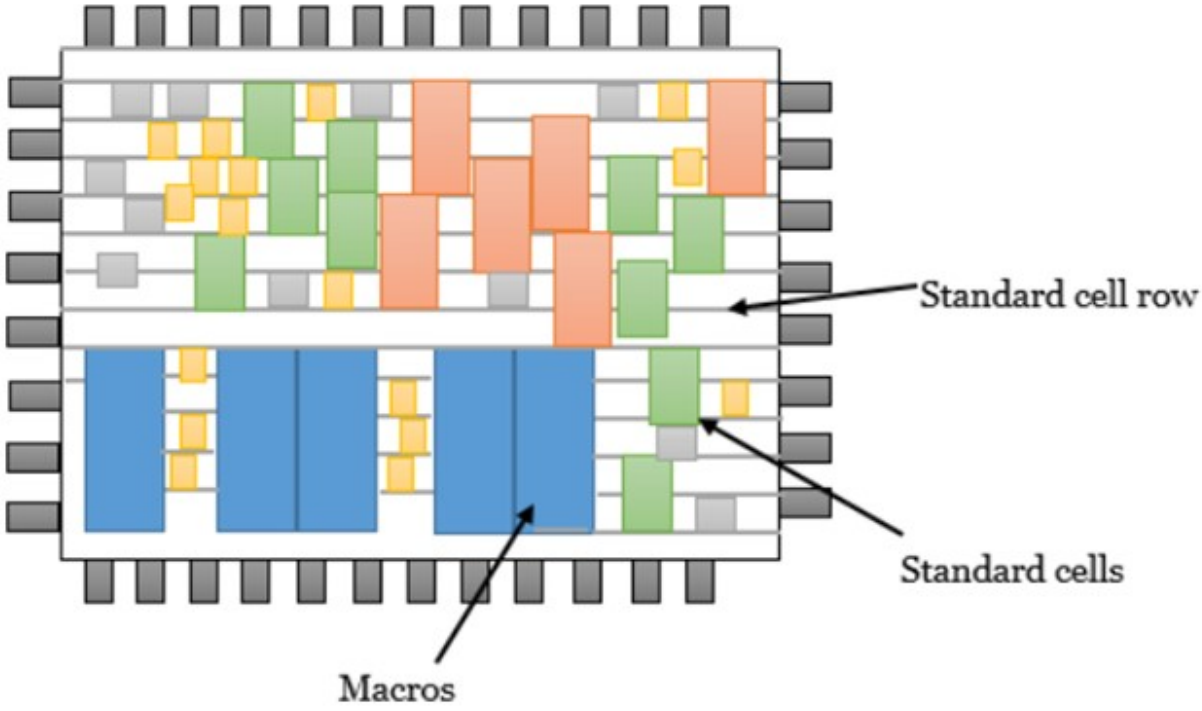


Fig : Legalization of standard cells

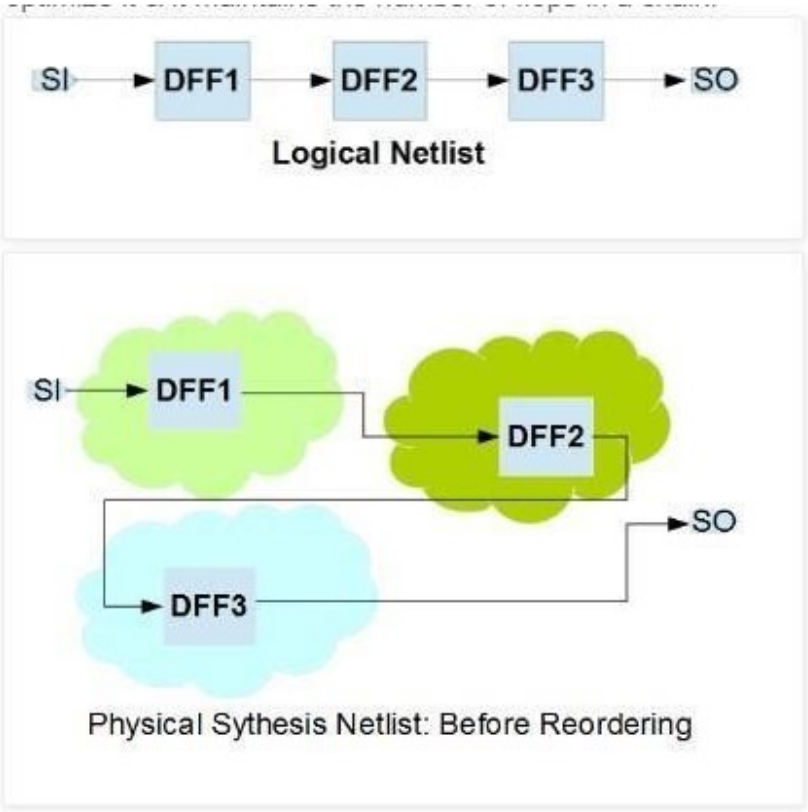
iv) Tie Cell insertion

Sometimes in the netlist some unused inputs are tied to VDD/VSS (logic1/logic0). It is not recommended to connect a gate directly to the power network, so we connect gate to TIEHI or TIELO cells.

Placement tool also does Tie-cell optimization, which places Tie cell near to parent cell.

viii) Scan-Chain Reorder

- If block contains scan chains by default Placement and CTS tools perform DFT optimization.
- During initial placement, the tool focuses on the QOR for the function nets by ignoring the scan chains. After initial placement, the tool further improves the QOR by repartitioning and reordering the scan chains based on the initial placement.
- Scan chains reordering reduces wire length so timing will improve.
- Scan chains reordering minimize congestions and improves routability.



v) **High fanout Net Synthesis (HFNS)**

- The process of buffering the high fan-out net to reduce the fanout load is called as High fanout net Synthesis. It is done because if design has too many loads then it affects delay and transition time.
- High fanout nets are mainly reset, preset, scan enable etc. these nets are not synthesized in the synthesis stage, also make sure you set an appropriate fan-out limit for your library

vi) **Placement Optimization stage:**

- In this stage tool tries to optimize placement to reduce congestion , improve timing and to fix timing DRVs
- Tool optimizes timing DRVs and setup violation by different methods like
  - Cell sizing
  - Vt swapping
  - Buffering
  - Cloning
  - Pin-swapping
  - Logical restructuring

Constraints for placement :

Placement constraints provide guidance during placement and placement optimization and legalization so that congestion and timing violations will be reduced.

i) Placement blockages
- It is the area where the cells are restricted during placement, optimization and legalization.
- It can be hard , soft and partial blockage.

ii) Placement bounds
- It is a constraint that controls the placement of groups of leaf cells and hierarchical cells.
- It allows you to group cells to minimize wire length and place the cells at most appropriate locations.
- When our timing is critical during placement then we create bounds in that area where two communicating cells are sitting far from another. It is a fixed region in which we place a set of cells.

Types of bounds:
- Soft move bound
- Hard move bound
- Exclusive move bound

Soft move bound:
In this tool tries to place the cells in the move bound within a specified region, however, there is no guarantee that the cells are placed inside the bounds.

Hard move bound:
In this tool must place the cells in the move bound within a specified region.

Exclusive move bound:
In this tool tries to place the cells in the group bound within a floating region, however, there is no guarantee that the cells are placed inside the bounds

iii) Density controls:
It means how the density of cells can be packed. We can control the overall placement density for the block or the cell density for specific regions. To control the cell density for specific regions we can also use partial placement blockages.

iv) Constraints for Max_fanout , max_transition and max_capacitance :
We can specify the Max fanout , max_transistion and max_capacitance constraints to placement tools , so that tool performs placement to meet given constraints

Checks after Placement:

i) Analyze and fix congestion

Congestion: It is a difference between available routing track and required routing track

To fix congestion

a) Apply soft blockage between macros

b) Apply keep-out margin to macros and for cells with more pin count

c) Apply partial blockage where there is more cell density .
d) Increase macro channel spacing
e) Use refine placement optimization to reduce congestion.

ii) Analyze max capacitance and max transition violation

Fix by using

a) Load splitting: Split fanout by using buffers
b) Cloning : Split fanout by cloning driver cell
c) Increase drive strength of driver
d) Split net length by inserting buffer

iii) Analyze and Fix Setup violation .

a) Upsize cells in combinational path
b) Vt swapping : Swap cell with lower Vt
c) Path grouping method: Assign weightage for most violating path so that tools gives more importance to path with more weightage during timing optimization

STATIC TIMING ANALYSIS

Difference between DTA & STA

Dynamic timing analysis [DTA]	Static timing analysis [STA]
Verifies functionality of the design by applying input vectors and checking for correct output vectors	Checks static delay requirements of the circuit without any input or output vectors, so analysis times are relatively short and STA does not check for logical correctness of the design
Quality increases with the increase of input test vectors	Clock related all information has to be fed to the design in the form of constraints and the correctness of the constraints decides the quality
Increased test vectors increase simulation time	Timing can be analysed for worst and best cases simultaneously and also all timing paths are considered
Can be used for synchronous as well as asynchronous designs	Not suitable for asynchronous designs
Also best suitable for designs having clocks crossing multiple domains	Not suitable for designs having clocks crossing multiple domains
Computational complexity involved in finding the input patterns/ vectors that produces maximum delay at the output	Has more pessimism and thus gives maximum delay of the design and it works with timing models.

Static Timing Analysis

- Effective methodology for verifying the timing characteristics of a design without the use of test vectors
- STA can be done only for register-transfer-logic (RTL) designs
- STA basically method of adding the net delays and cell delays to obtain path delays. Then STA tool analyses all paths from each and every start point to each and every end point and compares it against the constraints (timing specification) that exists for that path.

  Purpose of STA

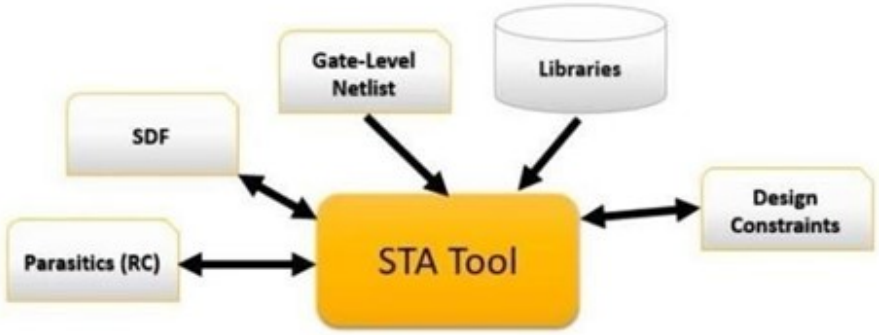
  - First, STA calculates the path delays for optimization tools. Then based on the path delays, the optimization tool chooses cells from the timing library to create a circuit that meets your timing requirements.

- Second, STA analyzes the timing of a circuit to verify that the circuit works at the specified frequency.

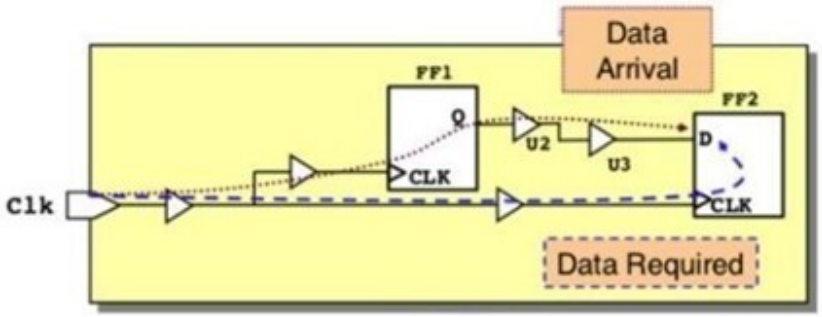
Steps in STA

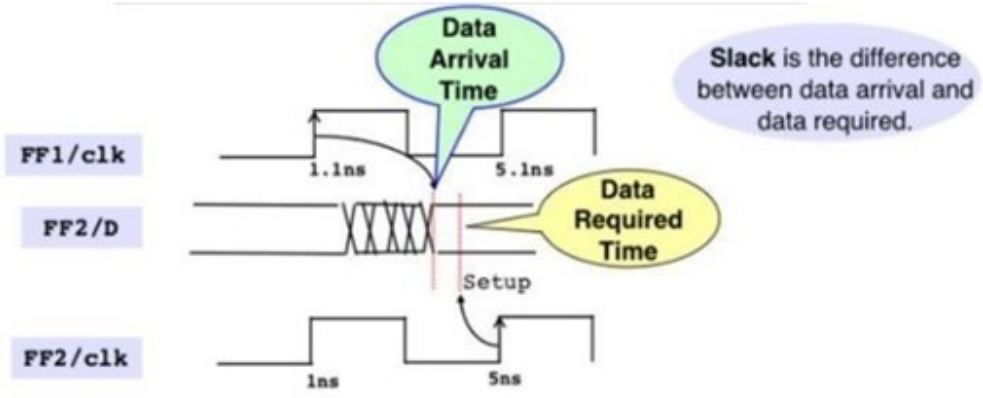
- Break the design into sets of timing paths
- Calculate the delay of each path
- Check all path delays to see if the given timing constraints are met

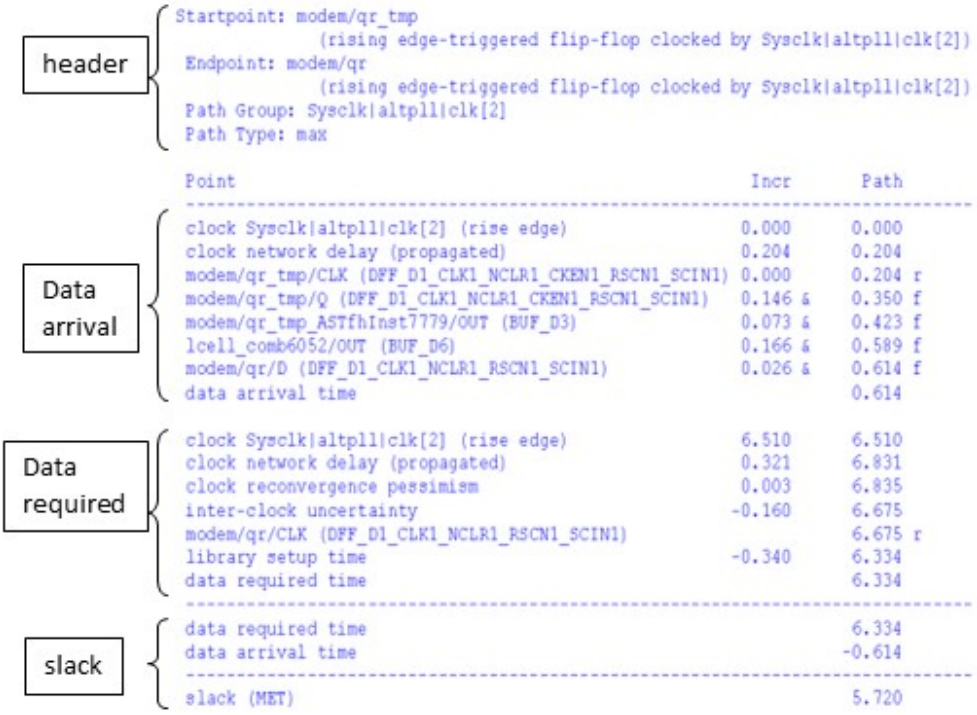
STA (input & output)



Timing report







Header

- It consists of start point (FF1) and end point (FF2)
- Path group which tells for which timing path group it belongs
- Path type: here it is max which states setup and if it was min then it is hold

Data arrival section

- Reports the total time taken to arrive at D pin of flip flop2.

Data required section

- Reports the total time taken to arrive the clock pin of FF1 – setup time of FF2

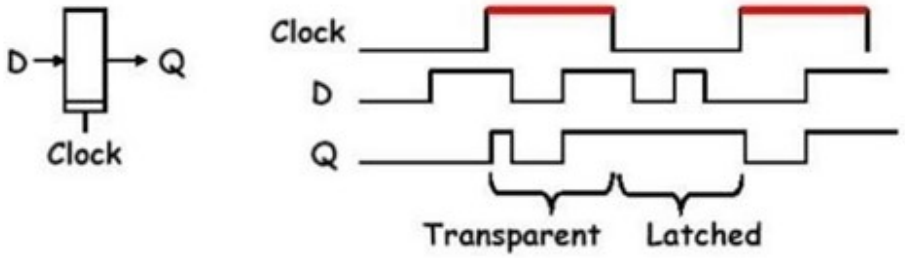
Slack

- Timing difference between required and arrival time i.e., RT-AT for setup analysis and AT – RT for hold analysis

Clocked storage elements

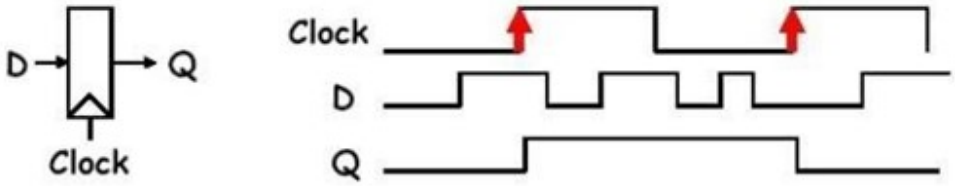
Transparent latch, Level sensitive

- Data passes through latch when clock high, latched when clock is low



D-type register of flip-flop, edge triggered

- Data captured on rising edge of lock,held for rest of the cycle



Delays

- Time taken by a signal to propagate through a cell or net
- Actual path delay is sum of net and cell delays along the timing path
- Cell delay is a function of input transition time (slew rate), total output load (net cap + sum of attached pin caps) and process parameters (temperature, power level)

Intrinsic delay

- Internal to the cell from input pin to output pin caused by internal capacitance

Propagation delay

- Delay by a cell for a change of input signal to result a change at output signal as a function of input slew and output load
- Propagation delay can be low to high (tPLH) and high to low (tPHL)
- Maximum propagation delay is considered for setup check

Contamination delay

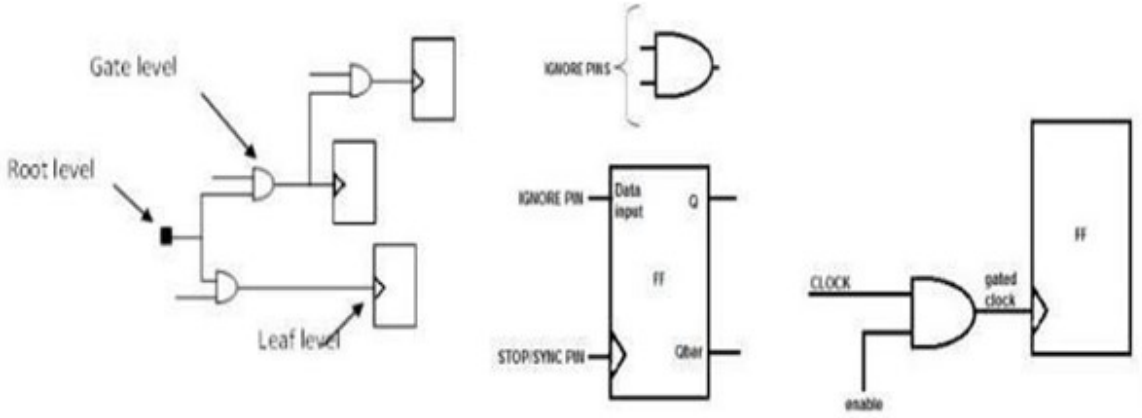
- Best case delay from valid input to output

- Minimum propagation delay which is called contamination delay is considered for hold check

Net delay

- Total time for charging/ discharging all the parasitic present in the given net

Pins related to clock



Start/ source/ root pins -

- Source pin of a clock

Stop/ sink/leaf pins –

- All clock pins of flip flops
- Clock won't propagate after this pin

Through pin –

- To make a clock pin of a flop not a CTS leaf pin

Preserved pin –

- If we need to preserve a pin with respect to location etc.

Exclude / implicit pins –

- All non-clock pins (D pin of flip flops or combo logic input pins)

Float pins (implicit stop / macro model) –

- Same as stop/sink pin but internal clock latency of it is considered for clock tree balancing
- It is actually entry pin of the hard macro

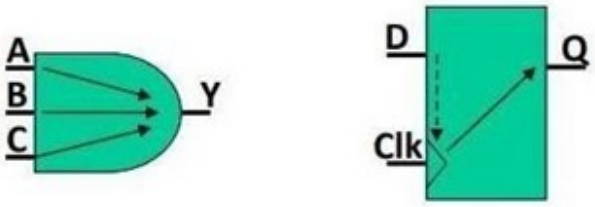
Explicit sync (stop) pin –

- Input of combinational logic considered for clock tree balancing
- Important while considering clock gating

Explicit exclude (ignore) sync pin –

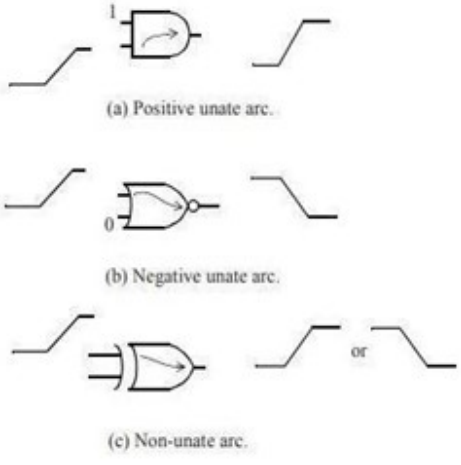
- Clock pin of flop which is not considered as sync/ stop pin
- This pin is due to clock gating concept
- In clock gating the signal will be given to AND gate

Timing Arc



- Timing arc is internal to the cell
- Combinational cells has timing arcs from each input to each output of the cell
- Flip-flops have timing arcs from the clock input pin to data output Q pin (propagation delay/ delay arc) and from clock input pin to data input D pin (setup, hold checks/ constraints arc)
- Latches have 2 timing arcs :
  - Clock pin to output Q pin, when D is stable
  - Data D pin to Q output Q pin when D changes

Timing Unate



Positive unate :
If a rising transition on an input causes the output to rise (or not to change) and a falling transition on an input causes the output to fall (or not to change). For example: the timing arcs for AND and OR type cells are positive unate.

Negative unate :

Timing arc is one where a rising transition on an input causes the output to have a falling transition and a falling transition on an input causes the output to have a rising transition. For example: the timing arcs for NAND and NOR type cells are negative unate.

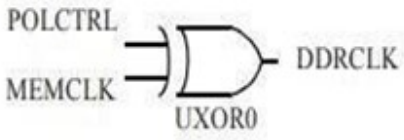
Non unate :

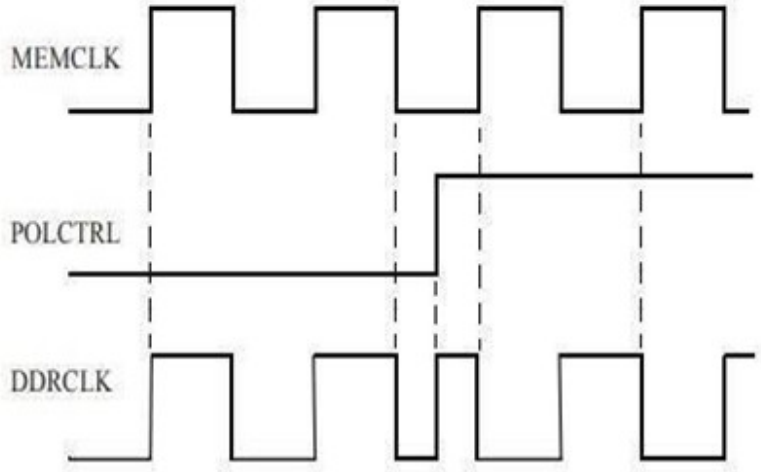
In a non-unate timing arcs, the output transition cannot be determined solely from the direction of change of an input but also depends upon the state of the other inputs. For example : the timing arc in an XOR cell are non-unate.

Unateness is important for timing as it specifies how the edges can propagate through acell and how they appear at the output of the cell.

One can take advantage of the non-unateness property of a timing arc,such as when an XOR cell is used, to invert the polarity of a clock.

For example : if input POLCTRL is a logic-0, the clock DDRCLK on output of the cell UXOR0 has the same polarity as the input clock MEMCLK. If POLCTRL is a logic-1, the clock on the output of the cell UXOR0 has the opposite polarity as the input clock MEMCLK.





Clock definitions in STA

Synchronous clocks :

- 2 clocks are synchronous with respect to each other
- Timing paths launched by one clock and can be captured by another

Asynchronous clocks :

- 2 clocks are asynchronous with respect to each other
- STA can't be applied for timing paths with asynchronous clocks , so the tool wont check the timing in these paths .

Master clocks :

- It is a source clock defined at input clock port of design

Generated clocks :

- Clock generated from a master clock as a multiple of the master clock frequency
- The generated clock frequency can be a multiple or can be a divided by master clock frequency

Virtual clocks :

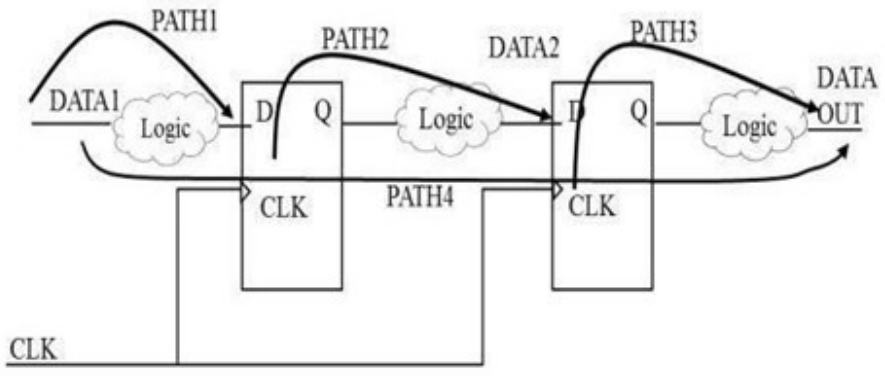
- This Clock is not associated with any pin or port of the design
- Used as a reference in STA to specify input delays and output delays relative to a clock

Different terms in STA

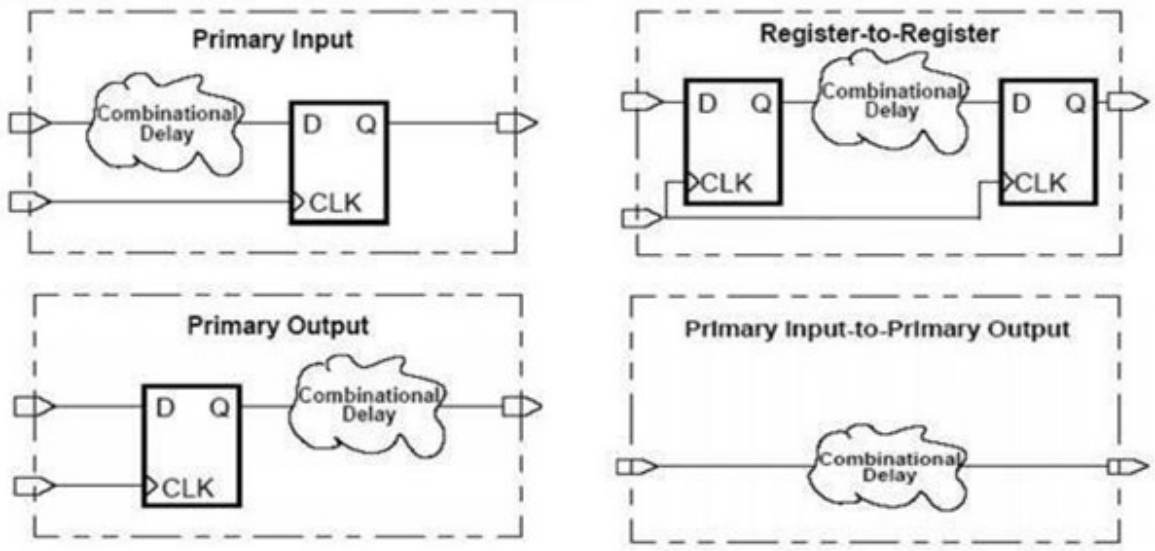
Timing paths

A timing path is a point-to-point path in a design which can propagated data from one flip-flop to another

- Each path has a start point and an end point
- Start point : input ports or clock pins of flip-flops
- End point : output ports or data input pins of flip-flops



Timing path groups



Timing paths are grouped into path groups by the clock controlling their endpoints

Input port to register path

Timing path between input port and to first register , Here external delay has to be specified On

Register to register path

Timing path between register to register in the design is called as reg to reg path

Register to output port path

Timing path between register to output port , Here external delay has to be specified On output ports

Input port to output port

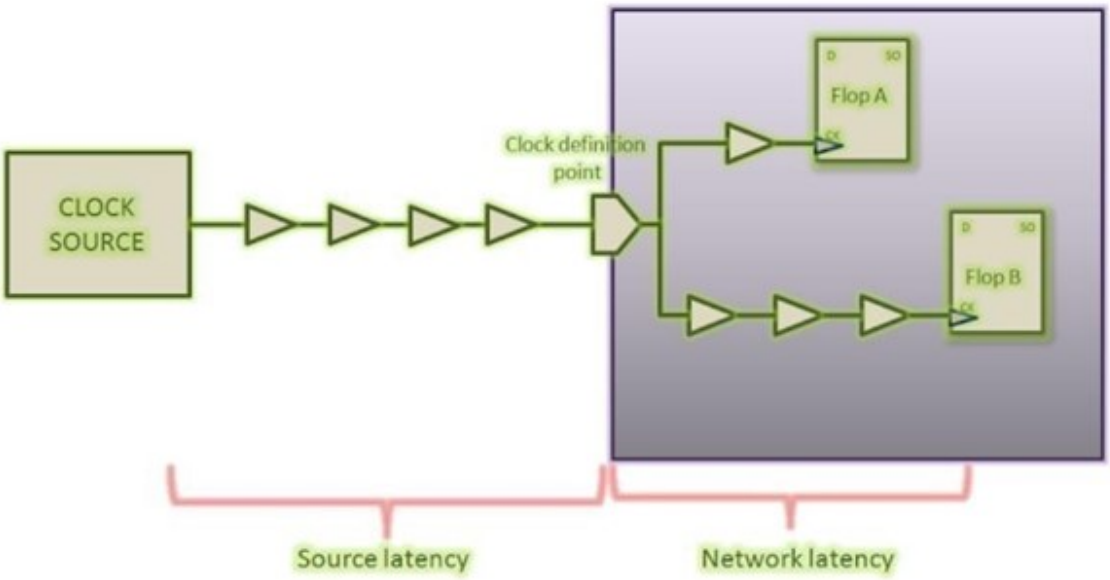
Timing path between input port and outport is called as in to out path, in between it may have combinational circuit delay

Clock latency

- Total time taken by the clock signal to reach the clock input pin of the register It includes source latency and network latency .
- Source latency is the time taken by clock to reach clock definition ports from clock Clock source point.
- Network latency is the time taken by clock to reach clock leaf pin in the design from clock definition port.

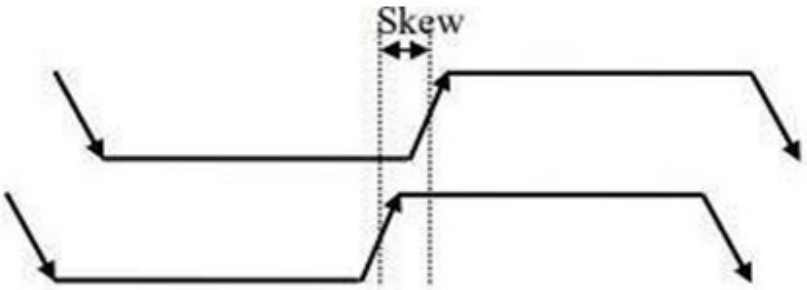
Insertion delay (ID)

- ID is the clock latency, but after clock tree is synthesized
- ID is the physical delay and clock latency is the virtual delay
- Latency is a target given to the tool through SDC file or clock tree attribute file and insertion delay is the achieved delay after CTS



Clock skew

- Clock skew is the time difference between the arrivals of clock signals at registers in one clock domain.



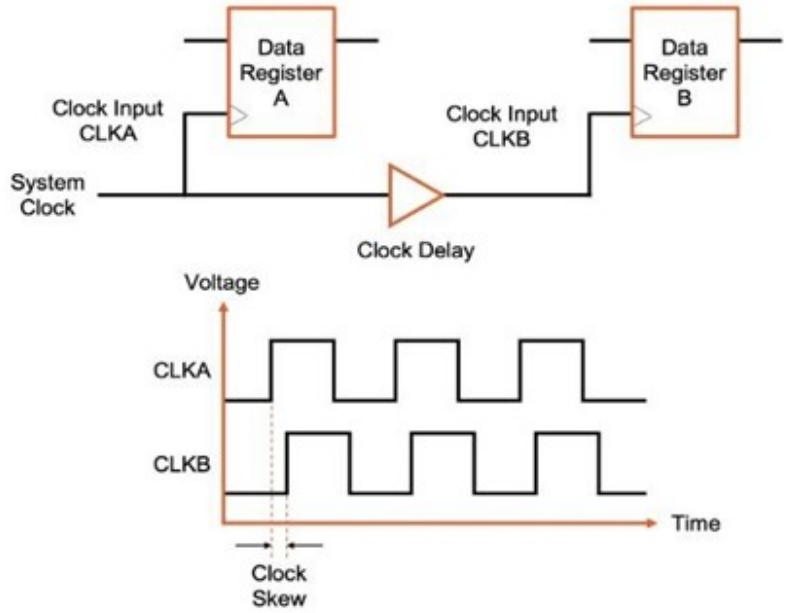
- **Local skew:** It is delay in arrival of clock at clock pins of two flip-flops which are the launch and capture flop of a timing path (source and destination flop)

$$T_{skew} = T_{capture_clock} - T_{launch_clock}$$

- **Positive skew:** Occurs when the capture clock is late with respect to launch clock

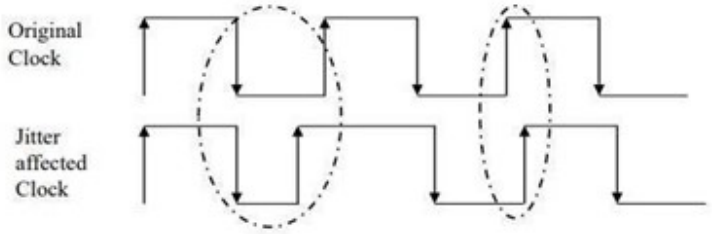
- **Negative skew :** Occurs when the capture clock is early with respect to launch clock

- **Global skew :** Difference between the longest clock path delay and the shortest clock path delay in the design is called global skew. (maximum insertion delay – minimum insertion delay)



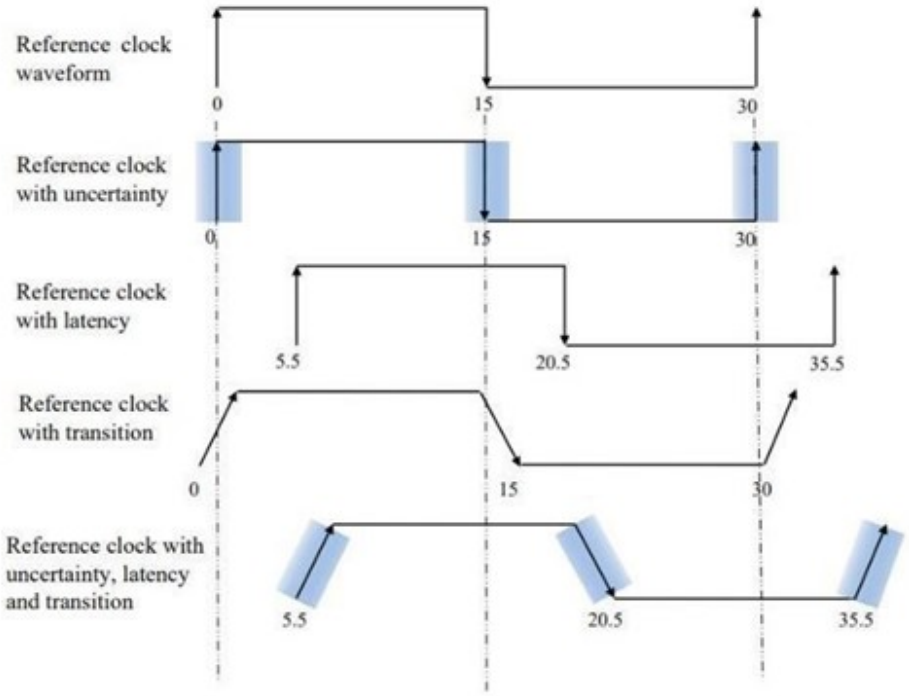
Clock jitter

- Jitter is the short-term variations of clock edge with respect to its ideal position in time.
- The two major components of jitter are random jitter and deterministic jitter
- Factors causing jitter includes imperfections in clock oscillator, supply voltage variations, temperature variations, crosstalk



Uncertainty:

Before CTS :
Uncertainty = skew + jitter + margin
Because clock is ideal before CTS so we specify skew in uncertainty
After CTS :
Uncertainty = jitter + margin
Because clock is after CTS so tool calculates skew from insertion dealay.



Pulse width

- Pulse width is the time between the active and inactive states of the same signal
- Minimum high pulse width is the amount of time after the rising edge of a clock that the clock is stable in high state.
- Minimum low pulse width is the amount of time after the falling edge of a clock, that the clock is stable in low state.

Duty cycle

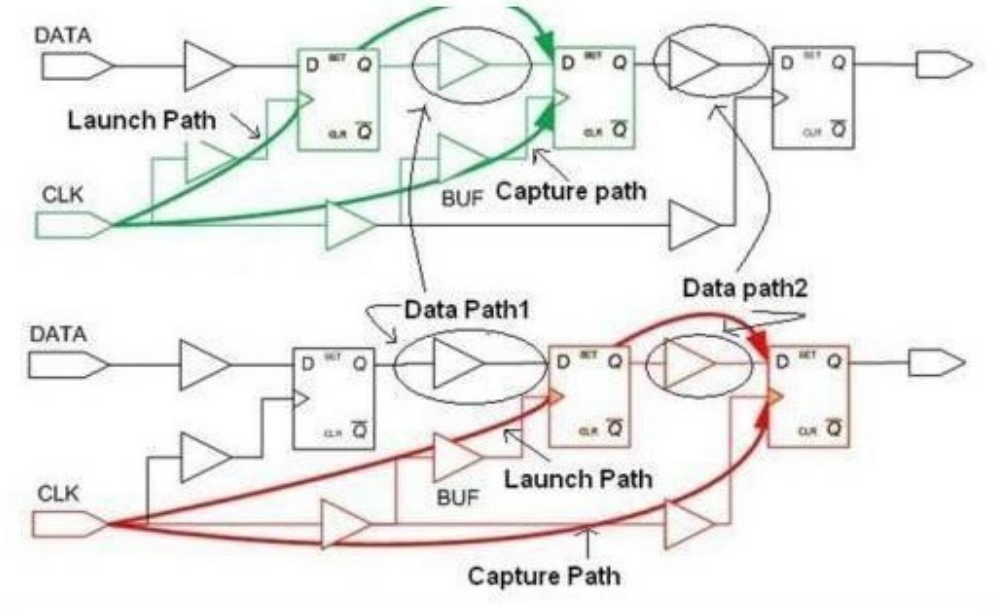
- Percentage of clock period having high pulse
- Typically clock waveforms are of 50% duty cycle
- Duty_cycle = on_time / (on_time + off_time)

Transition time/ Slew

- Time taken by a signal to change the state. (High to low or low to high)
- Rise slew (tr) is called rise time and fall slew (tf) is called fall time
- Minimum / maximum transition is the minimum/maximum slope allowed at leaf pins
- Transition affects power dissipation, latency , pulse width and delay.

Common Path Pessimism Removal (CPPR)

- Same clock path cells may be present in launch path and can be present in capture path .
- While doing OCV derating, same path may get both min/max delay
- But a path can have either as a maximum delay or a minimum delay (or anything in between) but never both delays at the same time
- STA tools will have techniques to remove artificially introduced pessimism between the launch clock path and the capture clock path that is called as CPPR/CRPR



Setup and hold time

Setup time: is the minimum amount of time the data signal should be held steady before the active clock edge so that the data are reliably sampled by the clock

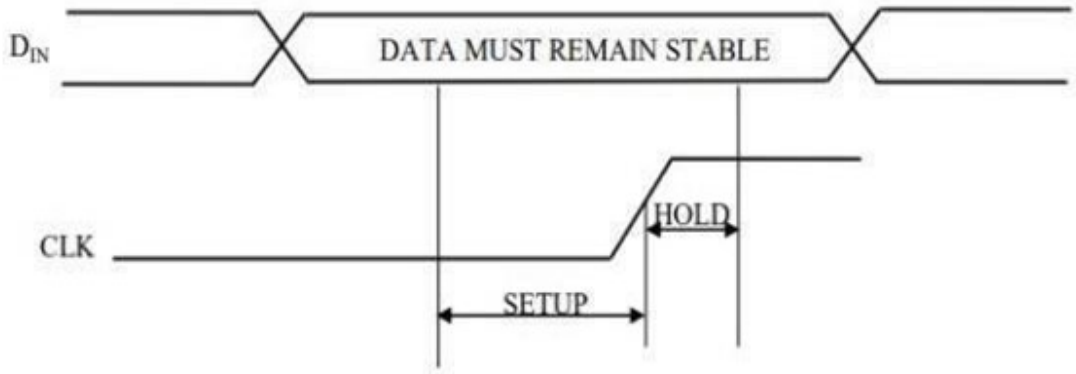
Tlaunch_clk + Tclkq_max + Tcomb_max <= T + Tcapture_clk – Tsetup – Tun

Slack = (T + Tcapture_clk – Tsetup – Tun) – (Tlaunch_clk + Tclkq_max + Tcomb_max)

Hold time : is the minimum amount of time the data signal should be held steady after the active clock edge so that the data are reliably sampled by the clock

Tlaunch_clk + Tclkq_max + Tcomb_min >= Tcapture_clk + Thold + Tun

Slack = (Tlaunch_clk + Tclkq_max + Tcomb_min) - (Tcapture_clk + Thold + Tun)



If Slack is positive timing is met

If Slack is Negative Timing is not met

Recovery time and Removal time

Recovery time

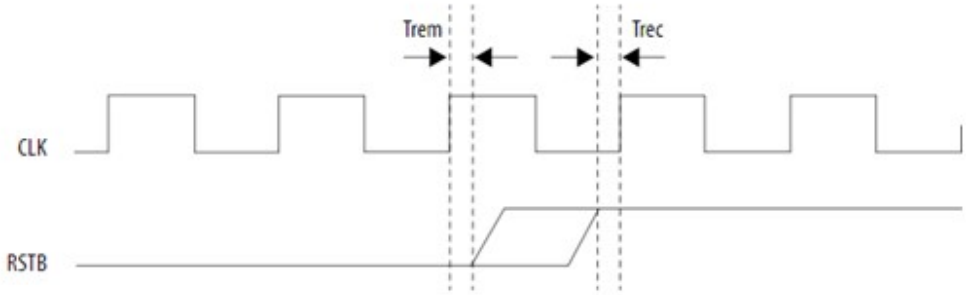
- Reset Recovery Time, Trec, is the minimum time between the de-assertion of a reset and the clock signal being high again. The reset Recovery Check ensures that the reset signal is stable for a minimum time after de-assertion, before the next active clock edge.

Removal time

- Removal Time, Trem, refers to the minimum time, after the active clock edge, that the reset must be stable before being de-asserted. The reset Removal Check ensures that the de-asserted reset signal is not captured by the same clock edge that launches the reset.

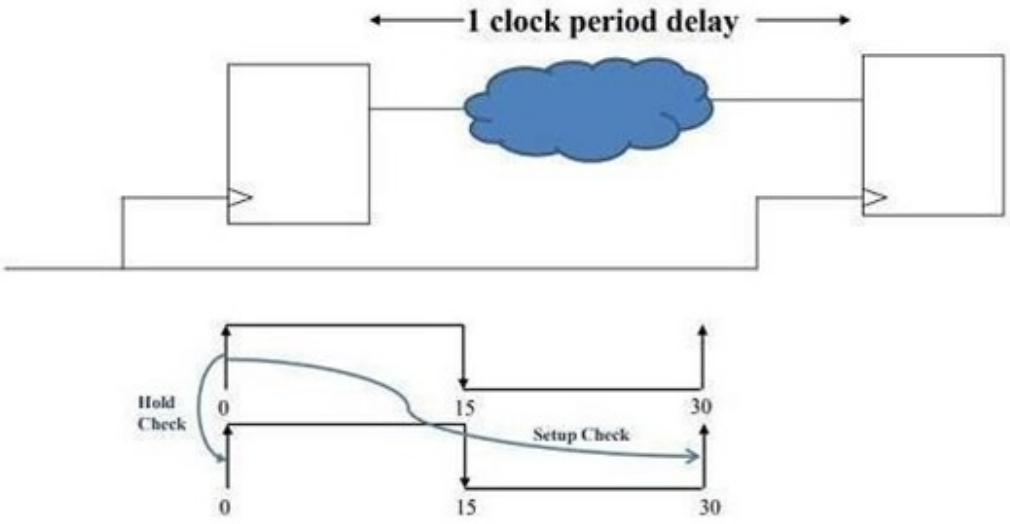
Recovery time and Removal time violations

- This check is to ensure that the asynchronously signal rise/fall edge is not occurring at the clock edge, it should be some time before or after the clock edge
- If that violates, then recovery time and removal time violations



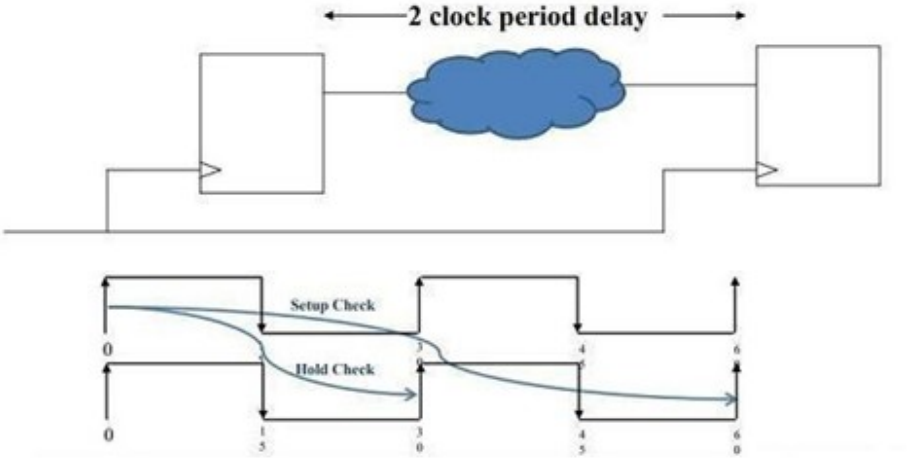
Single cycle path

- Timing path that is designed to take only one clock cycle for the data to propagate from the start point to the end point
- Start point and end point are flops clocked by the same clock
- By default tool will consider all timing paths as single cycle paths



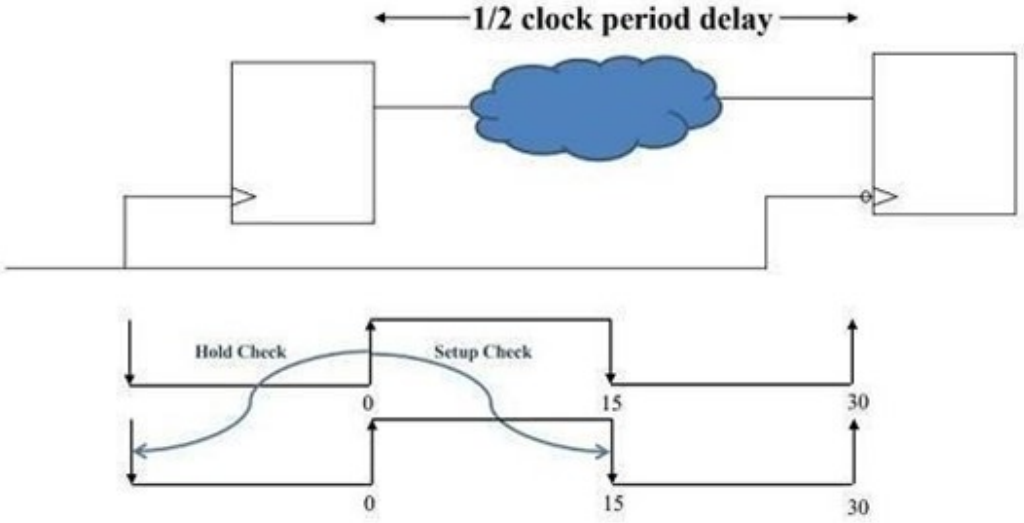
Multi cycle path

- Timing path that is designed to take more than one clock cycle for the data to propagate from the start point to the end point
- Start point and end point are flops clocked by the same clock
- Need to specify the launch edge and capturing edge in SDC



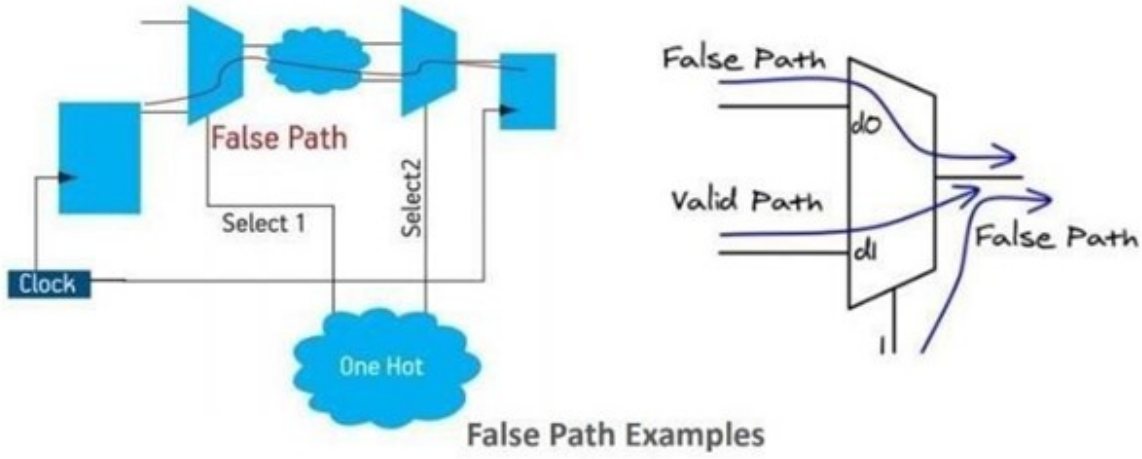
Half cycle path

- Timing path that is designed to take half clock cycle (both of the clock edges) for the data to propagate from the start point to the end point
- Start point and end point are flops clocked by the same clock.
- No need to specify the launch edge and capturing edge in SDC, since the tool can identify it from the netlist
- Half cycle path is critical for setup analysis and good for hold .



False path

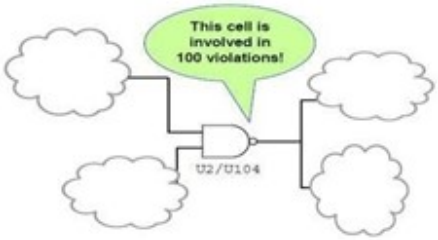
- Physically exist in the design but are logically/ functionally inactive/ incorrect path, means no data is transferred from start point to end point
- The goal in STA is to do timing analysis on all "true" timing paths, so these paths are excluded from timing analysis
- Similarly timing can be disable for a pin or port or cell where the delay will be computed but wont report it.



False Path Examples

Bottleneck analysis

- Lists the cells causing the timing violations on multiple paths
- By identifying and fixing the violation caused by a bottleneck cell , more timing paths may be fixed .



Multi-VT cells (HVT,SVT/RVT,LVT)

- Different threshold voltages are achieved by implanting dopants in different concentration
- To use multi VT cells design need to have multi-VT library
- Sub-threshold leakage increases with decrease in Vt .
- If the optimization target is power performance, first use the HVT cells library and then try LVT cells
- If the optimization target is to meet timing, then first use the LVT cells libraryand then HVT cells.
- If you swap launch flop from SVT to LVT or HVT, setup will improve and hold will be impacted correspondingly

High Voltage Threshold (HVT)

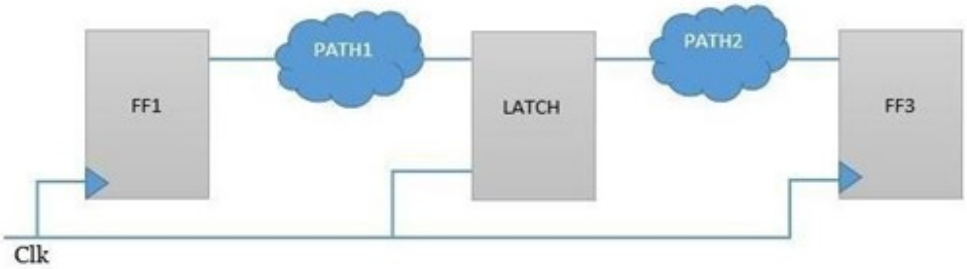
- Use in non-timing critical paths
- Use in power critical paths
- Has low leakage and low speed

Low Voltage Threshold (LVT)

- Use in timing critical paths
- Use in non-power critical paths
- Has high leakage and high speed

Time borrowing /stealing

- Edge-triggered flip-flops change states at the clock edges, whereas latches change states as long as the clock pin is enabled
- In latch based design longer combinational path can be compensated by shorter path delays in the subsequent logic stages
- The technique of borrowing time from the shortest paths of the subsequent logic stages to the longer path is called time borrowing or cycle stealing



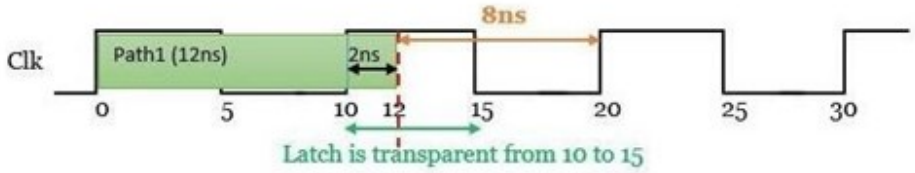
- Here we assume clock period T = 10ns and path1 is having delay of 12ns Path 2 is delay is 2 ns.

- Here setup violation happens if second register is a flip-flop.

- Here instead of flip flop, if we are using latch then there is no problem because the latch is transparent from 10 to 15, and no problem to receive the data so we can

receive our data by borrowing the time from the next cycle, this means latch provides the advantage over Flip flop of 2ns time.

- The maximum time we can borrow is 5ns from the latch. But this time is reduced for the latch to flip flop3 so Flip Flop3 must get the data before 20ns so propagation time limit between latch and ff3 is 8ns.

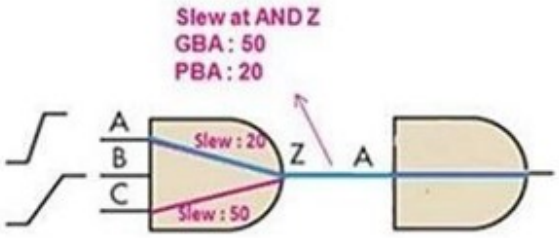


Types of STA (PBA, GBA)

Path based STA (PBA) – PBA chooses the propagation of the specific path

Graph based STA (GBA) – GBA chooses the worst case propagation delay from all inputs to output through a gate.

In PBA mode run time is more



SIGNAL INTEGRITY ISSUES

CROSS-TALK :

Effects of Crosstalk is due to coupling capacitance between two nets .

There are two effects of cross-talk

1. Crosstalk glitch or crosstalk noise
2. Crosstalk delta delay or crosstalk delay

Crosstalk glitch

In order to explain the crosstalk glitch, we will consider the following two cases.

Case-1: Aggressor net is switching low to high and victim net is at a constant low

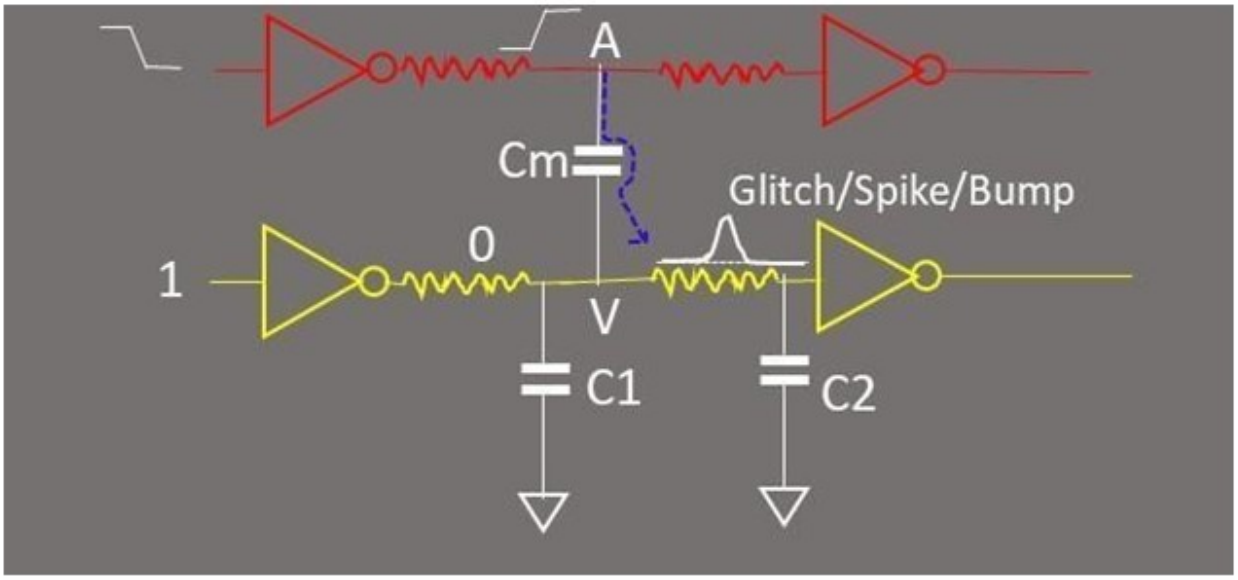


Figure-1: Crosstalk glitch (Raise)

In this case, the aggressor net switches from logic 0 to logic 1 and the victim net is at constant zero as shown in the figure-1. Now consider the node A, node V has Mutual capacitance Cm . As node A start switching from low to high, a potential difference across the mutual capacitance gets developed and the mutual capacitor Cm starts charging. During this event, there is some leakage current which starts flowing from node A to node V through the mutualcapacitance Cm due to the leaky nature of mutual capacitance. This leakage current will raise the potential of node V, which creates a rising spike

The magnitude of this voltage or height of the glitch will depend on the various factors which will be discussed later.

So, whenever one net switches from low to high and other neighboring net is supposed to remain constantly low will get affected by the switching net and have a glitch on it. Now let's discuss case-2 which is similar to case-1.

Case-2: Aggressor net is switching high to low and victim net is at a constant high

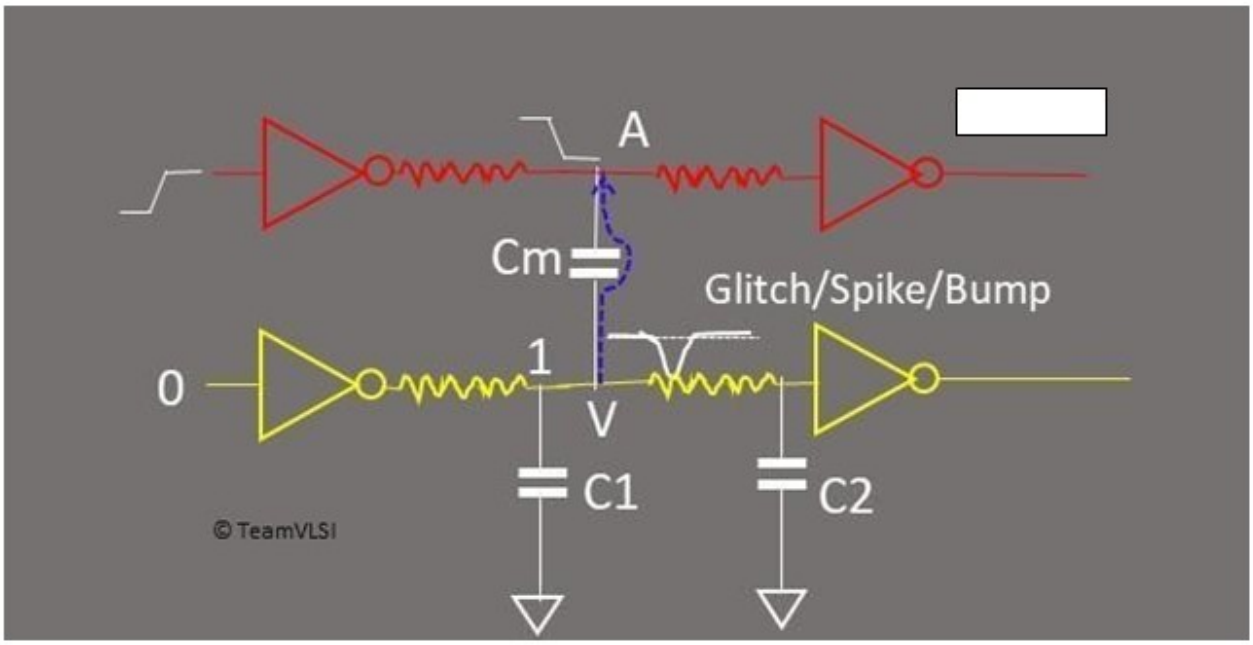


Figure-2: Crosstalk glitch (Fall)

In this case, the aggressor net switches from logic 1 to logic 0 and the victim net is at constant high logic as shown in the figure-2. Now consider the node A, node V, Mutual capacitance Cm and the path from V to A. As node A start switching from high to low, a potential difference across the mutual capacitance gets developed and the mutual capacitor Cm starts charging through node V to node A. During this event, there is a leakage current which starts flowing from node V to node A through the mutual capacitance Cm due to the leaky nature ofmutual capacitance. This leakage current will drop the potential of node V, which creates a falling spike or falling glitch on the victim net as shown in figure-2.

So, whenever one net switches from high to low and other neighboring net is supposed to remain constantly high will get affected by the switching net due to the mutual capacitance and have a falling glitch on it.

In case-1 and case-2 we have seen that if one net is switching and another neighboring net is at constant logic and if they have mutual capacitance between them, the other net may get affected and that net may have a sudden raising or falling bump or spike. such a spike on the

victim net is called a crosstalk glitch or crosstalk noise.

Effects of crosstalk glitch

Does every glitch unsafe? The answer is it depends on the height of the glitch and the logical connection of the victim net. If the height of the glitch is within the noise margin low (NML), Such a glitch is considered a safe glitch. If the glitch height is above the noise margin high (NMH), such a glitch is considered a potentially unsafe glitch. In the case of a glitch, height is in between NMH and NML, this is an unpredictable case

Crosstalk glitch height

Crosstalk glitch will be safe or unsafe depending on the height of the crosstalk glitch and the logic pin from which the victim net is connected. So let's investigate the factors on which the crosstalk glitch height depends. Crosstalk glitch height depends basically on three factors:

1. *Coupling capacitance*
2. *Aggressor's drive strength*
3. *Victim's drive strength*

Closer the nets will have greater coupling capacitance. More the capacitance will have a larger glitch height. Drive strength of the aggressor and victim driver will also affect the glitch height. The high drive strength of the aggressor net will impact more the victim net. If the drive strength of the victim net is high, then it will not be easy to change its value, which means lesser will be the effect of crosstalk.

It was all about the crosstalk glitch or crosstalk noise, Now let's move to the second effect which is crosstalk delta delay or crosstalk delay.

Crosstalk Delay

Crosstalk delay occurs when both aggressor and victim nets switch together. It has effects on the setup and hold timing of the design. Crosstalk delay may cause setup and hold timing violation. So it is important to do a crosstalk delay analysis and fix the timing considering the effect of crosstalk. Crosstalk could either increase or decrease the delay of a cell depending upon the switching direction of aggressor and victim nets. We will take two cases one when both nets switch in the same direction (high to low or low to high) and the other both the nets switch in opposite directions and will analyze the effect of crosstalk delay.

Case-1: Aggressor and victim net switch in opposite directions

Let's consider aggressor net switches from low to high logic and victim net switches from high to low (opposite). as shown in figure-6.

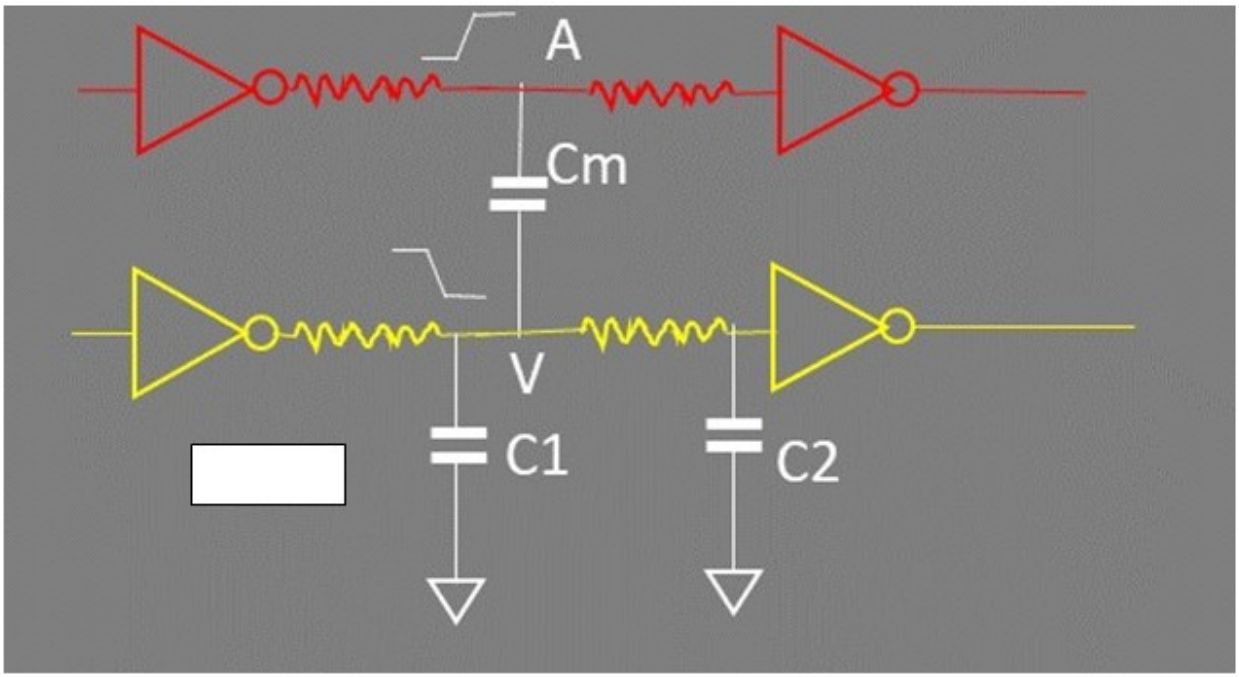


Figure-6: Crosstalk delay due to opposite direction switching

As node A starts to transition from low to high at the same time, node V starts switching from high to low. There will be a potential difference from node A to V as half of the transition happened. There is a coupling capacitance between A and V so the aggressor node will try to pull up the victim node. This will affect the smooth transition of the victim node from high to low and will have a bump after half of the transition and this will result in an increase in the transition time of the victim net. Figure-7 shows the transition of nets. After crosstalk, the delay of the cell will be increased by Δ and the new delay will be $D + \Delta$.

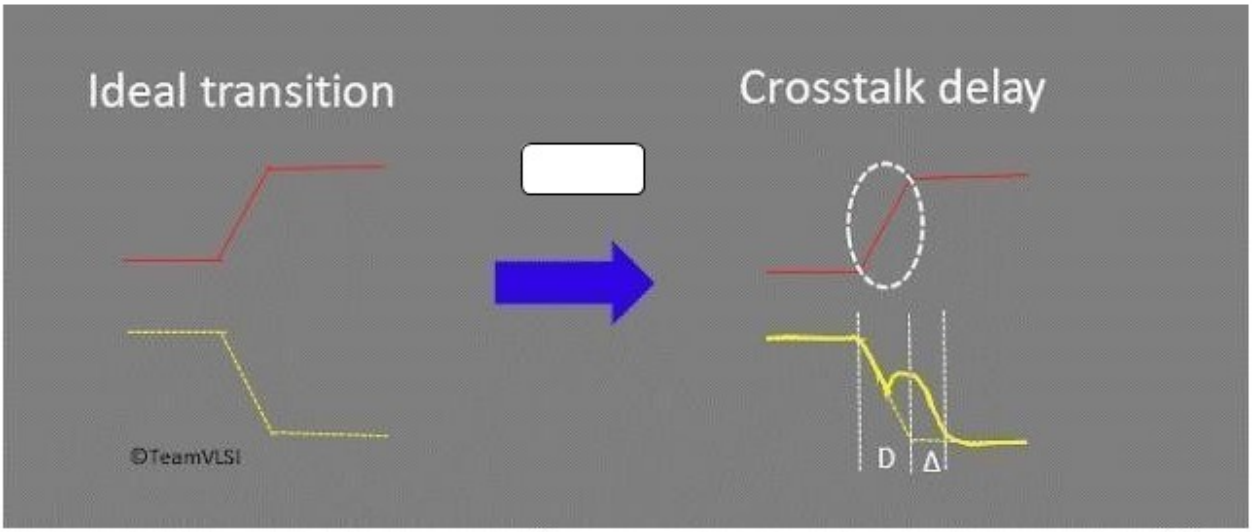


Figure-7: Crosstalk delay (increase)

Case-2: Aggressor and victim nets switch in the same direction

Let's consider the aggressor net switches from low to high logic and the victim net also switches from low to high (same direction). as shown in the figure-8.

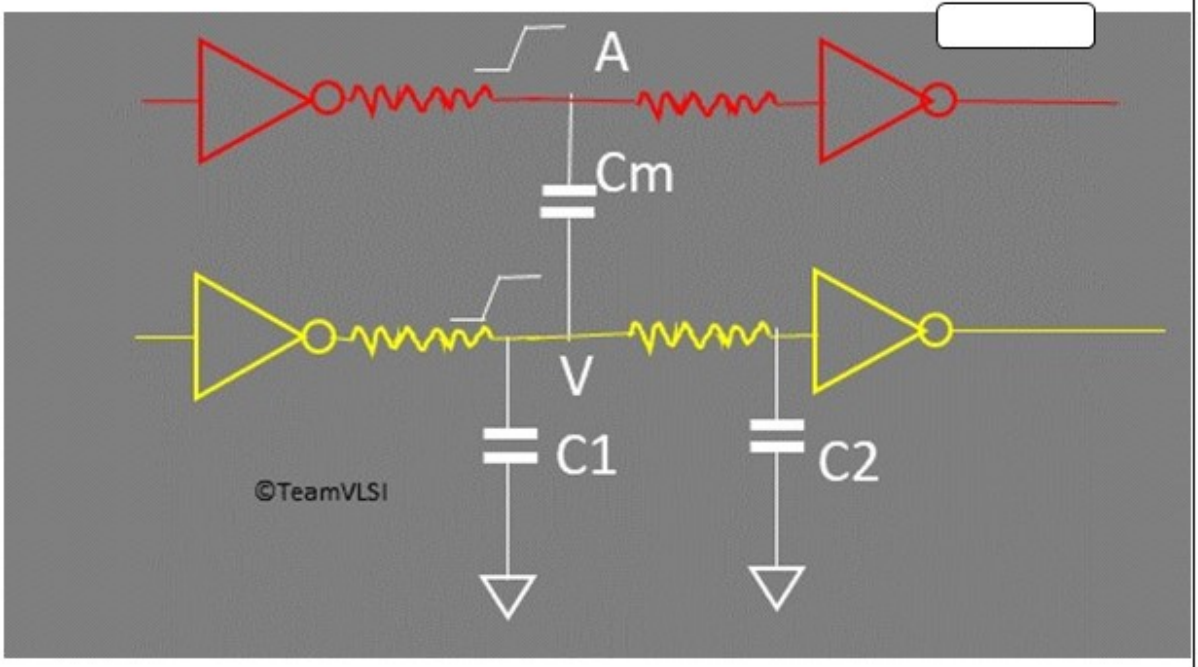


Figure-8: Crosstalk delay due to same direction switching

As node A starts to transition from low to high at the same time, node V also starts switching from low to high. Suppose the aggressor net has high drive strength and so fast transition, a potential difference from node A to V will be developed after half of the transition happened. There is a coupling capacitance between A and V so the aggressor node will try to fast pull up the victim node. This will affect the smooth transition of the victim node from low to high and will have a bump after half of the transition and this will result in a decrease in the transition time of the victim net. Figure-9 shows the transition of nets. After crosstalk, the delay of the cell will be decreased by Δ and the new delay will be $(D - \Delta)$.

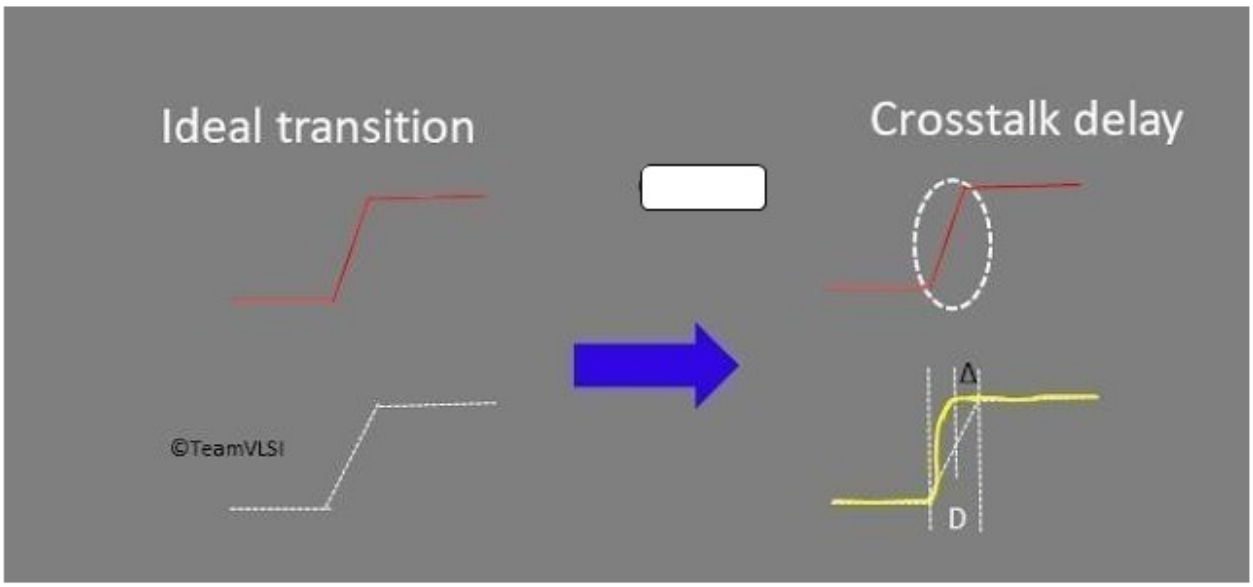


Figure-9: Crosstalk delay (decrease)

Effects of crosstalk delay

There are various effects of crosstalk delay on the timing of design. It could make unbalance a balanced clock tree, could violate the setup and hold timing. In this section, we will discuss some of them.

Effect on setup and hold timing:

Crosstalk delay can violate the setup timing. Figure-11, shows the data path, launch clock path and capture clock path.

For setup timing, data should reach the capture flop before the required time of capture flop. So if there is an increase of delay in the data path or launch clock path it may cause a setup violation. Setup violation may also happen if there is a decrease in delay on the capture clock path. These effects of crosstalk delay must be considered and fixed the timing.

Hold timing may be violated due to crosstalk delay. Figure-12, explains the situations where the hold time could violate due to crosstalk delay.

If there is a decrease in the delay of any cells in the data path and launch clock or there is an increase of delay of cells in the capture clock path due to crosstalk delay, It may result in the hold timing violation. Such cases must be considered and fix the timing.

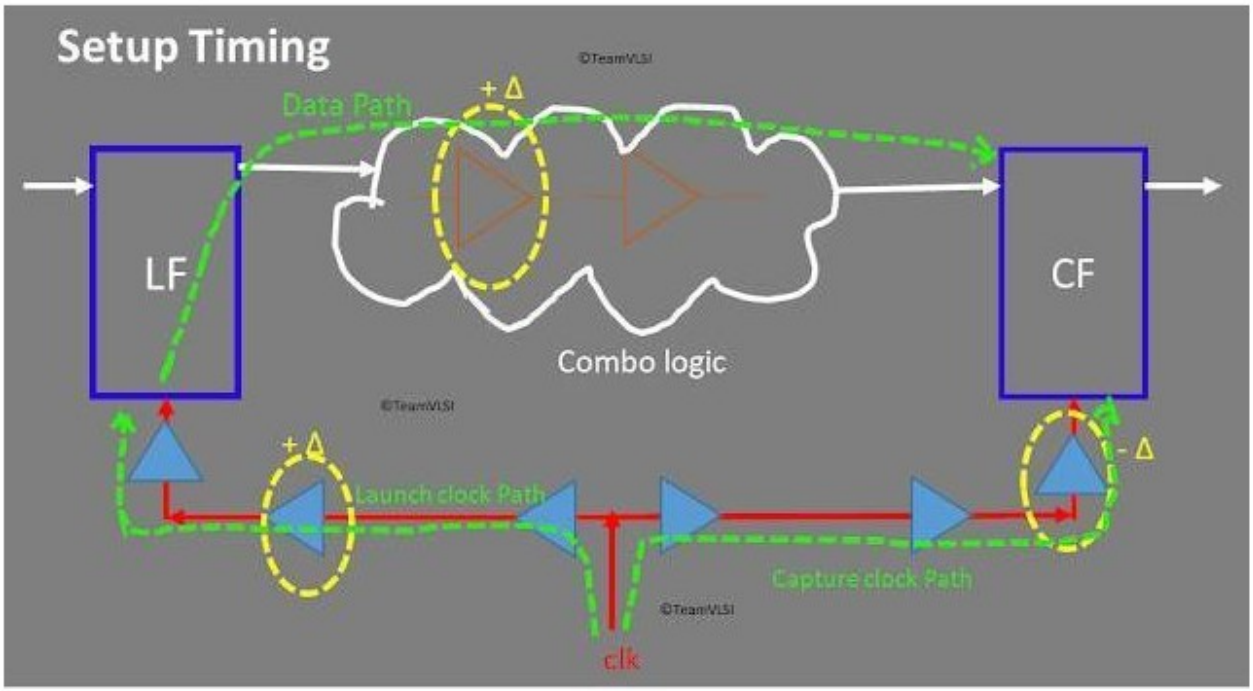


Figure-11: Effect of crosstalk delay on setup timing

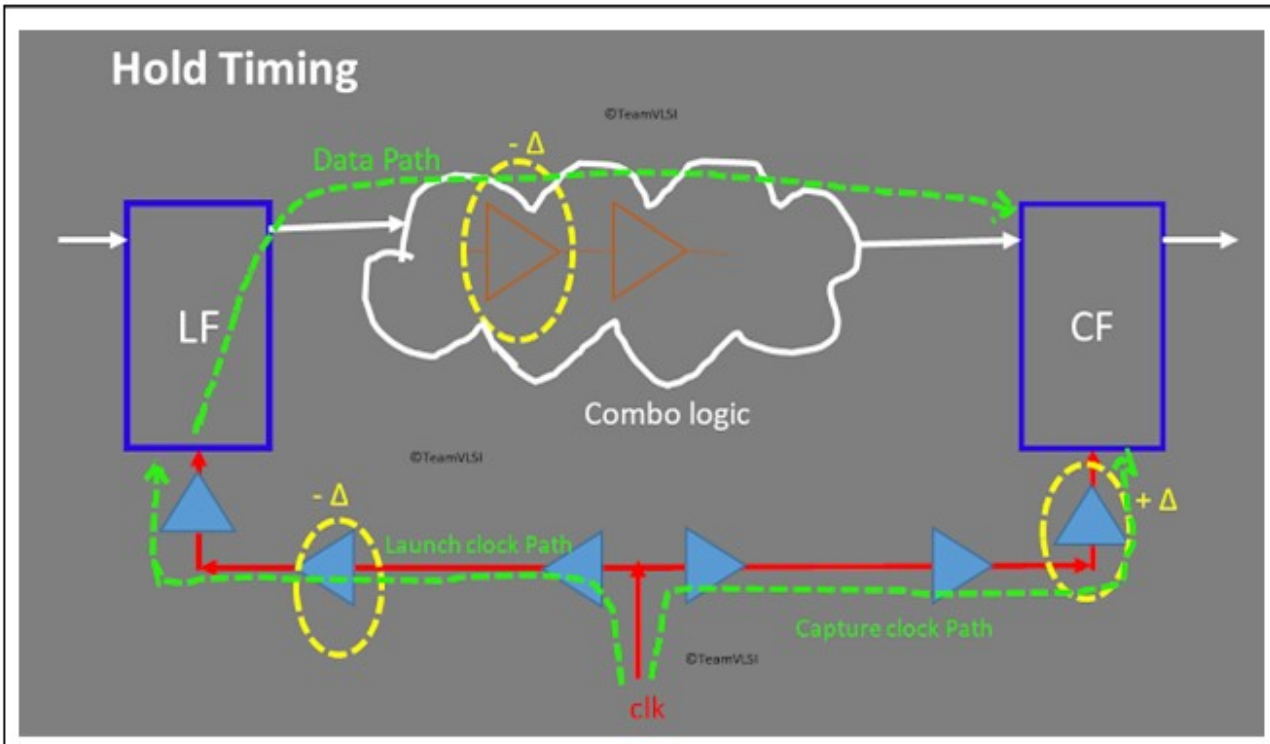


Figure-12: Effect of crosstalk delay on hold timing

Crosstalk prevention techniques

There are various ways to prevent crosstalk, some of the well-known techniques are as follow.

1. Increase the spacing between aggressor and victim net:

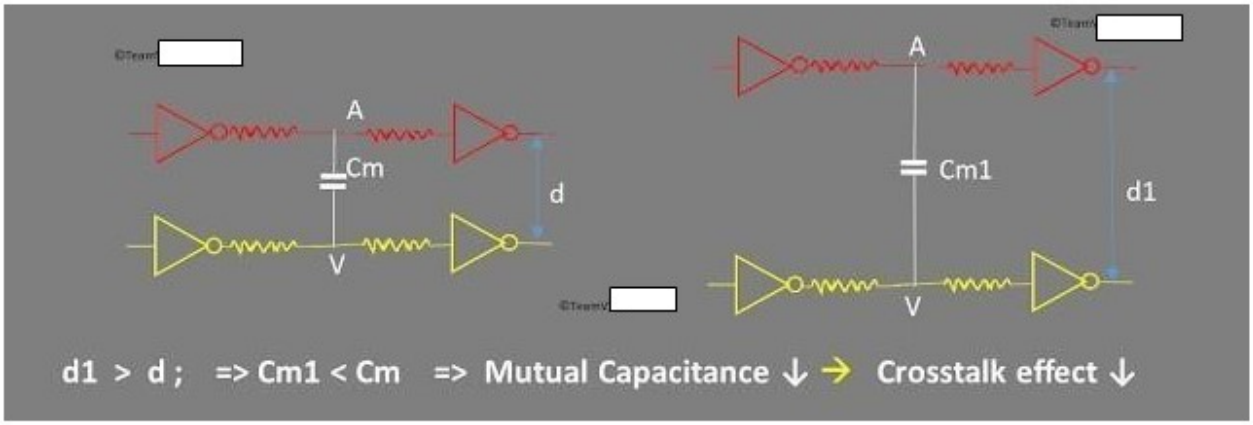


Figure-2: Effect of net spacing on crosstalk

Figure-2 shows that by increasing the spacing between aggressor and victim net we are ultimately reducing the coupling capacitance between them as the capacitance is inversely proportional to the distance between them. So by increasing the spacing crosstalk will decrease.

2. Shielding of nets:

Figure-3 shows the shielding technique used to prevent crosstalk. Generally, we insert a shielding net between the victim and the aggressor net. The shielding net is connected to strong VDD or VSS.

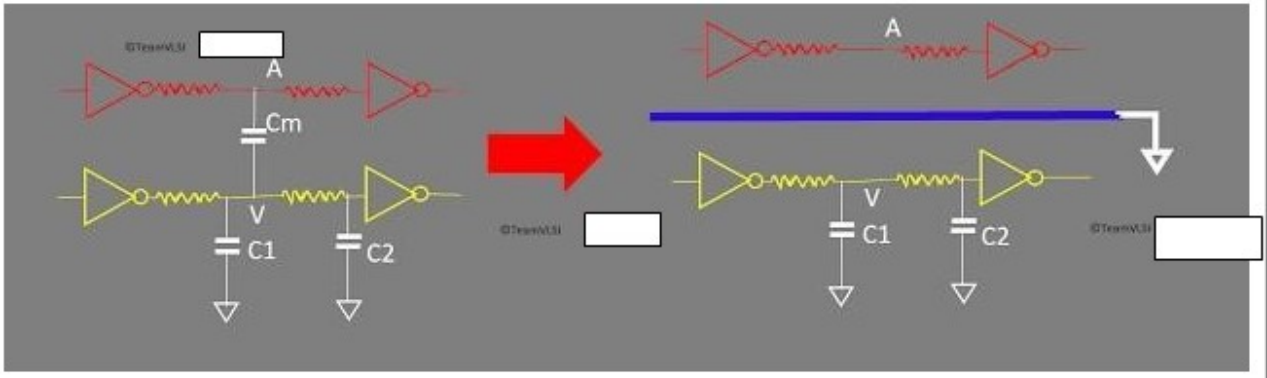


Figure-3: Shielding of a net

By shielding a net the two things will happen, one is the direct coupling capacitance between the aggressor and victim net will vanish and secondly the shielding net will remain at a constant logic so there are no chances of crosstalk.

The above two techniques will prevent the crosstalk but it has an impact on the area. Both techniques will require more area to route them.

3. Upsizing the victim cell:

If we increase the drive strength of the victim cell it will not be easy to affect by the aggressor net.

4. Downsize the aggressor cell:

Higher the drive strength of aggressor cell, higher is the impact of crosstalk on the victim. So by reducing the drive strength we can reduce the crosstalk effect.

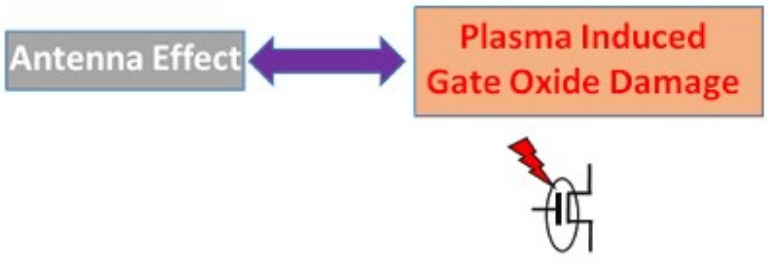
Crosstalk timing window analysis is based on the concept that we need to consider a timing window on which the aggressor has an effect on victim net.

Antenna Violation :

Gate Oxide of a MOS transistor is the most sensitive part of a MOS device. Special care needs to be taken during fabrication of ASIC to protect it from any damages during fabrication steps and ASIC operation too. The antenna effect is a phenomenon that may causedamage to the gate oxide of MOS during the fabrication process especially due to the plasma etching process. In this article, we will investigate the antenna effect phenomena in detail and the reasons which are responsible for this effect.

What is the Antenna Effect?

The term Antenna Effect might not give you the right intuition about the actual effect it may lead you to think about electromagnetic radiation or transmitter-receiver concepts but here the case is different. So It has another popular name which is called "Plasma Induced Gate Oxide Damage" which provides the right intuition about the effect. As this name itself indicates that this is an effect caused by the Gate Oxide Damage due to the Plasma Etching process during the fabrication process of VLSI chips.



Although the antenna effect occurs during the fabrication stage of the chip especially at the time of plasma etching but the prevention mechanism should be set in the physical design stage. The fabrication laboratory provides the antenna rule file which must be checked and designed should be cleaned as per the antenna rule during the physical signoff stage.

In fabrication flow first FEOL (Front End Of Line) is fabricated which involves the fabrication of all MOS transistors. Once the FEOL fabrication is done BEOL (Back End Of Line) fabrication starts which involves the fabrication of metal interconnects. Antenna effect comes into the picture while BEOL fabrication. In IC manufacturing process Plasma etching process is used to fabricate the metal interconnects. Plasma etching is a dry and anisotropic etching process, used for selective etching. Plasma contains high energetic ions and radicals which get collected by the metal interconnects while the etching process of metals. Figure-1 shows the structure of MOS and collection of plasma by the interconnect.

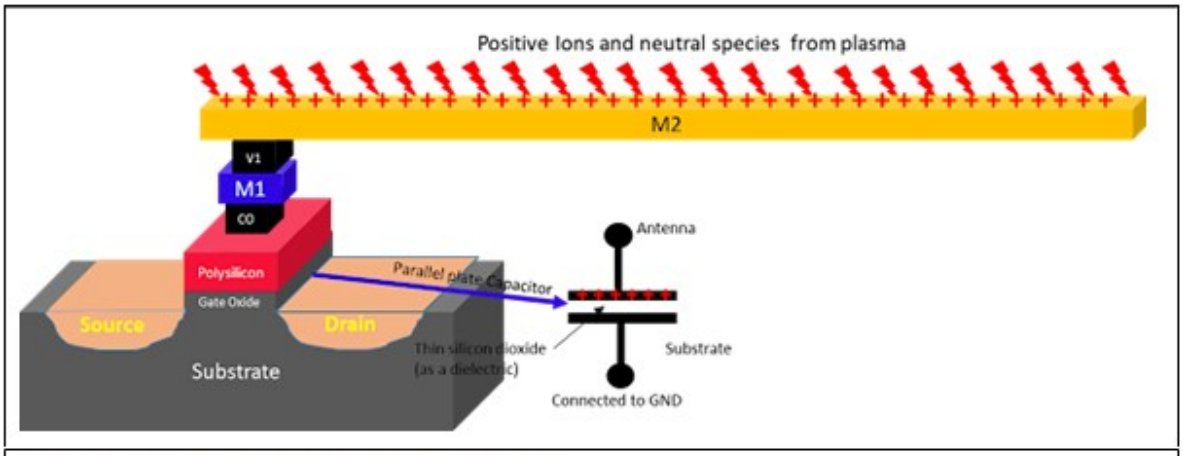


Figure-1: MOS structure and plasma etching

The amount of charge accumulation depends on the surface area of interconnect. These collected ions increase the potential of the interconnect and if the interconnect is connected to the poly gate, ultimately the potential of the gate will increase. Due to this increased potential of the gate, a drainage path may be formed through the gate oxide to substrate to balance this extra accumulated charge on the gate. If the amount of charge accumulation is high, this drainage path through the gate oxide may either breakdown the gate oxide which leads to permanent damage of MOSFET or may create charge trapping in the gate oxide which further leads to many side effects like early gate oxide breakdown, mobility degradation and threshold voltage shift.

Gate oxide damage occurs basically due to plasma etching of interconnects connected to the gate, that's why this effect is also called "Plasma Induced Gate Oxide Damage" or "Antenna Effect". The metal interconnect which collects the plasma (ions) and is connected to the gate is basically termed as the antenna.

Ways to remove antenna violations

1. **Antenna diode** – To avoid this deposition of charge at the gate of a transistor, a diode is generally used in reverse biased mode which can drain out the charge without affecting the transistor circuitry.

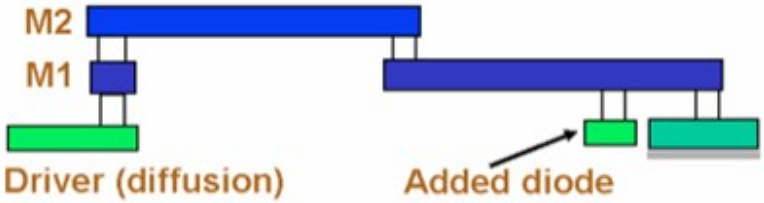


Figure 1 An antenna diode is used to remove antenna violation near the receiver.

2. **Metal hopping** – When the metal connected to a gate is long and there is space for a higher metal hop , it is always advisable to do so instead of using an antenna diode in order toavoid antenna violations.

Suppose we jog a metal 2 net with metal 3. Then while etching metal 2, the part of the net which is drawn in metal 2 only comes into picture because metal 3 has not been manufactured yet. So the effective charge reduces. This is the reason for using a higher metal as a jumper.

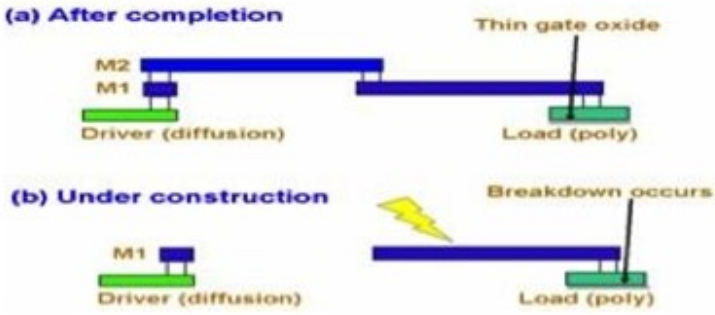


Figure 2 Use metal hopping to avoid antenna violations.

Electromigration:

When a high current density passes through a metal interconnect, the momentum of current-carrying electrons may get transferred to the metal ions during the collision between them. Due to the momentum transfer, the metal ions may get drifted in the direction of motion of electrons. Such drift of metal ions from its original position is called the electromigration effect

Effects of EM:

Once the metal ions get started to shifting from its original position, these will create problems in the interconnect. It could result in an excess of ions accumulation in a particular location of deficient of ions. So either Hillocks or Void could occur in the metal interconnect.

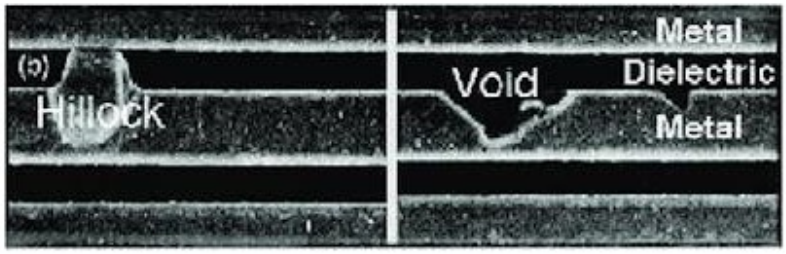


Figure-1: Hillock and Void formation in Interconnect

Void: If the incoming ion flux is lesser than the outgoing ion flux, It will create a void in interconnect. A void can lead a discontinuity in the interconnect and result an open circuit.

Hillocks: If incoming ion flux is greater than the outgoing ion flux, It will cause the accumulation of ions and create a hillock in the interconnect. A hillock can increase the width of a metal interconnect and touch the neighbouring metal interconnect which may result in a short circuit.

Prevention techniques for EM:

With the scaling of the technology node, the interconnect used is also changed. Initially, pure Aluminium was used as interconnect then the industry started using the Al-Cu alloy and later shifted to Copper interconnects. Copper interconnects can withstand approximately 5 times more current as compared to Aluminium interconnects while maintaining similar reliability requirements.

During the physical design, the following techniques could be used to prevent the EM issue

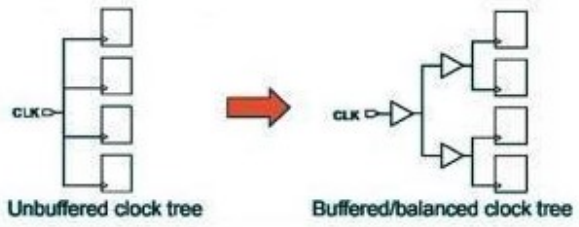
- Increase the metal width to reduce the current density
- Reduce the frequency
- Lower the supply voltage
- Keep the wire length sort
- Reduce the buffer size in clock lines

To prevent the EM issue, EM Checks is performed during the physical signoff stage with respect to the EM rules provided by the foundry.

CLOCK-TREE SYNTHESIS

Clock tree synthesis

- CTS is one of the most important stages in PnR. CTS QOR decides timing convergence and power. In most of the ICs clock consumes 30 to 40% of total power. So efficient clock architecture, clock gating and clock tree implementation helps to reduce power
- The process of distributing the clock and balancing the load is called CTS. Basically, delivering the clock to all sequential elements. CTS is the process of insertion of buffers and inverters along the clock paths of ASIC design in order to achieve zero or minimum skew or balanced skew. Before CTS, all clock pins are driven by a single clock source. CTS starting point is clock source and CTS ending point is clock pin of sequential cells.



Difference between HFNS and CTS

- Buffers and clock inverter with equal rise and fall times are used. Whereas HFNS uses buffers and inverters with a relaxed rise and fall times.
- HFNS are used mostly for reset, scan enable and other static singles having fan-outs. There is not stringent requirements of balancing and power reduction.
- Clock tree power is given special attention as it is a constantly switching signal. HFNS are mostly performed for static signals and hence not much attention to power is needed.
- NDR rules are used for clock tree routing.

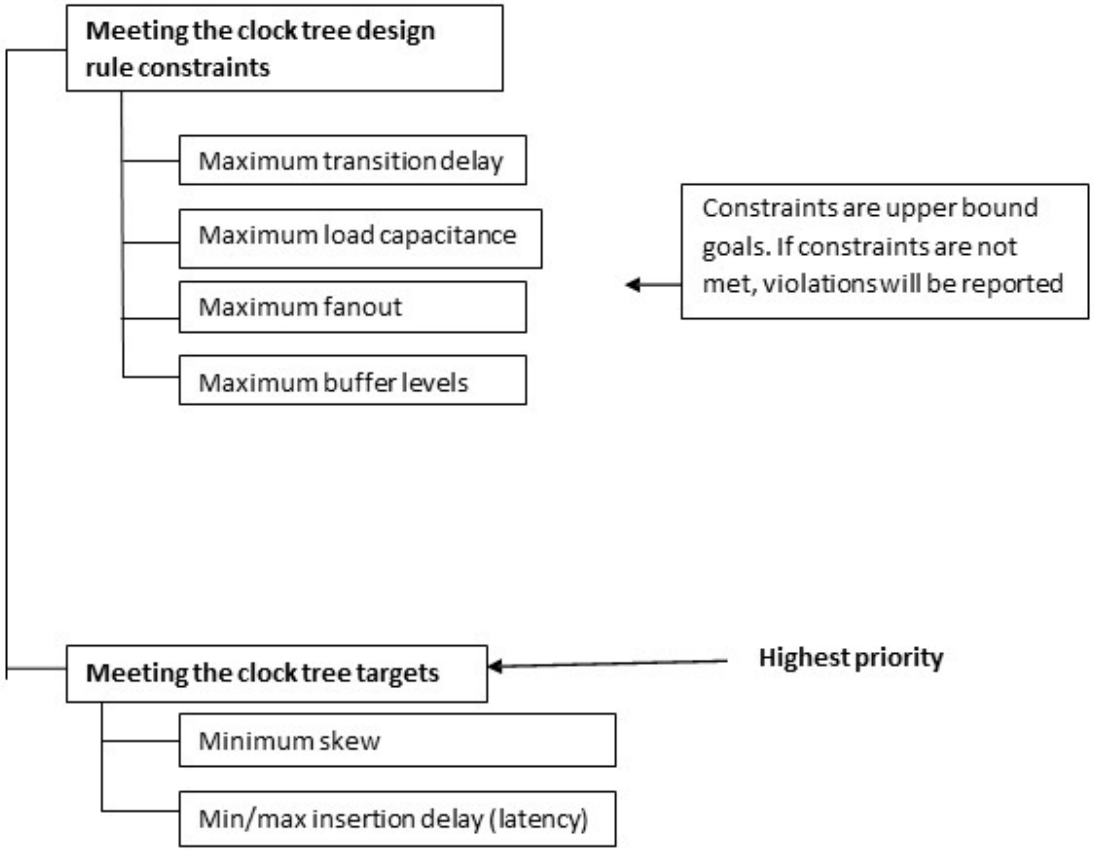
Difference between clock buffer and normal buffer

- Clock buffer have equal rise time and fall time therefore pulse width violation is avoided. In clock buffer BETA ratio is adjusted such that rise and fall time are matched. This may increase size of clock buffer compared to normal buffer.
- Normal buffer may not have equal rise and fall time. The clock buffers are usually designed such that an a input signal with 50% duty cycle produces an output with 50% duty cycle.

Inputs of CTS

- Technology file(.tf)
- Net list
- SDC
- Library file(.lib, .lef) and TLU+ file
- Placement DEF .
- Clock specification file which content insertion delay skew ,clock transition, clock cells, NDR, CTS tree type, CTS exception, list of buffers or inverters etc.

Goals of CTS

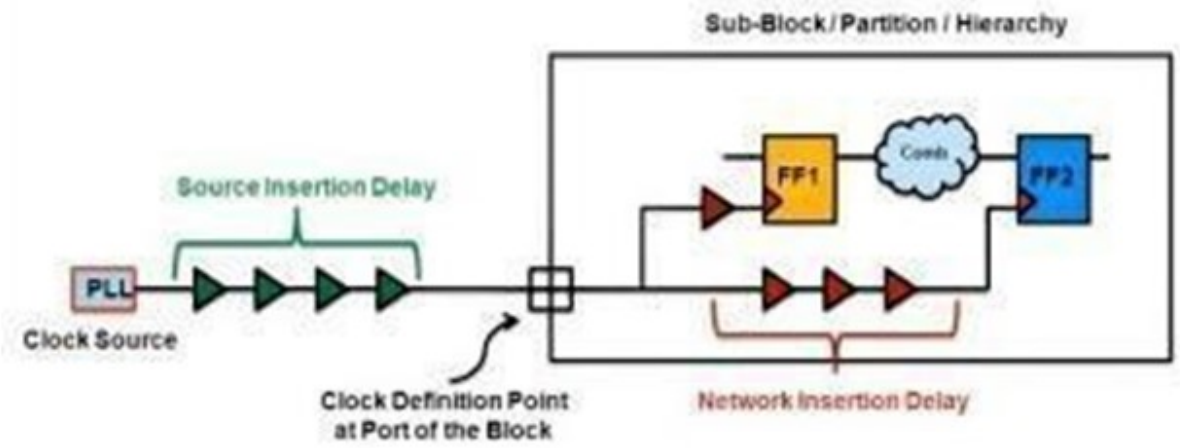


Sanity checks need to be done before CTS

- Check legality
- Check power stripes, standard cell rails and also verify PG connections.
- Timing QOR (setup should be under control)
- Timing DRVs
- High fan out nets (Like scan enable/any static signal)
- Congestions {Running CTS on congestion design/ design with congestion hot spot can create more congestion and other issues(Noise/IR)}
- Remove don't _use attribute on clock buffers and inverter
- Check whether all pre existing cells in clock path are balanced cells.
- Check and qualify don't_touch, don't size attribute on clock components.

Different Clock Terms:

a) Clock latency/insertion delay



- Total time taken by the clock signal to reach the inputs of the register
- Source latency is the time between clock sources to clock definition ports
- Network latency is the time between clock definition ports to clock leaf cells in the design

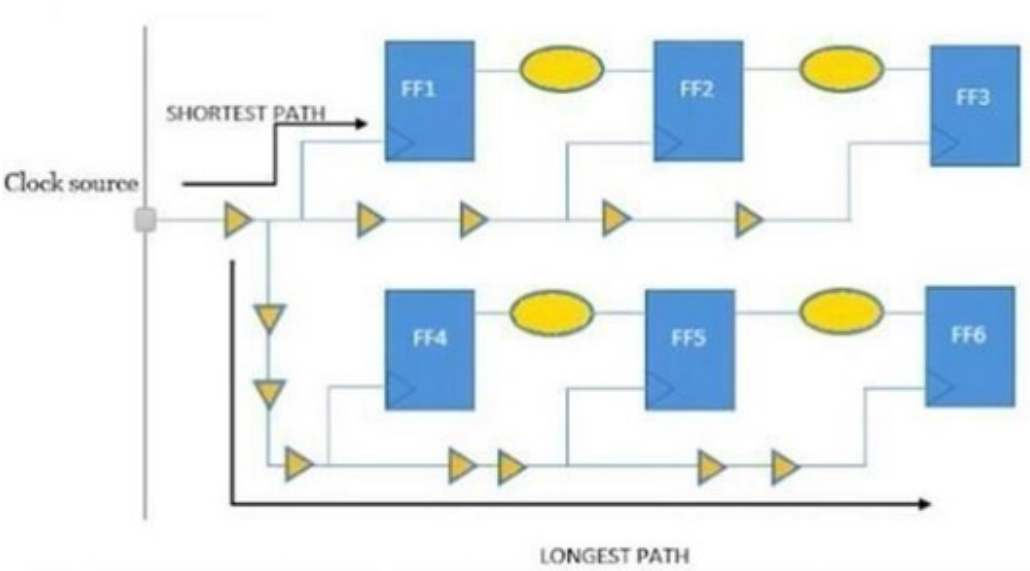
b) Insertion delay (ID)

- ID is the clock latency, but it is after clock tress is synthesized.
- ID is the physical delay and clock latency is the virtual delay.
- Latency is a target/constraint given to the tool through SDC life or clock tree attribute file and insertion delay is the achieved delay value in clock path after CTS.

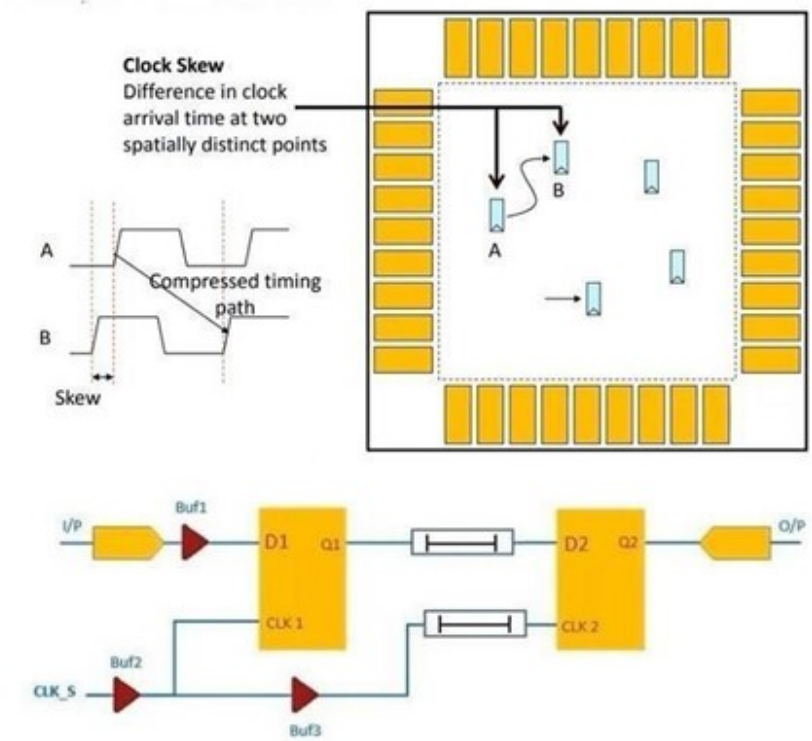
c) Clock skew : Clock skew between two flip-flops represents the difference in arrival times of clock signal at the respective clock pins

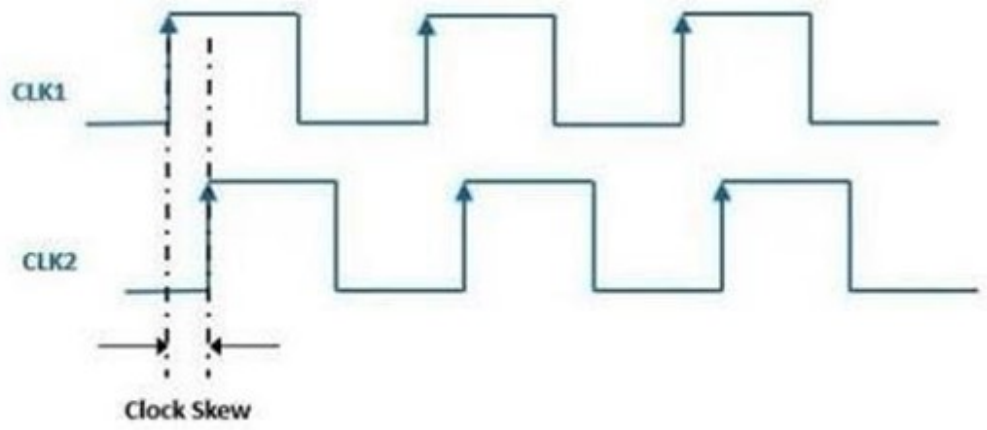
Local skew: It is the difference in arrival of clock at two related flip-flops .

Global skew: It is defined as the difference between max insertion delay and min insertion delay of any flops. It is also defined as the difference between shortest clock path delay and longest clock path delay reaching two sequential elements.

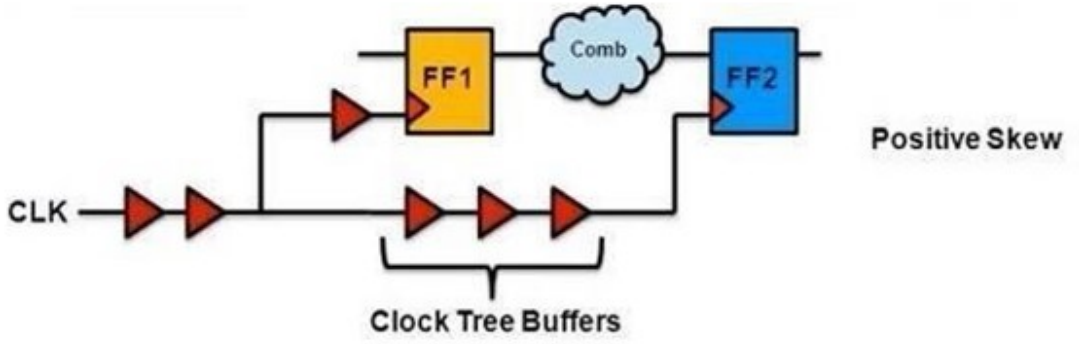


Useful skew : If clock is skewed intentionally to resolve violations. It is called useful skew.

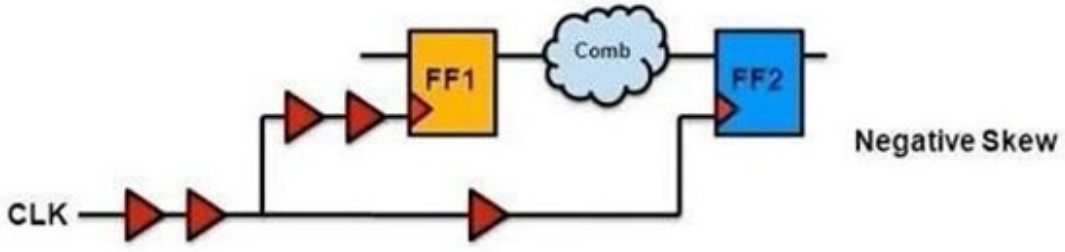




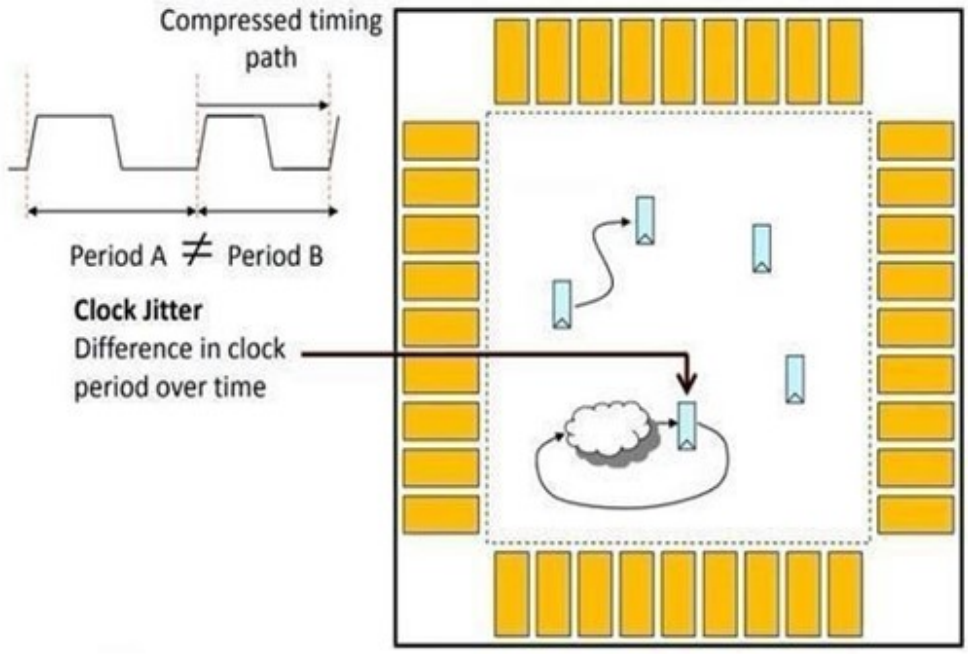
Positive skew: If capture clock comes late than launch clock then it is called positive clock.



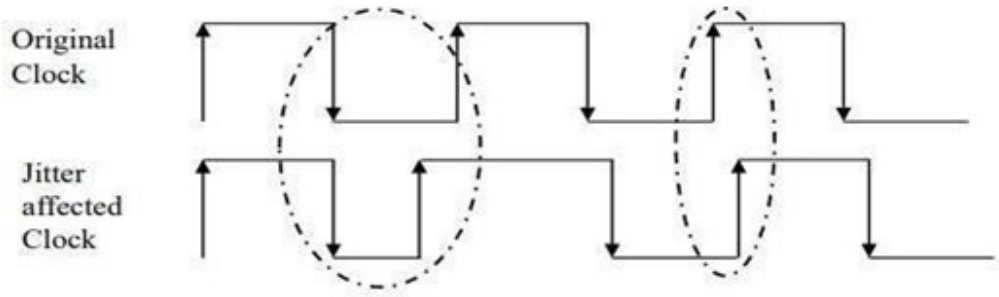
Negative skew : If capture clock comes early than launch clock then it is called negative clock.



d) Clock Jitter: Temporal Clock Variation



- Jitter is the short-term variations of clock edge from ideal position in time.
- The two major components of Jitter are random Jitter and deterministic Jitter.
- Factors causing Jitter includes imperfections in Clock oscillator, supply voltage variations, Temperature variations, Crosstalk.

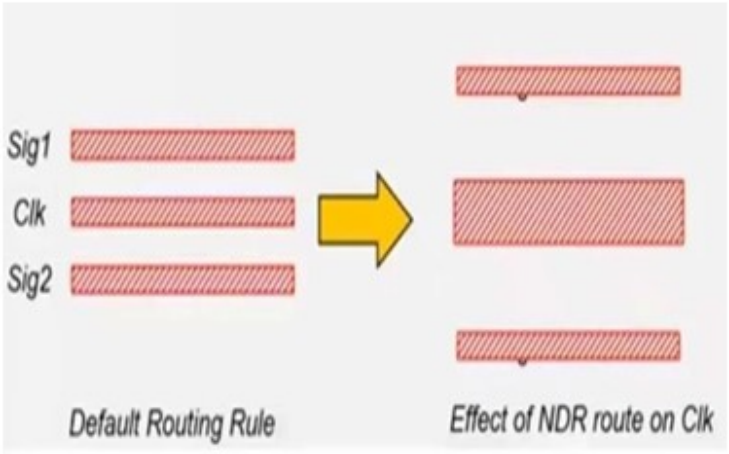


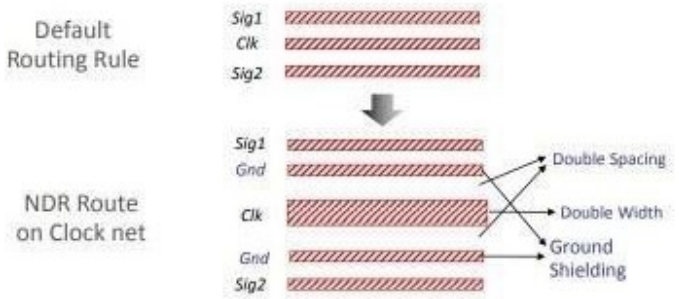
CONTENTS OF CTS SPECIFICATION :

- Cells to be used to balance clock trees
- CTS exceptions
- NDR rules (double width and double spacing)
- Max clock cap limit
- Clock inverters and buffers types
- Target skew (Different Skew groups)
- Max transistion
- Preferred routing layers

Non default rule

- This is user defined routing rules apart from the default routing rule
- NDR s make the clock routes less sensitive to cross talk or EM effects
- Double /triple width and spacing is used to avoid EM and cross talk
- NDRs will improve insertion delay

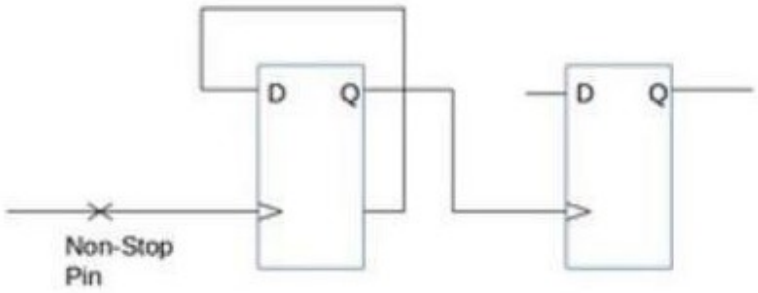




Clock tree exceptions:

- Non-stop pin
- Exclude pin
- Float pin
- Stop pin
- Don't touch sub-tree
- Don't buffer net
- Don't size net

Non stop pin :

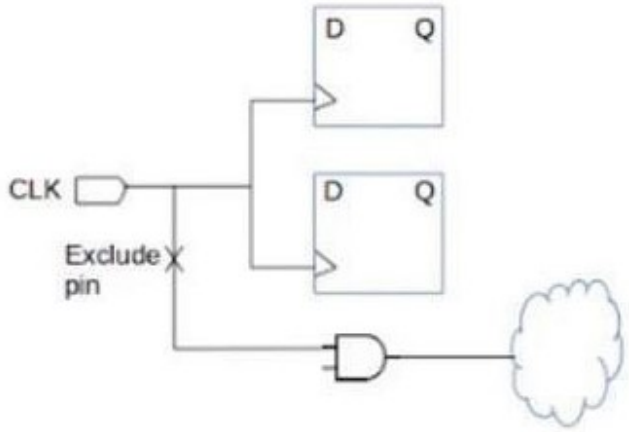


Non-stop pins trace through the endpoints that are normally considered as endpoints of the clock tree.

Example:

- The clock pin of sequential cells driving generated clock are implicit non stop pins
- Clock pin of ICG cells

Exclude pin :



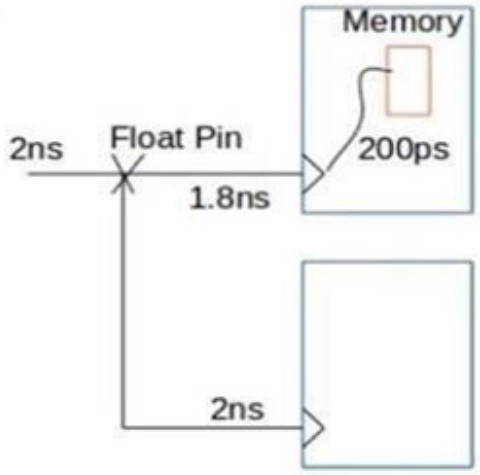
Exclude pins are non clock tree endpoints that are excluded from clock tree timing calculation and optimization. The tool considers exclude pins only in calculation and optimization of design rule constraints. During CTS the tool isolates exclude pins from the clock tree by inserting a guide buffer before the pin or these pins are need not be considered during the clock tree propagation.

Example :

- Non clock input pin of sequential cell
- Multiplexer select pin
- Three state enable pin
- Output port
- Incorrectly defined clock pin (if pin don't have trigger edge info)
- Cascade clock

In the above figure, beyond the exclude pin of the tool never perform skew or insertion delay optimization but does perform design rule fixing.

Float pin :

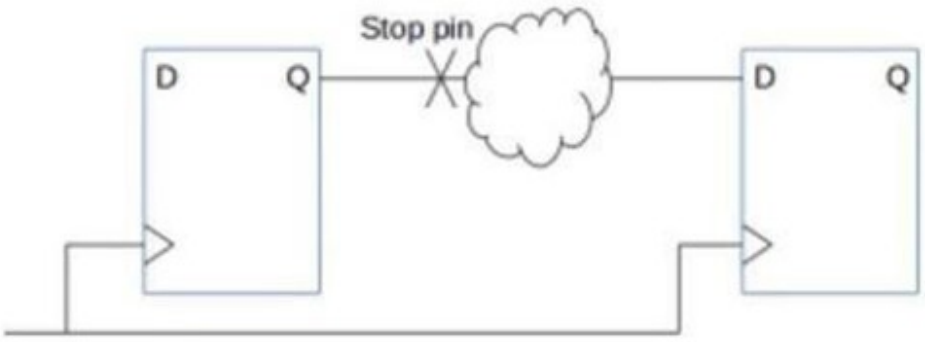


Float pins are pins that have special insertion delay requirements and balancing is done according to the internal pin delay. This is same as sync pin but internal clock latency of the pin is taken into consideration while building the clock tree. To adjust the clock arrival for specific endpoints with respect to all other endpoints.

Example :

- Clock entry pin of hard macros

Stop pin:



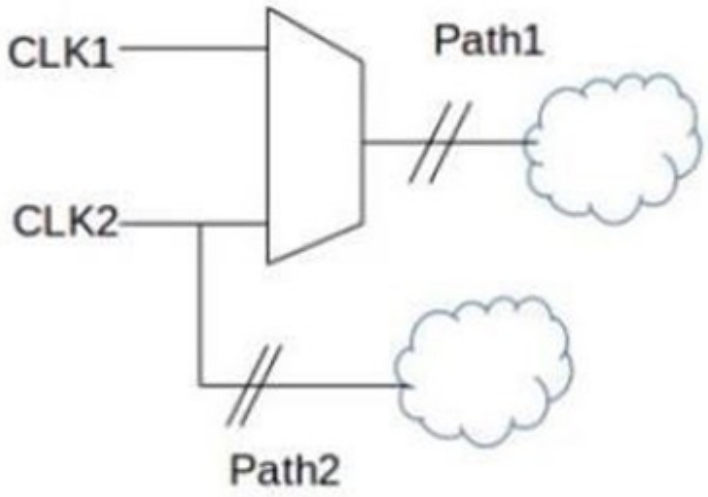
Stop pins are the endpoints of clock tree that are used for delay balancing. In CTS, the tool uses stop pins in calculation and optimization for both DRC and clock tree timing.

Example:

- Clock sink are implicit stop pins

The optimization is done only up to the stop pins as shown in above fig. The clock signal should not propagate after reaching the stop pin. This pin needs to be considered for building the clock tree.

Don't touch sub-tree :



If we want to preserve a portion of an existing clock tree, we put don't touch exception on the sub-tree.

- CLK1 is the pre existing clock and path 1 is optimized with respect to CLK1.
- CLK2 is the new generated clock. Don't touch sub-tree attribute is set with respect to CLK1.

Example:

- If path1 is 300ps and path2 is 200ps, during balancing delay are added in path2.
- If path1 is 200ps and path2 is 300ps, during balancing delay cant be added on path 1 because on path1 don't touch attribute is set and we get violation.

Don't buffer nets:

It is used in order to improve the results, by preventing the tool from buffering certain nets. Don't buffer nets have high priority than DRC. CTS do not add buffers on such nets.

Example:

- If the path is a false path, then no need of balancing the path. So set don't buffer net attribute.

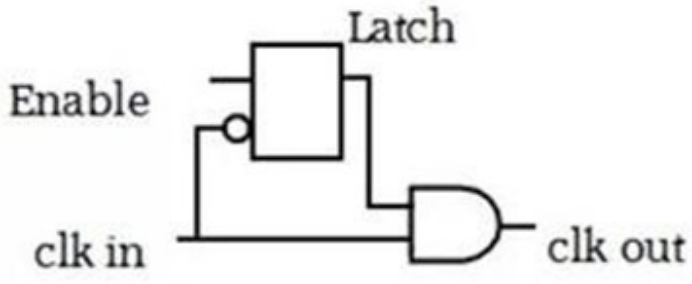
Don't size cell:

To prevent sizing of cells on the clock path during CTS and optimization, we must identify the cells as don't size cells.

Specifying size only cells:

During CTS and optimization, size only cells only be sized not moved or split.After sizing, if the cells overlap with an adjacent cell after sizing , the size-only cell might be moved during the legalization step.

ICG (Integrated clock gating cell) :



We use the ICG cell to stop the clock signal propagation to a big group of logic cells when the group is not required to operate.

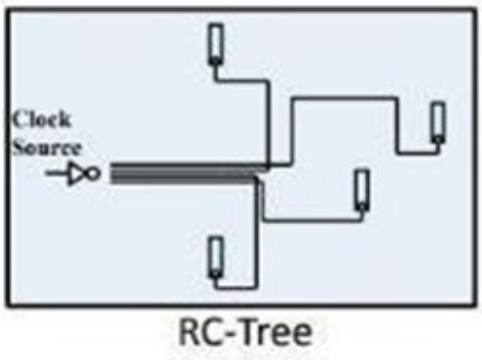
This is done through a clock enable signal generated internally in the block and applied to the EN pin of the ICG cell.

We know that the total power consumption of an SoC is the sum of dynamic power and static power. The clock tree is a major contributor to dynamic power as the clock signal has maximum switching activities.

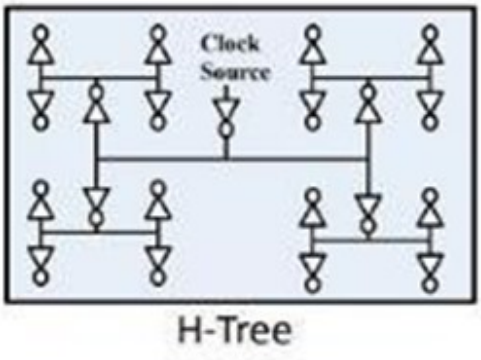
The ICG cell allows to stop the clock signal propagation beyond it and it helps to reduce dynamic power consumption in the design.

CTS algorithms:

- RC tree based CTS.
- H tree based algorithm.
- X tree based algorithm.
- Method of mean and medium (MMM)
- Geometric matching algorithm (GMA)
- Pi configuration

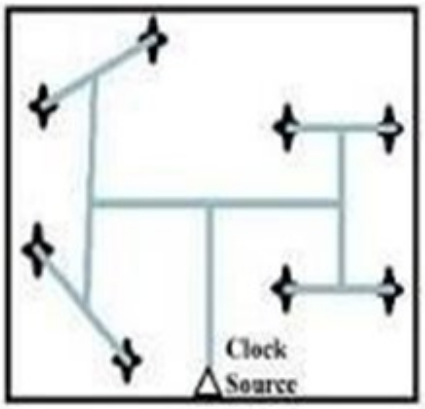


RC-Tree

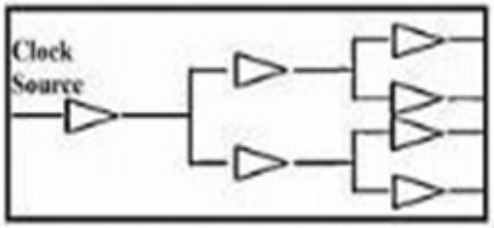


H-Tree

-

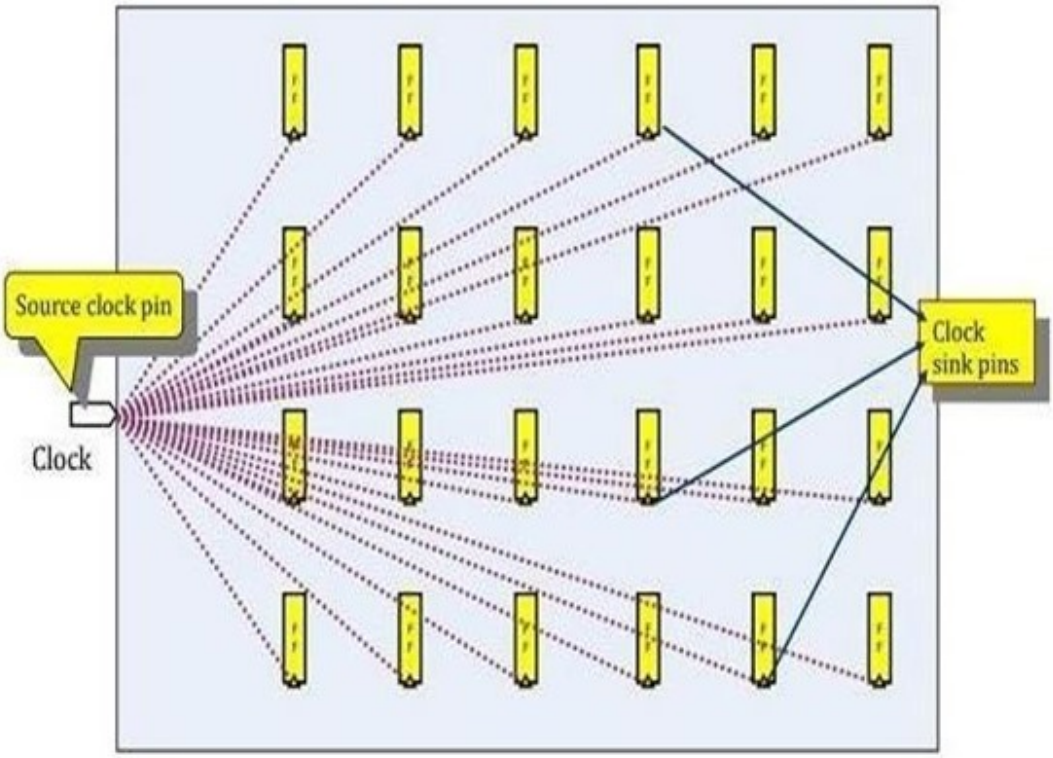


GMA

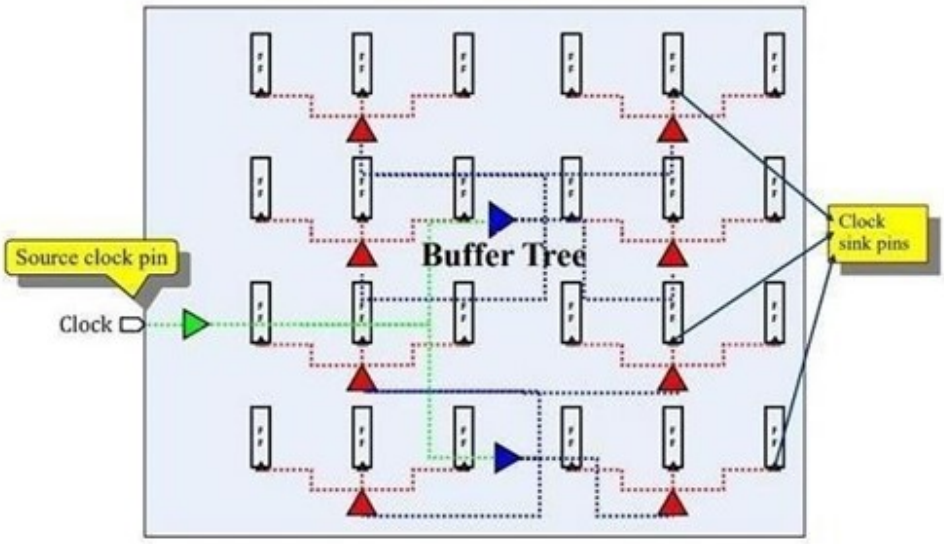


Pi Configuration

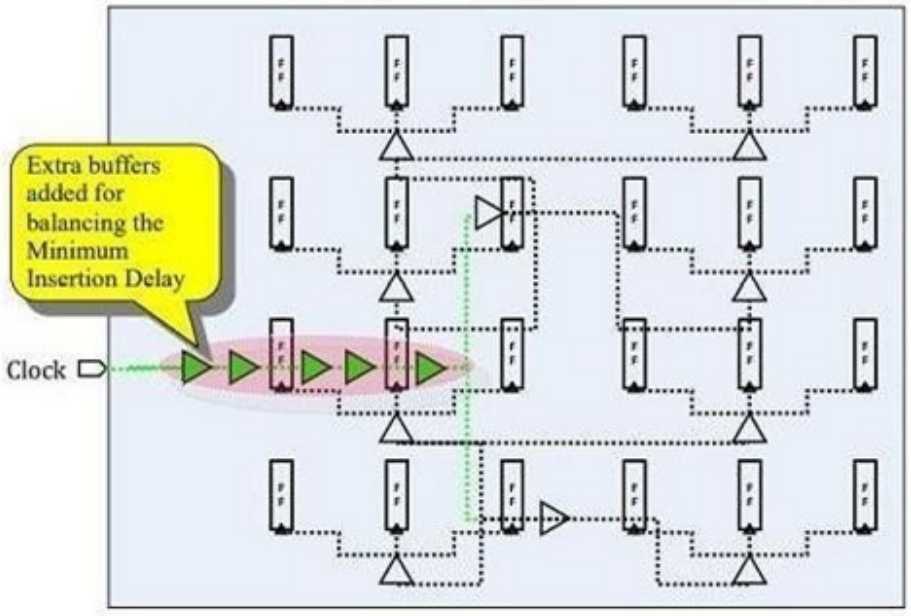
- Before CTS all clock pins are driven by a single clock source



- During CTS the buffer tree is built to balance the loads and minimize the skew



- After CTS a delay line is added to meet the minimum insertion delay (ID)



Analyze the clock tree:

- Report timing (both setup and hold)
- If timing not met then check clocks to be grouped (balanced together)
- Report insertion delay and skew and verify that the targets are achieved
- Report DRV targets (fanout, capacitance and transition)
- Check the intended leaf cell (clock sinks) is reached
- Check the clock tree exceptions are not in the clock tree
- Report the pre-existing cells, such as clock gating cells

- Do quality of report (QoR)
- Check clock tree converges either with itself or with another clock tree
- Clock tree has timing relationship with other clock trees for inter clock skew balancing
- Check design rule constraints
- Report power and area

Post CTS optimization:

- Optimization with useful skew to meet setup and hold time
- Optimization with total negative slack (TNS)
- Post CTS optimization techniques
    - Shielding
    - Sizing
    - Buffer re-location
    - Level adjustment
- Optimize the design for Hold time.
    - Hold violations should be fixed first in best corner and then in worst corner

CTS output

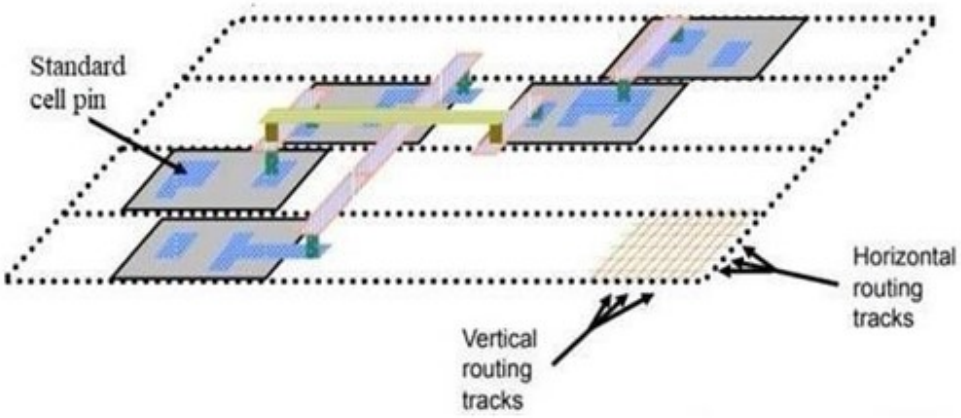
- Timing report
- Congestion report
- Skew report
- Insertion delay report
- CTS DEF file

Checks after CTS:

- In latency report check is skew is minimum? And insertion delay is balance or not.
- In QoR report check is timing (hold) met, if not why?
- In utilization report check standard cell utilization is acceptable or not?
- Check global route congestion?
- Check placement legality of cells.
- Check whether the timing violations are related to the constrained paths or not like defining false paths, asynchronous paths, half cycle paths, multi-cycle paths in the design.

ROUTING

- Making physical connections between signal pins using metal layers are called routing.
- Electrical connections using metals and vias are created in the layout, defined by the logical connections present in the netlist (i.e., logical connectivity is converted into physical connectivity)
- After CTS, we have information of all the placed cells, blockages, clock tree buffers / inverters and I/O pins. The tool relies on this information to electrically complete all connections defined in the netlist such that :
  - There are minimal DRC violations while routing
  - The design is 100% routed with minimal LVS violations
  - There are minimal SI (signal integrity) related violations
  - There must be no or minimal congestion hot spots
  - The timing DRCs and QoR are met and good respectively



Routing inputs

- Netlist
- All cells & ports should be legally placed with clock tree structure
- NDRs (Non Default Routing) rules
- Routing blockages
- Technology data (metal layers (.lef, .tf etc.,) DRC rules, via creation rules, grid rules

Routing goals

- Minimize the total interconnect / wire length
- Minimize the critical path delays
- Minimize the number of layer changes that the connections have to make (minimizing the number of vias)
- Complete the connections without increasing the total area of the block

- Meeting the congestion hotspots
- SI driven : reduction in cross-talk noise , delta delays and antenna violation.

Routing constraints

- Set constraints to number of layer to be used during routing
- Setting limits on routing to specific regions
- Setting the maximum length for the routing wires
- Blocking routing in specific regions
- Set stringent guidelines for minimum width and minimum spacing
- Set preferred routing directions to specific metal layers during routing
- Constraining the routing density
- Constraining the pin connections

Routing flow

The different tasks that are performed in the routing stage are as follows :

Trial/ Global Routing

- Identifying routable path for the nets driving / driven pins in a shortest distance
- Does not consider DRC rules, which gives an overall view of routing and congested nets
- Assign layers to nets
- Identify and assign net segments over the specific routable window called Global Route Cell (GRC)
- Avoid congested areas and also long detours

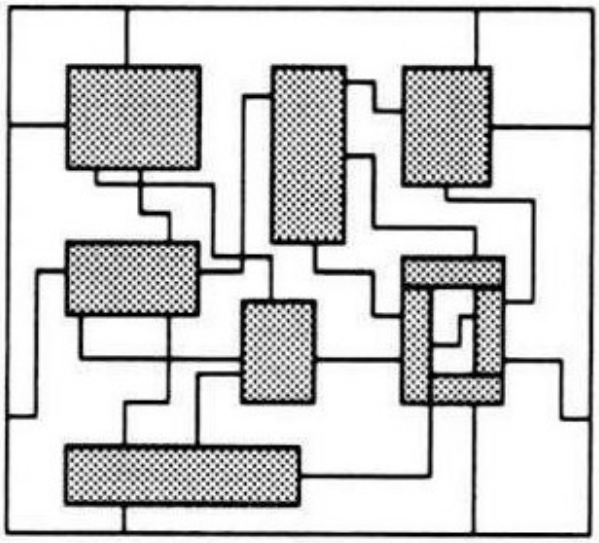
- Avoid routing over routing blockages
- Avoid routing for pre-route nets such as rings/stripes/rails
- Uses Steiner Tree and Maze algorithm

Track assignment

- Takes the global routed layout and assigns each nets to the specific tracks and layer geometry
- It does not follow the physical DRC rules
- It will do the timing aware track assignment
- It helps in via minimization

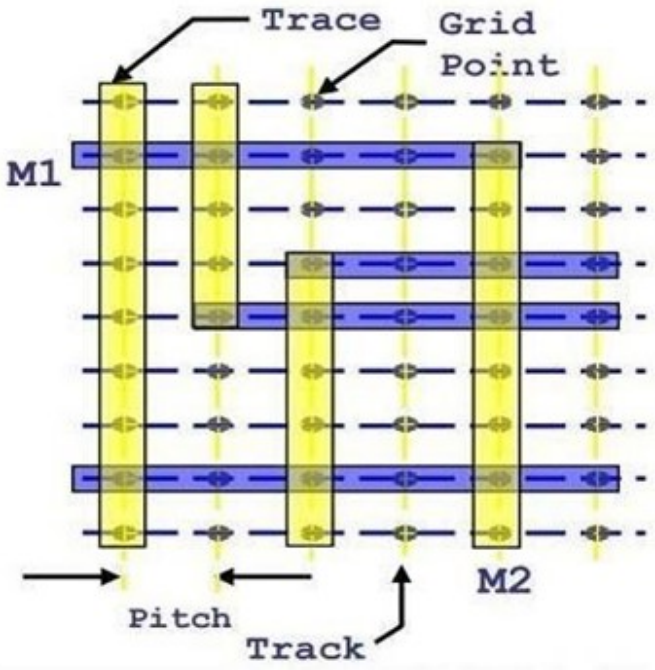
Detail / Nano Routing

- Detailed routing follows up with the track routed net segments and performs the complete DRC aware and timing driven routing.
- It is the final routing for the design, built after the CTS and timing is freeze
- It routes Metal layers on tracks so that it maintains minimum spacing
- During routing it continuously checks for DRC violations and fixes it in iterative steps



Grid based Routing

- Metal traces (routes) are built along and centered upon routing tracks on the grid points
- Various types of grids are manufacturing grid, routing grid (pitch) and placement grid
- Grid dimension should be multiple of manufacturing grid

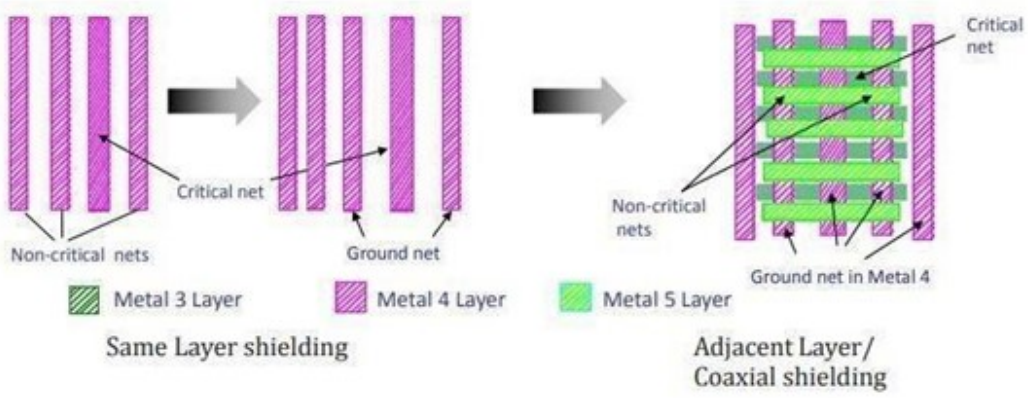


Search and repair :

- The search and repair stage is performed during detailed routing after the first iteration. In search and repair, shorts and spacing violations are located and rerouting of affected areas to fix all possible violation is executed

Post routing optimization

- Re-routing of time critical nets
- Signal integrity (SI) optimization by NDRs and shielding for the sensitive nets
- Types of shielding for sensitive nets
  - Same layers shielding
  - Adjacent layer / coaxial shielding



Filler cell insertion

- Filler cells can be inserted before or after detailed routing
- If fillers contain metal routing other than pre-routing then fillers should be inserted before routing
- Width of the smallest filler cell is the placement grid width
- Once fillers are inserted then the placement is fixed and tool can't move cells for further optimization

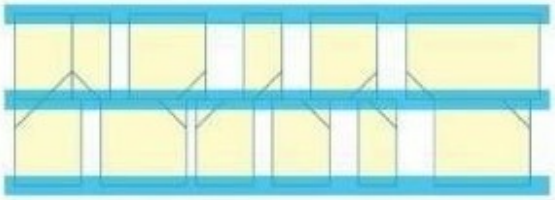


Fig. Before filler cell insertion

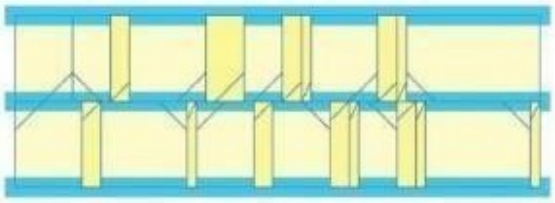


Fig. After filler cells are inserted

Metal fill

- Filling up the empty metal tracks with metal shapes to met metal density rules
- Two types of metal fill :
  - Floating metal fill : doesn't completely shield the aggressor nets, so SI will be prominent
  - Grounded metal fill : completely shields the aggressor nets, so less SI impact. This is complex as compared to floating metal fill
- Metal density rule helps to avoid over etching / metal erosion

PHYSICAL DESIGN VERIFICATION

Design Rule Check (DRC)

- Design Rule Check (DRC) is the process of checking physical layout data against fabrication-specific rules specified by the foundry to ensure successful fabrication.
- Process specific design rules must be followed when drawing layouts to avoid any manufacturing defects during the fabrication of an IC.
- Violating a design rule might result in a non-functional circuit or low Yield.

There are many design rules at different technology nodes, a few of which are mentioned below.

Types of DRC

i) Base level DRC : Here DRC is checked for geometries inside transistors

Main Checks are

- Well spacing, Poly spacing and poly width check
- Tap cell requirement check
- Well continuity check

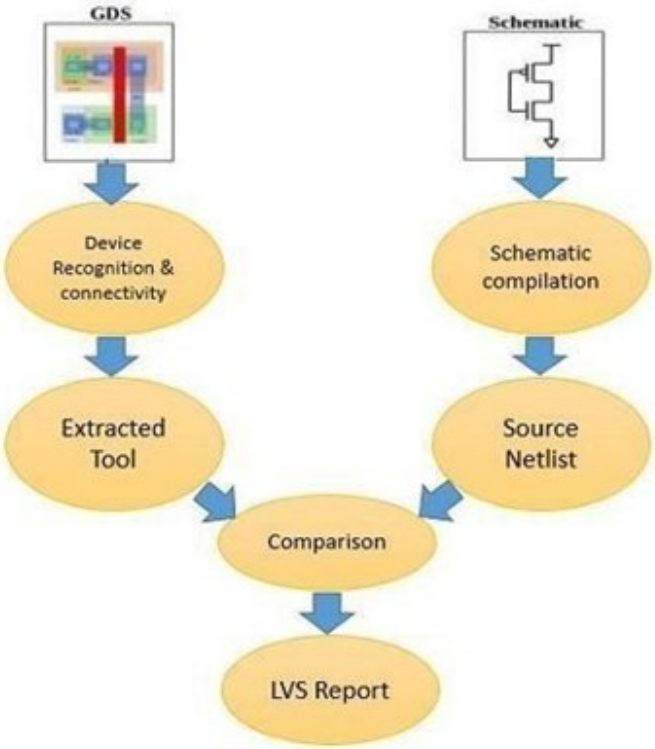
ii) Metal level DRC : It is checked on all routing layers and vias

Types of DRCs:

- Minimum width and spacing for metal
- Minimum width and spacing for via
- Fat wire Via keep out Enclosure
- End of Line spacing
- Minimum area
- Different net spacing
- Shorts violation
- Antenna Violation

Layout Versus Schematic (LVS)

- Layout Versus Schematic (LVS) verifies the connectivity of a Verilog Netlist and Layout Netlist (Extracted Netlist from GDS)
- Tool extracts circuit devices and interconnects from the layout and saved as Layout Netlist (SPICE format)
- As LVS performs comparison between 2 Netlist, it does not compare the functionalities of both the Netlist.
- Input Requirements
  - LVS Rule deck
  - Verilog Netlist
  - Physical layout database (GDS)
  - Spice Netlist (Extracted by the tool from GDS)



LVS Flow

LVS rule deck is a set of code written in Standard Verification Rule Format (SVRF) or TCL Verification Format (TVF). It guides the tool to extract the devices and the connectivity of IC's. It contains the layer definition to identify the layers used in layout file and to match it with the location of layer in GDS. It also contains device structure definitions.

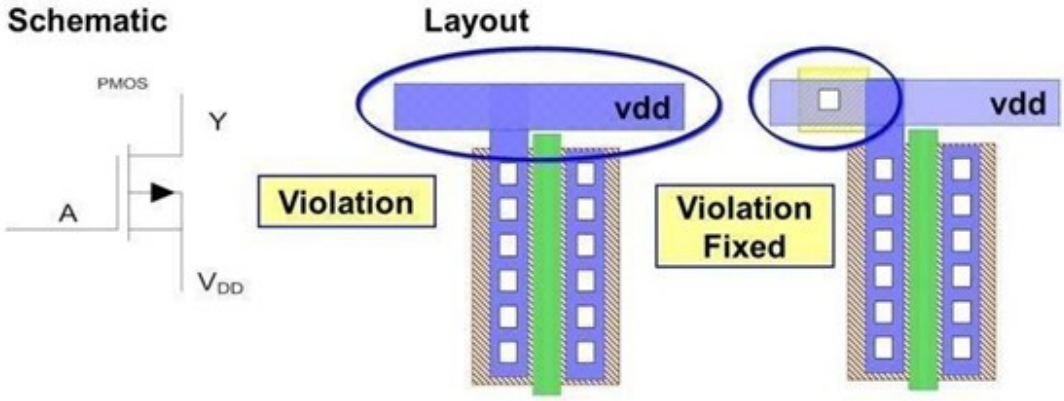
- LVS checks examples
  Short Net Error, Open Net Error, Extract errors, Compare errors

Electrical Rule Check (ERC)

- Electrical Rule Check (ERC) is used to analyze or confirm the electrical connectivity of an IC design

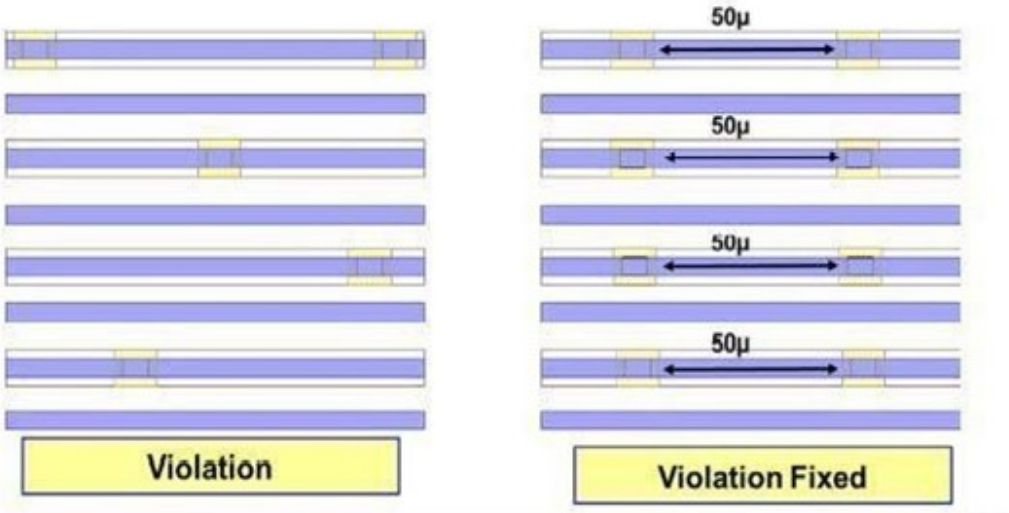
- ERC checks are run to identify the following errors in layout
  - To locate devices connected directly between Power and Ground
  - To locate floating Devices, Substrates and Wells
  - To locate devices which are shorted
  - To locate devices with missing connections

- Well Tap connection error: The Well Taps should bias the Wells as specified in the schematics



- Well Tap Density Error: If there is no enough Taps for a given area then this error is flagged
- Taps need to be placed regularly which biases the Well to prevent Latch-up

  e.g., In typical 28nm process the Well Tap Density Rule require Well-taps to be placed every 50 microns



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VLSI PHYSICAL DESIGN

ASSIGNMENTS

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2023

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PHYSICAL DESIGN ASSIGNMENTS

(Total: 428 Questions)

I. PD INPUT FILES

SUBJECTIVE TYPE QUESTIONS

1. What is the difference between a Milky way design library and reference library? What do they have in common?
2. What are different types of delay models. Explain.
3. Match the following.
   A. Partitioning
   B. Floorplanning
   C. Placement
   D. Routing
      (i) Takes a set of blocks of defined shapes and pin locations, and assigns the blocks to exact locations on the layout surface.
      (ii) Takes the netlists in some particular order, and provides interconnection among them.
      (iii) Tries to define exact shapes of the blocks to minimize layout area.
      (iv) Breaks a netlist into several pieces that can be laid out separately on the layout surface.
         a. A-(iv), B-(iii), C-(i), D-(ii)
         b. A-(iv), B-(i), C-(iii), D-(ii)
         c. A-(iv), B-(ii), C-(i), D-(iii)
         d. None of the above
4. what is LEF what is the significance of it in PD?
5. What is .lib , SDC file ?What information it contains?
6. what are the input of pd ? Explain all inputs and outputs of each stage in detail?

7. what are the delay models. Explain.
8. What are the initial sanity checks in the design before going to floorplan?
9. What is meant by .Tf file and what is presented in that?
10. What is the info in every input files like in .V , .TF, .lef, fp.def, .lib, etc?
11. In which format GDSII file will present?
12. What is instance count
13. What is gate count
14. What are the inputs required for floorplan..?
15. What is there in logical and physical library..?
16. Why you need technology file..?
17. What are Pre-placed cells
18. What is Std cell lef file and technology lef file?
19. What do you mean by color pattern
20. Why we are giving color for metal layers
21. Can we change the color of metal layers ?
22. What is current density
23. What does SCANDEF file contains ?
24. In which input file, cell delay information will be there?
25. Advantages of 28nm
26. What is there in .lib file
27. What is there in .lef file (or) .mw file?
28. What is there in .tf file
29. What is sdc file
30. How you will check the given Netlist is proper or not ?
31. On which factors cell delay depends?
32. 14) Pitch of the wire is____.
    a. Min width
    b. Min spacing
    c. Min width – min spacing
    d. Min width + min spacing

LABS TYPE QUESTIONS

1. Calculate the delay from below table for(0.08,0.2).
   Index1: input Trans
   Index2: output Load

```
                    {
index_1         ("0.04,
                 0.08,      0.08, 0.12,   0.16,   0.224,    0.28");
index_2         ("0        0.1  0.2,   0.28,   0.35,    0.45");
  values        (
      "0.079673,  0.207946,  0.334592,  0.425146, 0.569867,  0.696376,  0.877286",  \
      "0.091084,  0.219494,  0.346136,  0.436651, 0.581313,  0.708057,  0.888938",  \
      "0.106128,  0.234604,  0.361261,  0.451879, 0.596688,  0.722824,  0.904232",  \
      "0.121243,  0.249789,  0.376525,  0.466972, 0.611897,  0.738429,  0.919149",  \
      "0.136158,  0.264597,  0.39127,   0.481763, 0.62675,   0.753135,  0.934101",  \
      "0.15929,   0.287829,  0.414537,  0.505057, 0.64982,   0.776425,  0.957331",  \
      "0.178436,  0.30697,   0.433641,  0.524151, 0.668842,  0.795608,  0.976339");
```

2. Calculate the spacing requirement for parallel length of 1.5 and width of 0.75

```
SPACINGTABLE
   PARALLELRUNLENGTH 0 0.32 0.75 1.5 2.5 3.5
   WIDTH 0 0.06 0.06 0.06 0.06 0.06 0.06
   WIDTH 0.1 0.06 0.1 0.1 0.1 0.1 0.1
WIDTH 0.75 0.06 0.1 0.25 0.25 0.25 0.25
   WIDTH 1.5  0.06  0.1  0.25  0.45  0.45  0.45
   WIDTH 2.5  0.06  0.1  0.25  0.45  0.75  0.75
   WIDTH 3.5  0.06  0.1  0.25  0.45  0.75  1.25 ;
```

II. FLOORPLAN

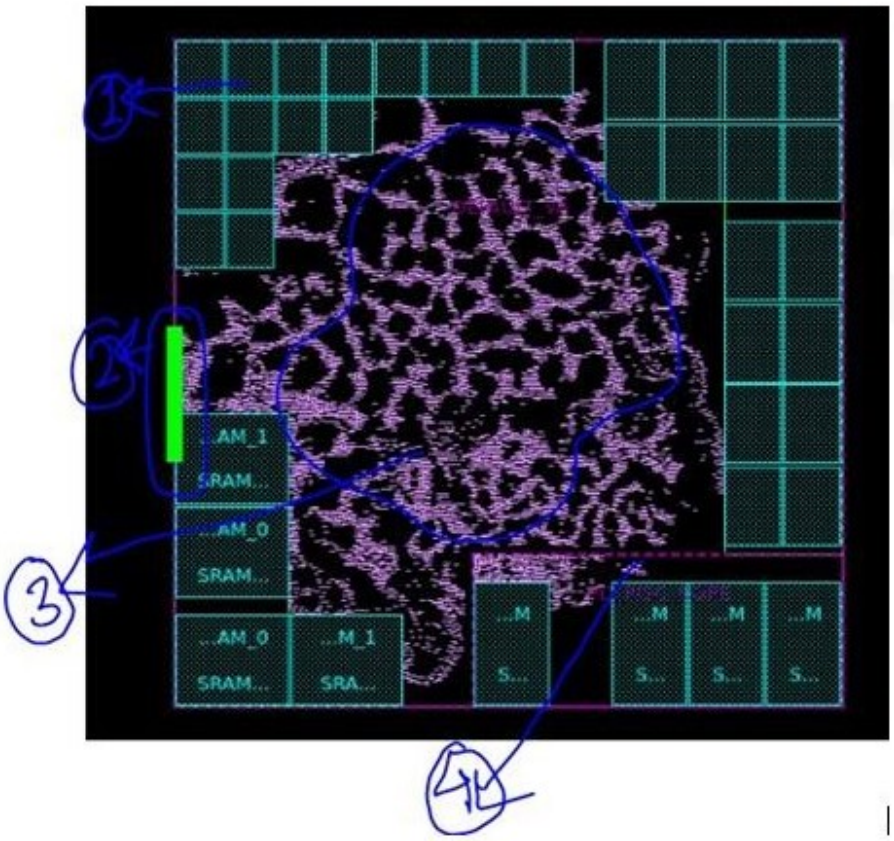
SUBJECTIVE TYPE QUESTIONS

1. What are input files in floorplanning stage?
2. What is latch up and reducing technique?
3. If there are two macros whose pins on not in same side then if I abut it ,what kind of DRC violation will I get.
4. How addition of tap cells help in CMOS Latchup.
5. So on which factors cell delay is dependent.if a cell has multiple pin then how is i/p transition calculated.
6. what is latch up. Explain with Diagram ? how to avoid latchp?
7. How will you decide the die size?
8. what guidlines u will keep in mind to place macro?
9. what are the physical cells? At what stage in pd it is used?
10. tool related question
11. If u have 9 metal layer in your design . Which metal layer u will use for power planning and CTS ? EXplain.
12. write down the formula to calculate the channel width.
13. How level shifters will help in power reduction ?
14. What are the low power design techniques ?
15. What is meant by Macro Modelling?
16. What are the physical cells u used in design ur design ?which purpose u used them?
17. Movement of macros depends on horizontal and vertical resources available or required. True or false?
18. What is Keepout margin or Hallows
19. What is difference between SOFT and HARD MACRO.?
20. What is IR-drop
21. What is channel length
22. What will happen if you apply HARD Blockage around macros.?
23. Why END-CAP cells are used for each block and not for full chip?
24. How to place macros when the no.of macros are around 500

25. What is channel spacing?
26. How do you qualify Floorplan
27. With respect to block-level fp arrange the following in right order.
28. {Place macros, creat die & core area, Place ports/pins, create p/g grid }.
29. What is difference b/w a Halo & a Blockage?
30. Explain various kind of preplaced cells.
31. Which of the following metal layer has Maximum resistance?
    a. Metal1
    b. Metal2
    c. Metal3
    d. Metal4
32. The minimum height and width a cell can occupy in the design is called as____.
    a. Unit Tile cell
    b. Multi heighten cell
    c. LVT cell d. HVT cell
33. What does initialize_floorplan do?
    1. Defines placement rows within the core area
    2. Defines the chip boundary or periphery area
    3. Places IO pad cells in their defined locations
    4. Places macro cells per their placement constraints A. 1,2,4
       B. 1,3,4
       C. 1,2,3
       D. All of the above
34. Explain the following terms :
    1.Die 2.Core 3.Pin 4.Ports 5.Terminals 6.Macros 7.Standard cells.
35. What are guidelines of macro placement .Explain need of flyline analysis

LABS TYPE QUESTIONS

1. Name all objects denoted By 1,2,3,4 What is use of Volatage area.



2. If there are Two macros Facing each other each have 100 pins and Pins are in M4 whose Pitch is 0.08, If it's a 9 metal later Desi Then how much spacing need to be given.

3. Create a keep_out margin of 1 micron across all the macros in the design Using set_keepout_margin

4. What does get_attribute [get_core_area] boundary do ? Analyze every pair of co-ordinates. Why is there seven pair of co-ordinates for six edges. Check in GUI.

5. Create a new floorplan using initialize_floorplan ? Right click on macros and see the fly lines. Re-arrange the macros to your best knowledge to reduce flylines crossing. See how the pins are arranged. What is the distance between the ports that tool has by default taken. Use various switches of initialize_floorplan to get the best floorplan structure.

6. Create a rectilinear floorplan using initialize_rectilinear_block Change values in the switches. How are the pin locations done. What is the distance between every pin ? Are they same. Locate the VDD and VSS pins using change_selection command

7. Write a TCl program to find total area of macros and cells belonging to I_RISC_CORE or I_PCI_TOP

8. Create core area (Write commands for following)
   a) For different co-ordinates mentioned
   b) Write core_area to DEF file and observe what is present in DEF file , and read DEF file
   c) Create core area for L shape for utilization 0.7
   d) Obtain co-ordinates for core area with utilization 0.6 and side ratio {1 2 1 1} for L shape

9. Theoritacally calculate core height and core witdth for utilization 0.75 and Aspect ratio {2 1} for R shape (TCL progaram) and comapare with practical value

10. Write TCL program to place ports at placement blockage location from left to right as below
    - Port_type : All input ports
      - layer : M6
      - Side : 2

- spacing : 2 track spacing

- Port_type : All output ports
  - layer : M6
  - Side : 4
  - spacing : 2 track spacing

11. Write TCL program to calculate co-ordinates for creating voltage area for PD_RISC_CORE power domain

12. Write commands to place macros and mention the guidelines for macro placement
13. Write a script to create hard blockages automatically around Macros by taking lu left & lu right to the Macros.

III. POWERPLAN

SUBJECTIVE TYPE QUESTIONS

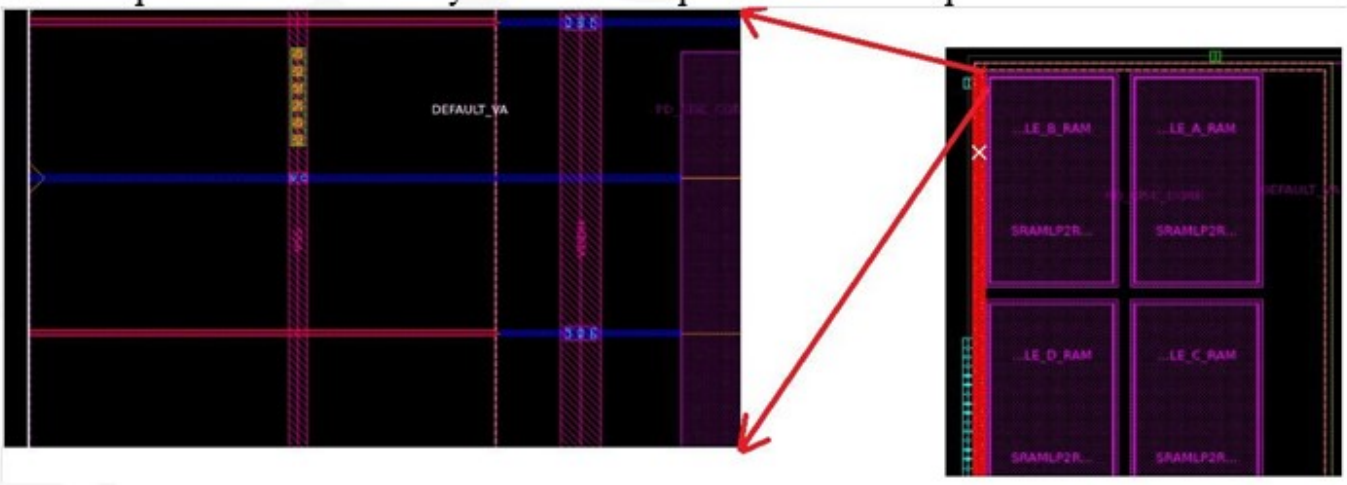
1. what is difference between Retention cells and isolation cells
2. How to calculate core ring and stripe widths?
3. Difference between Single Voltage Design and Multi Voltage Design?
4. What is UPF file? Differnce between CPF and UPF?
5. What is there in UPF and use of it?
6. How to fix IR-drop
7. What is the use of cut_row.tcl that is being used by powerplan?

LABS TYPE QUESTIONS

1. Create net shape for PG Strap

2. Mention Problems faced during power planning and explain methods to fix it. Justify your each fixes.

3. Explain the violation you see in snapshot? Fix & explain the same.



IV. STATIC TIMING ANALYSIS

SUBJECTIVE TYPE QUESTIONS

1. What is SET UP time and HOLD time?
2. What is multicycle path explain with example?
3. What is setup and hold time of flop? What if setup and hold violates? Can u do tapeout with setup or hold violation? Explain in detail?
4. How delays vary with different PVT conditions? Show the graph.
5. write down the Setup slack equation by considering uncertainity, CPPR.
6. write down the hold slack equation by considering uncertanitiy and cppr.
7. how to fix setup and hold violation?
8. what is CPPR/CRPR and why u need to consider?
9. what is temperature inversion explain with current equation.
10. What is transition? What if transition time is more?
11. What is OCV , POCV and AOCV? Explain.
12. why delay decreses when upsizing is done?
13. Does crosstalk always causes setup and violation?
14. What u will do if you have very less amount of Timing Violation (Let say 1ps)?
15. Let say u met your timing? And there are some logical drc violations. What u will do?
16. What are OCV's in chip designing? How u will reduce this effect?
17. What are different types of DRC'S with some Examples?
18. If we increase the fanout of a cell, how it will effects the cell delay?
19. Difference between WLDM and NLDM?
20. What is meant by Multi cycle path? Bydefault if you give inly multicycle path for setup, is there any problem for hold? If problem is there how you will reduce? Explain with waveforms?

21. Timing is good will you leave the problem of max transition? If any problem with it what is the problem created by the max transition violation?
22. You got less setup violation like 1ps what you will do, you won't bother about it or u will resolve?
23. What is SETUP AND HOLD TIME WITH CIRCUIT AND WAVEFORMS? And draw the waveforms separately when the capture clock is negative edge triggered?
24. What do you mean by Setup Margin ?
25. how do they decide setup time and hold time for particular flop?
26. Propagation delay for a cell, and all the other delay definitions?
27. What is meant by Cell delay ? What will happen if load is increasing to the output pin of cell?
28. Setup and hold slack calculation and explain the resolving techniques if they violated?
29. CRPR mechanism in the tool how it will calculate ?
30. What is meant by pulse width and difference between normal pulse width and min pulse width?
31. Min width, max transition, setup, hold in which priority you will resolve these issues (give 1 to 4 Preference)?
32. What is ocv ?
33. What is pvt corners ?
34. What is slow corner..?
35. What is fast corner..?
36. Why you want to check setup at slow corner?
37. Why you want to check hold at fast corner?
38. Why is fast corner having more voltage ..?
39. Why is slow corner having less voltage.?
40. What is typical voltage..?
41. To fix, Timing violations which cell is better HVT or LVT ?
42. To fix, Power violations which cell is better HVT or LVT ?
43. In Timing report how you identify that the path is setup or hold
44. What is meant by Max path ?

45. What is meant by Min path ?
46. How to fix Setup violation ?
47. How to fix Hold violation ?
48. What is VT Swapping ?
49. How do you fix transition violation?
50. In which scenario, Setup and hold violations will occur together ?
51. What is Derate?
52. What is CRPR
53. What is multicycle path
54. What is false path
55. What is disable path
56. What is crosstalk
57. How you will fix the crosstalk?
58. What are the different types delays on which we focus mainly during STA?
59. Explain mmmc. Which parasitic interconnect corners are checked for setup and hold?
60. Explain OCV, Derate and CRPR.
61. What is setup & hold timing? Derive equation to find setup & hold slack.
62. What are timing exceptions? Why they should be given as constraints?
63. What is uncertainty? Explain inter clock uncertainty.
64. Can there be setup & hold violation for same path? Justify your answer.

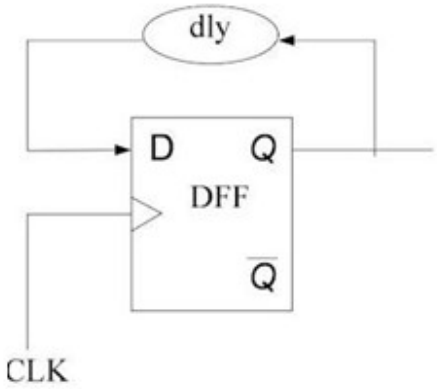
65. To achieve better timing_____cells are placed in the critical path.
a. HVT
b. LVT
c. RVT
d. SVT

66. In OCV timing check, for setup time,____.

a. Max delay is used for launch path and Min delay for capture path
b. Min delay is used for launch path and Max delay for capture path
c. Both Max delay is used for launch and Capture path
d. Both Min delay is used for both Capture and Launch paths

67. Which of the following is having highest priority at final stage (post routed) of the design____?
e. Setup violation
f. Hold violation
g. Skew
h. None
68. What happens to delay if you increase load capacitance?
69. What is cell-delay?
70. What is net-delay?
71. What is timing-arc?
72. What is setup time & Hold time?
73. What are the components you need to include while doing setup analysis?
74. What are the components you need to include while doing Hold analysis?

75. What is clock-skew?
76. Explain in2reg, reg2reg, reg2out & in2out paths in STA.
77. What are all the different components you see in a reg2reg timing report?
78. Can you tape-out the chip with setup & hold violation?
79. Lets say Setup & Hold is clean, but you've DRV violations. Can we tape-out the design?
80. Can there be setup and hold violation on same path?
81. What is setup and hold violation and reducing techniques
82. Given the following flip-flop circuit with a delay dly between input and output. And the clock CLK is applied to the flip flop

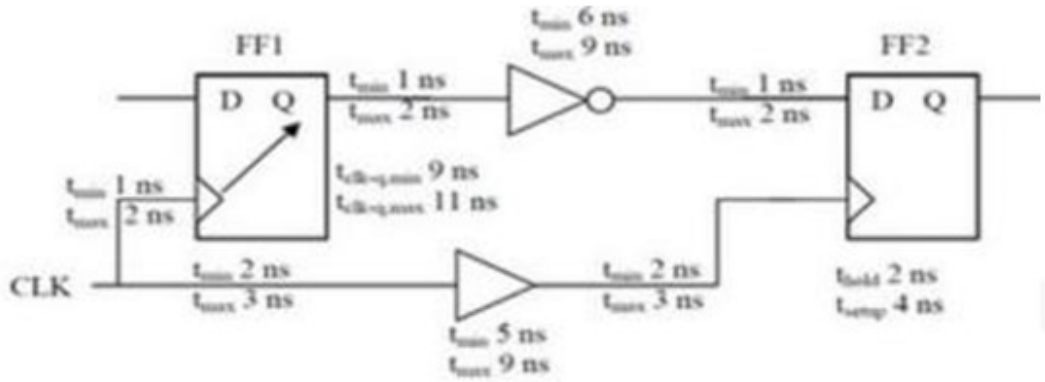


a. What will be the expression for the minimum time period (Tminimum) or maximum clock frequency (fmaximum)? Derive it by considering clock to Q delay (Tclock_Q), setup time (Tsetup_time), and hold time (Thold_time) of the flipflop.

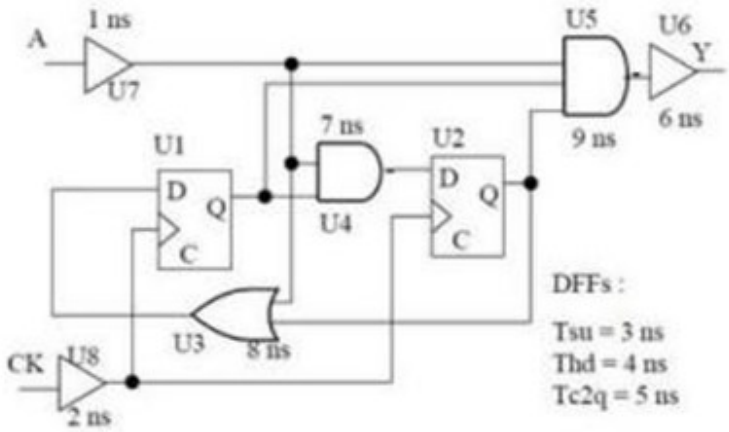
b. If the value of dly = 0ns, and given three flipflop values as per the table below, Which of the flip flops are a good fit when the minimum required time period is 5ns, 8ns, and 15ns.

Flip Flops	FF1	FF2	FF3
Tclock_Q	5	6	8
Tsetup_time	3	4	2
Thold_time	2	1	1

83. In the following Circuit, Find out whether there is any Setup Or Hold Violation? Given: Clock period is 15ns



84. In order to work correctly, what should be the Setup and Hold time at Input A in the following Circuit. Also find out the maximum operating frequency for this circuit. (Note: Ignore Wire delay). Where Tsu- Setup time; Thd-Hold Time; Te2q- Clock-to-Q delay



LABS TYPE QUESTIONS

1. Below Snippet is from which file explain the file in detail;

```
create_clock [get_ports pclk]  -name PCI_CLK  -period 7.5  -waveform {0 3.75}
set_clock_uncertainty -setup 0.3  [get_clocks PCI_CLK]
set_clock_uncertainty -hold 0.1  [get_clocks PCI_CLK]
set_clock_transition -max -rise 0.2 [get_clocks PCI_CLK]
set_clock_transition -max -fall 0.2 [get_clocks PCI_CLK]
set_clock_transition -min -rise 0.2 [get_clocks PCI_CLK]
set_clock_transition -min -fall 0.2 [get_clocks PCI_CLK]
```

2. Write TCL program to find Driver cell of a DRV violating net and to upsize it automatically to fix timing vioalation.
3. Write TCL program to find Driver cell of a DRV violating net and to Downsize it automatically
4. Write TCL program to find Driver cell of a DRV violating net and to insert buffer at driver pin

V. PLACEMENT

SUBJECTIVE TYPE QUESTIONS

1. What is the significance of "set_zero_interconn_delay_mode"____?
2. NDR rules define non-default rules for routing. Why apply these before placement.
3. What is the benefit of having separate path groups for I/O logic paths?
4. What does applying a timing critical range do, and what is the benefit of this?
5. Explain Set_Driving_cel?
6. Explain Set_max_delay command ?
7. Applying a weight > 1 to a path group may increase a design's critical path delay. True or False?
8. How to reduce setup violations in placement stage?
9. What is insertion delay and different types?
10. Will addition of buffer can help in Setup violation. If yes then how.
11. Which is better buffer and inverters?
12. What is done to solve congestion in lower and higher metal?
13. How cell spreading helps in routing congestion?
14. Cell spreading is done on what basis?
15. What is threshold voltage? How it affect timing?
16. What is scan chain? What if scan chain not detached and reordered? Is it compulsory?
17. what is congestion? How to fix it?
18. What is difference between normal buffer and clock buffer?
19. If the routing congestion exists between two macros, then what will you do?
20. how tool does area recovery ?
21. what are the inputs for placement ?
22. what checks u will do after placement?
23. How to reduce dynamic IR drop and Static IR Drop?
24. What is cell delay and net delay? On which factor it depends?
25. what are the placement blockages. Explain the types of blockages.

26. The propagation delay of a long interconnection line can be reduced by:
5. Making the line thinner.
6. Making the line wider.
7. Inserting one or more buffers along the line.
8. All of the above

27. Cloning of a gate can reduce the delay when:
A. When the number of fanout connections from the gate is large.
B. All the fanout targets are located close to one another.
C. The input interconnection delay of the gate is large.
D. None of the above.

28.Suppose that a long interconnection line is split into four shorter segments by inserting repeaters. Which of the following statement is true?
i. The total delay increases by an amount equal to the delay of three repeaters.
j. The RC delay reduces sqrt(4) = 2 times.
k. The RC delay reduces by 4 times. d. None of the above.

29. What are the constraints present in .SDC?
30.What is meant by Path Grouping? What will happened if we keep extra weight on that groups?
31. What do you meant by scan chain reorder , what is the purpose of it, at which stages it will be done,
32. Techniques u applied to remove congestion in placement stage
33. For asynchronous circuit how you perform timing ?
34.What will happen if you reverse the Pmos & Nmos in the inverter circuit
35. What is placement
36. What is magnetic placement
37. What is Placer max rule

38. What is Cell padding
39. How Bound is used to fix the violations
40. What is Partial blockage
41. What is Multi-VT cells
42. What is Multi-driven net
43. In placement stage, how will you fix the setup violations
44. What is scan chain reordering?
45. Explain following terms and ways to fix them:-
    - Congestion
    - Electro migration
    - Crosstalk
    - CMOS Latch-up

46. Why we have to remove scan chains before placement?
    a. Because scan chains are group of flip flop
    b. It does not have timing critical path
    c. It is series of flip flop connected in FIFO
    d. None

47. Routing congestion can be avoided by____.
    a. placing cells closer
    b. Placing cells at corners
    c. Distributing cells
    d. None

48. What is the effect of high drive strength buffer when added in long net ?
    a. Delay on the net increases
    b. Capacitance on the net increases
    c. Delay on the net decreases
    d. Resistance on the net increases.

49. A "soft" placement blockage will:

a. Allow only timing-critical cells to be placed
b. Allow only non timing-critical cells to be placed
c. Allow cell placement only during initial coarse placement
d. Prevent cell placement during initial coarse placement

50. Circle the correct statement(s) about what create_fp_placement does by default:
    a. Optimizes logic (cell sizing, buffering) to improve timing
    b. Legally places std cells and non-fixed macros to minimize wire length
    c. Optimizes placement to improve timing
    d. Clumps cells from the same logical hierarchy

51. what is significance of following clocks:

```
create_generated_clock -name CLK_INV \
-source [get_ports CLK] -divide_by 1 get_pins A/Y
create_generated_clock -name CLK_INV \
-source [get_ports CLK] -divide_by 1 [get_pins A/Y] \
-combinational
```

52. Explain mmmc. Which parasitic interconnect corners are checked for setup and hold?
53. What is virtual clock? How we can say any clock is virtual/ generated/ mastered by looking into sdc. File?

54. What is cloning?
55. What is significance of bounds and placement blockages?
56. Command to check quality of your placement.
57. Write command to general configuration

LABS TYPE QUESTIONS

1. Write TCL program to spit high fanout net by clonning method .

2. a) For any one cells find all these attributes height , width , area , is_hard_macro , is_sequential , bbox
   b) Find Site heigth , site width and compare in GUI
   c) Find relation between pitch and tracks

3. Write commands to Filter as follows
   a) Filter only cells with attribute "is_sequential == true"
   b) Filter only ports with "direction == in"
   c) Filter metal layers with pitch more than 0.5
   d) filter cells with pin_count more than 5
   e) Filter only OAI gate with 2 X 2 inputs with drive strength X1 and HVT flavour

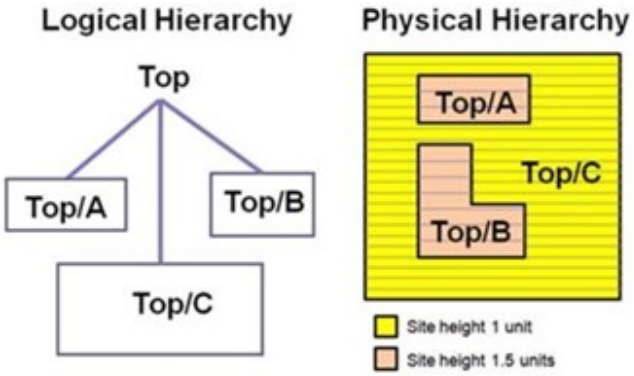
4. Read mcmm file and note
   a) Differnt clocks with thier time period
   b) Different False paths
   c) Differnt PVT conditions in each corner
   d) Derates in each corner

5. Write TCL program to filter cells with pin count more than 6 and to apply keepout margin.

6. What is magnet placement ?
   A. Do a placement with magnet placement
   B. Perform timing analysis to see the problem.
   C. Verify that the critical cells are correct
   D. Verify that the correct magnet cells moved and that the timing is better.

7. Write a script to get site rows within a rectangle Specify rectangle bounding box {x1 y1} {x2 y2}
   A. Get all rows inside the {x1 y1} {x2 y2} rectangle
   B. Use cut_rows to remove site rows in that specified area. of the site rows removed area.
8. Create bounding box list of site rows Print first and last bounding box

9. create exclusive bound for the given top/A.



10. In flat_dp stage during virtual flat placement, Use the commands to place the macros as an array or relative location.

11. Create blockages for cells. Write procedure provided that it should be able to get the below conditions.
    Create_blcokages -cells <cell type> -grow <by any value> -type <blockage type> -name <blkg_name>
12. Create Congestion Map from shell and save snapshot

13. Analyze congestion map layer based. & save snapshot
14. Check the scan chain cris-cross. Check the scan chain after place_opt –optimize dft. Check if there is any difference in the scan chain routing. Provide snapshot with explanation

15. Do Lab on following:
    1. Check timing after place_opt

2. Check if timing is met
3. Create big placement blockages as well as around macros.
4. Place_opt command to be run
5. Check timing. There should be degradation in timing.
6. Fix these violations.

16. Report timing for a path group
    a. Do get_clocks
    b. See number of clocks in the design
    c. Do get_path_groups
    d. See the number of path_groups in the design
    e. Check if the number of clocks listed and the number of path_groups match.
    f. Do get_scenarios
    g. Create group_path with I2R, R2R, R2O, I2O
    h. Use switches –weight and –critical_range for path_group.
    i. Report_timing –nosplit –group <> -from <> -to <> Use switches –trans, –cap, –crosstalk_delta, –derate, –input_pins, –nworst, –max_paths, –scenarios, –path_type full_clock_expanded
    j. Analyze AT and RT.

VI. CLOCK TREE SYNTHESIS (CTS)

SUBJECTIVE TYPE QUESTIONS

1. What is goal of CTS? what is false path explain with diagram.
2. What are clock tree exceptions?Explain.
3. What is skew,insertion delay,latency?
4. What are the two main goals of CTS?
5. What is the difference between stop and exclude pins? List some examples of implicit stop/exclude pins.
6. How is a float pin different from a stop pin?
7. What happens when building the CT using min_max?
8. Explain Clock tree optimization. Explain different ways of min. skew.
9. How do you set a clock skew target of 0.1 for clk1, and a minimum insertion delay of 0.7 for clk2? What is the skew target for clk2?
10. Write the command(s) to balance the two clock trees clk1 and clk2, so clk2 arrives 0.3 earlier!
11. Why is it important to remove or adjust the clock uncertainty before executing clock_opt
12. what is clock gating and use of clock gating?
13. what is electron migration and how to reduce EM?
14. Antenna violation and reducing techniques?
15. What is crosstalk.How to avoid cross talk problems?
16. what is uncertainity and types of uncertainity
17. Why we define Generated clock why can't we use same master clock on place of it.
18. At which stage we fix hold violations and why?
19. what is content of clockspec.tcl,Explain mining of target skew,global skew,local skew.
20. What is local skew and global skew?
21. What is latency? Give the types?
22. What is signal integrity? How it affects Timing?
23. What is cross talk? How can you avoid crosstalk?
24. What is difference between HFN and CTS?

25. what are the clock tree exceptions explain .
26. why after CTS hold fixed is done?
27. What is uncertainty? Why it occurs write down the reason?
28. why jitter is not considered post cts? write down the uncertainty equation pre CTS and Post CTS.
29. what is clock gating ? Explain with diagram.
30. what is power gating ? Explain difference between clock and power gating.
31. What is Spare cell ? Which stage you will use spare cell?

32. One of the naïve approaches to handle the clock skew problem is:
    A. Increase the time period of the clock.
    B. Balance the delays of the critical paths in the circuit.
    C. Equalize the worst-case delays of every combinational block between pairs of register stages.
    D. None of the above.

33. One of the naïve approaches to handle the clock skew problem is:
    1. Increase the time period of the clock.
    2. Balance the delays of the critical paths in the circuit.
    3. Equalize the worst-case delays of every combinational block between pairs of register stages.
    4. None of the above.

34. Why do we use buffers in a clock tree?
    ii. To reduce the delay.
    iii. To reduce the area requirement.
    iv. To generate multiple phases of a clock.
    v. None of the above.

35. It is sometimes necessary to deliberately add clock skew so that:
    1. The total delay of the combinational sections between sequential storage elements are made equal.

2. The clock frequency can be increased without moving gates across storage elements.
3. The power consumption is reduced.
4. None of the above.

36. when clocks nets are routed at which stage ?
37. what is latency? Explain the types of latency?
38. What is virtual and real clock ? why virtual clock is used?
39. Why unequal rise and fall times in normal buff/inv and Equal rise and fall times in clk buff/inv?
40. What is meant by Clock gating and which power is reduced by this technique ?
41. What is meant by Clock gating and which power is reduced by this technique ?
42. What is meant by source latency and insertion delay?
43. Difference between CTS Exceptions and Timing Exceptions?
44. Why you done CTS separately ?why you can't do along with in the placement stage
45. why u applied NDR in CTS , What are the exception you see in CTS .tcl file?
46. Types of buffers and inverter used in clock tree building? And difference between normal buffers and these buffers?
47. Why cross talk will come and techniques to reduce?
48. What is ESD and EM and techniques to resolve it?
49. How uncertainty value will change from starting stage to ending stage… whether it increase or decrease?
50. What is Clock gating technique
51. Advantages of clock gating
52. Disadvantages of clock gating
53. If Clock spec file is not provided by top level, then can u do it by urself ?
54. What is preferred and non-prefered Routing layer..?
55. What is Clock – SPEC File

56. Difference between clock buffers and normal buffers
57. What is Target skew ?
58. What is the Goal of CTS
59. What is NDR rule?
60. In which case D-CAP cell is used
61. What is local skew and global skew
62. What is the physical difference between clock routing and signal routing..?
63. Why clock buffers are used instead of normal buffers in clock path
64. Why insertion delay should be less
65. Why can't we do CTS before placement
66. How to identify if NDR rule is wrongly assigned to other signals
67. In which case hold depends on frequency
68. What is useful skew
69. What is un-clocked flops?
70. What is master clock
71. What is meant by generated clock
72. What is virtual clock
73. What is virtual clock? How we can say any clock is virtual/ generated/ mastered by looking into sdc. File?
74. What is clock latency?Explain source & n/w delay.
75. What is crosstalk delay and Crosstalk noise?
76. What is NDR? Which stage of PD we apply NDR and why?
77. What are the main goals of CTS?

78. Delay between shortest path and longest path in the clock is called ____.
    i. Useful skew
    ii. Local skew
    iii. Global skew
    iv. Slack

79. Cross talk can be avoided by____.

b. Decreasing the spacing between the metal layers
c. Shielding the nets
d. Using lower metal layers
e. Using long nets

80. Which of the following is best suited for CTS?
f. CLKBUF
g. BUF
h. INV
i. CLKINV

81. Can a new master clock be defined at output of th flip flop instead of generated clock .Support your answer with valid explaination.
82. In placement and CTS stage we haven't done routing. How does report.timing calculate the timing always on the nets?

LABS TYPE QUESTIONS

5. Note max_latency and min_latency , max_skew , global skew for all clocks in all scenarios

6. Write Commands to fix Setup and Hold after CTS .

7. check Clock net attributes before and after CTS. Explain.

8. Calculate insertion_delay to clock pin of a flop and its associated transition . . .

9. Report clock cells inserted by CTS

10. By default, ICC does not balance paths between two clock groups. If they are talking to each other, then they are balanced. Use command to see inter clock paths and its reported values

11. balance the paths between two clocks

12. Do OCV aware CTS. Explain

13. Removing placement of all standard cells

14. Create skew group to add flop clock endpoints to a group and give –target_skew

15. Clock path stop pin which CTS should consider as balance endpoint.

Clock path stop pin which CTS should consider as a float point.

If float, exclude, stop is set on the same pin . . .what precedence you are going to follow

16. Use skew_opt (for reducing skew) command

1) Collects the worst timing path between every startpoint and endpoint pair

2) Collects clock latencies at every startpoint and endpoint.

3) Classifies each pin as adjustable, fragile, or fixed.

4)Optimizes clock latencies at adjustable pins to improve timing slack.

5) Converts the latencies to skew for clock-tree synthesis to implement.

6) Configures interclock balancing for clock tree synthesis.

7) Writes a solution file. . . skew_opt.tcl file

8) Sources the solution file.

Analyze the solution file skew_opt.tcl file.

17.Analyze clock trees for all the clocks one by one . . e.g..SDRAM_CLK.

18. Analyze clock health . . .

Choose New Interactive CTS Window

VII. ROUTING

SUBJECTIVE TYPE QUESTIONS

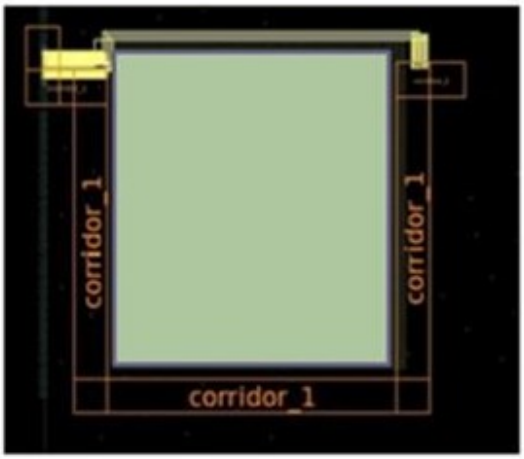
1. Prerouting means routing of_____.
    i. Clock nets
    ii. Signal nets
    iii. IO nets
    iv. PG nets
2. What is IR drop? How to avoid? how it affects timing?
3. What is EM and it effects?
4. What are types of routing?
5. what is antena effect? If you have antena violation in higher metal layer how you will fix?
6. What is EM how to reduce EM ?
7. What is meant by DRC and lvs rules ?
8. What is meant by ECO and what you will do there?
9. How shorts and opens formed and how to resolve?
10. What is meant by Preferred and non Preferred routing?
11. What is meant by Multi source CTS?
12. What is Antenna violations
13. Write short notes on LVS
14. Write short notes on FV
15. How to calculate net wire length of routed signals
16. When you are going to lower nodes, what complexity you will face
17. Filler cells are added____.
    a. Before Placement of std cells
    b. After Placement of Std Cells
    c. Before Floor planning
    d. Before Detail Routing
18. Search and Repair is used for____.
    a. Reducing IR Drop
    b. Reducing DRC
    c. Reducing EM violations

d. None

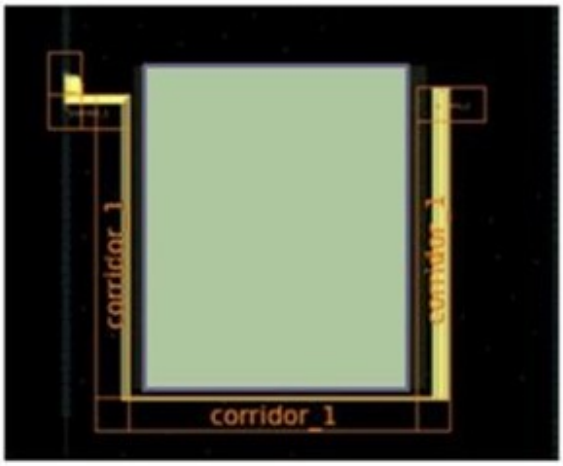
19. What is electron migration and how to reduce EM?
20. What is antenna effect?How to eliminate this?Is there any violation if (Max antenna ratio=400;gate area=1sq. unit;Metal area connecting to gate=500sq. unit)

LABS TYPE QUESTIONS

1. What is route guide ?
   Select one net say for e.g n1836 Find the layers on which these nets are routed. Create corridor for these nets within a corridor box for eg snapshot



Before routing



After routing

2. Write TCL program to find Min_spacing , min_width , pitch for all metal layers

3. Explain checks to be performed after routing and methods to fix it .

4. Write shortcuts for following operation
   a) For manual routing
   b) For cutting metal layer

c) For streching metal layer
d) For deleting metal layer
e) For enabling only metal 5
f) Creating via.

5. Mention checks to be performed after each steps
   a) Sanity checks after loading all input files
   b) Checks after port placement
   c) Checks after macro placement
   d) Checks after placement of end cap and tap cells
   e) Checks after power planning

VIII. PHYSICAL VERIFICATION

SUBJECTIVE TYPE QUESTIONS

1. What are the types in Physical Verification? What LVS errors you came across ?
2. What is DFM and why is it needed?
3. What happens if DFM is not met?
4. What are the inputs to GDSII file?
5. What is crosstalk, Which checker is used to detect the crosstalk?
6. What is the difference between gate level netlist and layout netlist?
7. What does the length of channel refer to?
8. Explain transfer characteristics of a CMOS Inverter.
9. What are lambda based design rules?
10. What is nm in 10nm technology node?
11. Do you know about under bum density?
12. Why ID layers have been given in lower nodes (10nm)?
13. Why nwell continuity is required?

DFM

21. Why Density needs to be maintained?
22. What kind of densities are there?
23. How do you know the density of each cell?
24. If there are 100 plus density windows, then how to fix it?
25. What happens if there are min or max density during fabrication?
26. Whats are the projects you have done related to synthesis and Physical Design?

DRC

27. What is DRC?
28. Will the design be clean if halos are places properly in design?
29. What are the types of DRC?
30. What kind of base DRCs have you encountered?

31. Have you seen DRCs between the standard cell and filler cell of the same height?
32. How did you solve DPT (double patterning)?
33. What metal DRC have you encountered?
34. What kind of via DRCs have you come across?
35. What will happen if GDSII has DRC unfixed and sent to Fab?
36. Is there any problem if the ID layers are interchanged?
37. Which ID layer is manufactured first?

LVS

38. What is LVS What it checks for?
39. What are the inputs and outputs of LVS?
40. What are hierarchical shorts?
41. What are the inputs to generate the OASIS file?
42. What is the full form of OASIS?
43. What is the difference between ICC LVS and ICV LVS?
44. In what format does the tool calculate the LVS?
45. What kind of issues are seen in LVS?
46. What is priority errors in LVS? Could there be false violations?
47. Where does the tool start to calculate LVS from?

ECO Implementation

48. What are types of ECOs?
49. Where the ECOs generated and given to you or you prepared ECOs?

Physical Only Cell

50. How does halo cell avoid DRCs t boundary? What is there inside the halo cell that prevents DRCs ? How it is different from standard cells?
51. What difference it makes when you add halo cell in boundary instead of standard cell?
52. Why is the standard cell nwell bigger in size and slightly coming out of the standard cell? It is not so in halo cell.

53.Suppose a tap cell is covering 10um distance, then where should the next tap cell be placed in the same row?

54. How do you calculate the distance between tap cells in a row?

_____END_____

VLSIGURU

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VLSI PHYSICAL DESIGN

INTERVIEW QUESTIONS

2023

#25, 2ND MAIN ROAD, VIJAYA BANK COLONY EXTENTION, HORMAVU, BENGALURU, KARNATAKA 560043.

SYNTHESIS and SDC
Explain What is Synthesis?
Explain input files for Synthesis
Explain the role of link library during synthesis
Explain the role of target library during Synthesis
Explain the Significance of wire-load model during synthesis
Explain on what factors cell delay depends.
Explain steps in synthesis
Explain Significance of synthetic library and design ware library
Explain difference between RTL and Gate level netlist
Write setup and Hold equation during Synthesis
Explain Why hold analysis is of not much important before CTS
Explain difference types of clock in design and how to declare it.
Write commands to declare generated clock from master clock (Divide by 2 etc)
Explain significance of set_input_delay and set_output_delay in SDC file
Explain how to declare set_input_delay and set_output_delay in SDC file
Explain why set_input_transition/set_driving_cell used in SDC file
Explain why set_load is used in sdc file and how to declare it .
Explain checks to be performed after synthesis
Explain how to check consistency of netlist after synthesis
Explain different methods used to improve timing after synthesis
Explain how to use UPF and how to add level shifters during synthesis
Explain how to add retention and isolation cells during synthesis.
Why clock is ideal during synthesis

Explain different methods to reduce power during synthesis
Explain how to decide max_operating frequency
In a circuit, for reg to reg path ...Tclktoq is 50 ps, Tcombo 50ps, Tsetup 50ps, tun is 10ps. Then what is the maximum operating frequency?
Explain what in intrapolation and extrapolation .
Explain why we use set_multicycle_path during synthesis
Whether we can synthesize without using SDC file ?
What is difference between logical synthesis and Physical aware synthesis
What are all inputs for physical aware synthesis
Why topographical mode dc_shell is used during synthesis
How to use physical information of cells during physical aware synthesis or What is milkyway library
Explain steps in physical aware synthesis
What is DEF file ? Why DEF file is used during Physical aware synthesis
What happens if floor plan DEF is not give during Physical aware synthesis
What are all checks to be done after physical aware synthesis
Why tlu+ file can be used only during physical aware synthesis and cannot be used during logical synthesis
Explain outputs of physical aware synthesis
Explain how to use results of Physical aware synthesis in floorplan
Explain how STA differs after synthesis, After placement , After CTS and After routing
Explain how to use macros during synthesis
How to handle unmapped components during synthesis
How to handle if we have assign statements in synthesized netlist.
Brief out the contents of SDC file

INPUT FILES FOR PD
Mention/Explain role of input files for PD
Explain difference between .db and .lib file
Why we have many .lib files and only one lef file
Explain on what factors delay of a cell depends in .lib file
Explain on what factors short circuit power and switching power depends in .lib file
On what factor setup time and hold time of a flip-flop depends on ?
What is difference between CCS model and NLDM model
Difference between wireload model, tlu+ files and SPEF file
What Is top, enclosed and segmented in wireload model
Explain what is difference between LEF and GDSII file
Why complete base layer information is not used in LEF file
If tool needs allowed orientation of cell it will get information from which file ?
max_cap and max_trans information is present in which file
tile /site row information is present in which file ?
If pitch of metal layes is 0.5um , min_spacing is 0.3um and min_width is 0.3um is this a valid metal layer rule
Why we have mapping file .
Why cmax and cmin tlu+ files
Explain difference between rtl and netlist
How to find a top module in a netlist
What is hierarchal design and flat design
What is difference between cell_name/instance and ref_name/lib_name in netlist
What do you mean by block box in design

Why thick metal layers have more spacing
Why upper metal layers have more pitch/width
What is difference between fanout and drive strength of a cell
Why HVT cells are prefer more compared to LVT cells ?
Explain how do you find fanout of a cell in netlist using gvim/unix commands
What do you mean by create_generated_clock -combination/inverter
In which file to declare multicycle path and why ?
Explain content of UPF for multivoltage design/power_gating_design
Whether we can do floorplan without SDC and UPF ? Justify your answer
Explain significance of scandef file during physical design
Whether we can do Physical design without scandef file ? Justify your answer
What is NDM . Explain library and block in ICC2
What are input files needed to create NDM .
If any ndm is missed during creation of lib how do you add it
Explain different sanity checks done after import design
If there is unresolved references after loading netlist. How do you handle it .
Explain checks performed by check_design command
Explain checks performed by check_mv_design
If any error displayed during chek_mv_design , how do you handle it ?
Explain checks performed by command check_scan_def
Explain checks performed by command check_timing
check_timing shows few flip-flop don't have clock in func mode? How do you handle it
Why check_timing shows false_path between Q to SI pin of all flip-flop in func mode
If launch and capture flip-flop is using different non related clock how you handle it . How it is reported in check timing

FLOORPLAN PORT_PLACEMENT MACRO_PLACEMENT

Questions
Explain difference between Full chip design floorplan and block level floorplan
What are the initial sanity checks in the design before going to floorplan (it is already discussed Sanity checks after import design)
Explain terms Utilization, Core-offset and aspect ratio
Why we use io-pads in full chip design and ports in block level design
Write ICC2 command to create core area for utilization 0.75, for L-shape and with all side lengths same
Write ICC2 command to create core area for given co-ordinates
Write TCL program to find utilization for already created core area
If utilization is very high? What problem we face in future
Whether block owner has flexibility of changing core and die area? Justify your answer
Why we use io-pads in full chip design and ports in block level design
Why die to core spacing required in all sides
Whether we can use -boundary and -utilization at same time in initialize_floorplan command
What is DEF file and why it is used during floor-planning
What are site_rows and routing tracks
Why we have preferred and non-preferred routing directions for any metal layer
What are the inputs for floor-planning
What are all the checks performed after port-placement
Why we have to place ports only on routing tracks
What is pad-limited design and core limited design
During IO pad placement, If Length of Side is 1185 um and you have to place 9 io-pads with height 65um and 2 corner cells with height 250 um . Find spacing between io pads

Why ESD protection is used in input ports
What are macros?
What are different types of macros
What are different Analog and memory macros used in design
What is between hard macro and soft macro
Why Macros should be placed at the periphery of the block?
What factors determine spacing between macros
Explain guidelines to be followed during macro placement
Explain different types of placement blockages in the design
What type of placement blockage is used in macro channel and why.
What is keep out margin and why we use around macros
What is difference between keep out margin and Placement blockage
What factors determine the keep out margin distance
What are notches Macro placement

CMOS , LOW POWER TECHNIQUES AND PHYSICAL ONLY CELLS
Questions
Explain short channel effect and Why Finfets are used at lower technology
Explain what is leakage power and on what parameters it depends on?
Explain What is short-circuit power and on what parameters it depends on?
Explain what is switching power and on what parameters it depends on?
What is UPF File and why we need it ?
Mention different power reduction techniques used in design
Explain Why we use Multi-voltage design
Explain Why we use Level shifter in Multi-voltage design
Explain in what power domain we place low_to_high and high_to_low level shifter
Explain contents present in UPF for multi-voltage design
Why guard band is used while creating voltage area
What is power gating design and why it is used
What are power switches and where we place these power switches
What are retention cells and in which power domain we place these cells
What are isolation cells and in which power domain we place these cells
What is clock gating design
What type of power we address in multivoltage , clock gating and power gating design
Why we use different types of Vt cells in design
What check check_mv_design will do
Why we use end cap cells and where we place them

At what stage we have to place end cap cells
Why we use TAP cells and where we place them
At what stage we place TAP cells
Why we use TIE cells and where we place them
At what stage we place TIE cells
Why we cannot place TIE cells before placement
Why we use Decap cells and at what stage we place these cells
What is the advantage of using spare cells
At what stage we can add spare cells
What are filler cells
Why we add filler cells and at what stage we add filler cells in design

POWER PLANNING and STA-PART 1 (Skew Latency Uncertainty setup and Hold)

SI NO	Questions
1	Explain Power Planning at Full chip level design
2	Explain Power Planning at Block level design
3	Explain How to decide width power rings
4	Explain who to decide width and spacing of power straps
5	Explain Why power straps are formed by using higher metal layers
6	Explain how double height cells get power
7	What is IR drop ? What are steps taken to reduce IR drop during power planning
8	How you will fix EM violations. What are the step taken to minimize Electromigration
9	After adding power straps if you have hot spot what to do?
10	How to do power planning for multi voltage design?
11	How to do power planning for multi voltage design?
12	Why gate of standard cells cannot be connected to rail voltage and how to eliminate this problem
13	Explain how to do power planning to macros
14	Explain how to fix floating rail violations
15	Explain how to fix floating macro power pin violations?
16	If rails are touching macro pins how to avoid it?
17	Explain What is local skew and global skew
18	Explain Setup and hold violation with equation
19	Explain how skew affects setup and hold
20	Explain what is Jitter and how it affects set up hold
21	Write equations for setup and hold using latency and uncertainty
22	Write equation for uncertainty before and after CTS and Explain why?

23	Explain Why hold analysis is given more importance after CTS
24	Explain How to do STA for in to reg and reg to out path
25	Explain how HVT LVT and RVT cells affect timing and power
26	Explain how drive strength of cells affects timing
27	Explain on what parameters setup and hold time of flip-flop depends
28	Whether setup and hold time can be negative? Justify your answer
29	Explain why virtual clock used in SDC file
30	Explain what is generated clock? What circuit is used to increase or decrease frequency.

STA

Questions
Explain what is max_trans and max_cap violation
Explain methods to fix max_cap and max_trans violation
Explain PVT conditions, How it affects cell delay
Explain parasitic corners, cmax , cmin , rcmax and rcmin
Explain modes, corners and scenarios
Explain corner used for power analysis
Explain setup and hold corners
Explain what is false path and in which scenarios we use them
Explain what is half cycle path
Explain how half-cycle path affects set and hold analysis
Explain multicycle path
Explain how to define multicycle path for setup and hold
What is PBA and GBA analysis and at what stages we use it
What are methods to fix max_trans and max_cap violations
DRV, setup and hold which has highest priority
Explain What is path grouping
Explain methods to fix setup and hold at each stage of PD
Explain What is derate and explain flat derate, AOCV and POCV derating techniques
Explain cross talk glitch and cross talk delay
Explain how to reduce cross talk
Explain what do you mean by useful skew method for fixing setup
Explain time borrowing technique to fix setup

Explain CCD flow to fix set and hold violations
Explain why scan chain path is more prone to hold violation
Explain how RC values are calculated during synthesis, during PnR and during signoff
Explain inputs for STARRC for SPEF extraction
Explain inputs for prime_time
Explain ECO flow for timing
Explain what is physical aware ECO
What is bottle neck cell
Explain where we add clock gating cell either near source or sink
Explain what is CRPR
Explain what is advantage of multipoint CTS
In same path if setup and hold is violating by small margin. How to fix it ?
If violation is not sent in ICC2 but more slack is violated in prime time. Explain what is reason for that

	Placement
SI NO	Questions
1	Explain what is placement
2	If tool cannot do legalize placement? What are the reasons for that ?
3	Explain what is coarse placement and checks done after coarse placement
4	Explain what is legalize placement will do ?
5	If tool cannot do legalize placement ? What are the reasons for that ?
6	Explain why we do zero interconnect timing check before placement
7	Explain what is scan chain reordering
8	Explain at what stage TIE cells are inserted
9	Explain TIE cell optimization
10	Explain different inputs for placement
11	Whether we can do coarse placement without scandef file ?
12	Explain different app options used/ constraints for placement
13	Explain what is cell density and pin density
14	Explain what is congestion and Explain methods to reduce for congestion
15	Explain what is High fanout net synthesis
16	Why we should not give low fanout limit : Ans : tool will insert more buffers to split fanout
17	Explain what is load splitting and cloning
18	Explain what is module splitting and how it can be avoided
19	Explain What are methods used to max cap and max trans violations
20	Explain what is global routing
21	What is magnet placement

22	Explain advantages of bounds in Placement stage
23	What is multibit banking and why we use it and what are the advantages and disadvantages of that
24	Can we consider clock in HFN synthesis during placement stages?
25	Write commands to upsize driver or insert buffer manually
26	Explain why hold scenario is not considered during placement
27	Explain methods to fix setup violations after placement
28	Explain what is path grouping and how to use it during placement
29	Explain different checks after placement
30	Explain what is SAIF file and why we use during placement
31	Explain placement of double height cell

CTS

Questions
Explain checks to be done before CTS
What are contents of CTS specification file
What are clock tree exceptions
Why clock nets are given more priority over other nets
What do you mean by skew group
Why we use NDR Rule definition.
What are clock buffers ? How they differ from normal buffers
What circuits are used to increase clock frequency
What circuits are used to decrease clock frequency
How many clocks you have in your design? Why we use virtual clock
What is insertion delay? How to have minimum insertion delay
What is clock skew ? How to reduce clock skew ?
Why we use clock gating and What Is clock gating violation
What is pulse width violation
What CCD flow
What is useful skew method to fix Setup violation
How insertion delay affects setup and hold
Where we have to add clock gating cells? Near source end or destination end for timing to be good
Where we have to add clock gating cells ? Near source end or destination end for power to be good
How to fix hold violations
What is multipoint CTS

If same path is violating for setup and hold ? How to fix it .
Explain What is CRPR ?

Routing
Questions
What are pre-route checks
What is global routing
Why do we use alternate layer horizontal/vertical/horizontal (HVH or VHV) routing?
What is routing pitch
What is grid based routing
What is the basic difference between 9 track and 12 track in standard cells?
Explain steps in routing
How do you take care electromigration, crosstalk etc during routing
What is antenna effect and how to reduce antenna violations
What is search and repair in routing
What is routing blockage? Why do we need them?
Explain how to fix cross talk after routing
What are the checks done after routing
How to fix open and shorts after routing
Explain cross talk and What are the methods used to reduce cross talk
What are different DRC violations after routing and how to fix it
What is ECO routing
What are secondary PG connections in routing
How Level shifters or AON cells get secondary power connections
What is different between net and shape
What is LVS and what checks are done during LVS check
What may be the reasons for more shorts after routing
What is buffer on routes?

Questions
REDHAWK - IR Drop/Power and Electromigration , ECO and TCL
Explain on what factors swithcing power depends
Explain on what factors leakage power depends
Explain on what factors internal power depends
At what stages we do IR drop analysis
What is APL . What are contents of APL
What is CMM File
How current limit is obtained for a metal layer
What is VCD and Fsbd file
What is timing window file
What is Static Analysis
What are different steps in obtaining static IR drop
Explain types of Dynamic analysis
What is Dynamic vectorless analysis
What is Dynamic vectored analysis
What are input files for IR drop analysis
What is true time and non-true time fsdb
What is dynamic simulation time and dynamic time step in vectored analysis
What are states for power switch
Explain about timing ECO
Explain input for Prime time
Explain DMSA mode in pt_shell
Write TCL script for finding high fan out nets

Write TCL script on ECO fixing like inserting buffers on high fan out nets
Write TCL script on inserting antenna diodes to fix antenna violations.
Write TCL script for port placement based on given co-ordinates or location
Write TCL script for Upsizing driver for DRV violated net
Write TCL script for inserting buffer for DRV violated net
Write TCL script for getting common instance in most violated paths
Write TCL script for Extracting Start point and end point from violated timing paths
Write TCL Script for filtering Arrival time and Required time from violated paths
Write TCL script for cloning to split fanout of a net
Write TCL Script to filter warning and error messages in log files
Write TCL script to find number of layers in net
Write TCL script to automate fixing of violated nets after routing
Write TCL script to print flip-flops with and without clock gating cells